



Statistical Inferences

(Properties of Good Estimator)

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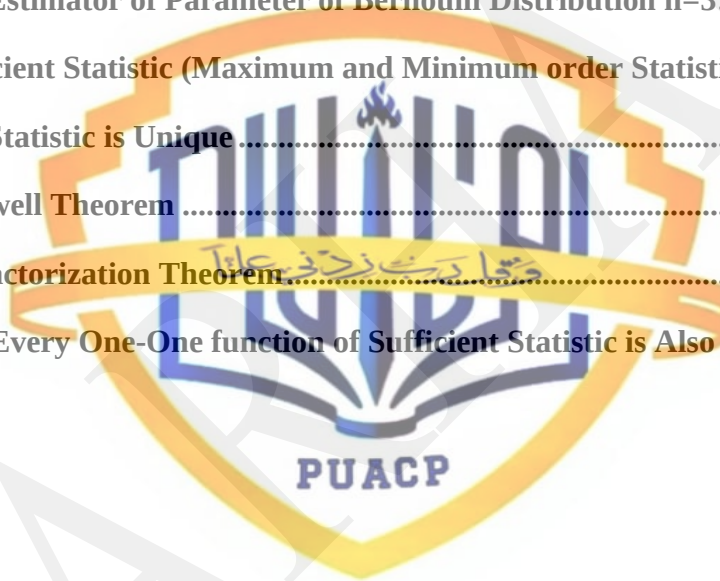
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1.0 Statistical Inferences

The process of finding drawing the conclusion about the population in the basis of information contained in the sample, taken from the population is called statistical inferences.

1.0.1 Types of Statistical Inferences

Statistical inference is traditionally divided into two major areas.

(1) Estimation

(2) Testing of Hypothesis

1.1 Estimation

- There are few occasions when population is studied as a whole.
- Generally, a sample is drawn from the population and population constants are determined.
- Population constants are usually those which occur in population density/mass/moments/ or some other constants like mean or median.
- Various sampling procedures do exist to determine the population constants.
- Hence, estimation is a procedure by which we obtain an estimate of the true but unknown value of population parameter by using the sample observation from the population.

1.1.1 Estimator

“A known function $T = t(X_1, X_2, \dots, X_n)$ of the observed variates of a random sample $X_1, X_2, \dots, X_n$, whose values are used to obtain the estimates of a parameter or function of parameter is called estimator”.

Stated in simple words, “An estimator stands for the rule or method that is used to estimate the parameter”.

For example: sample mean $\bar{X} = \frac{\sum X}{n}$ is the estimator of population mean μ .

- An Estimator itself is a random variable.

1.1.2 Estimate

“If $x_1, x_2, \dots, x_n$ are the values of the random sample $X_1, X_2, \dots, X_n$, the value $t(x_1, x_2, \dots, x_n)$ of the estimator T_n is known as an estimate of the parameter θ .”

Stated differently, “An estimate stands for the numerical value obtained by substituting the sample observations in the rule or formula”

For example, $\frac{\sum x}{n}$ is the estimator and $\bar{X}$ is the estimate.

1.1.3 Point Estimator

With the help of sample observations we find a value, which is taken as the value of the parameter θ .

1.1.4 Point Estimate

The value of the point estimator obtained by using the random sample is known as point estimate. For example, the value $\bar{x}$ of the statistic $\bar{X}$ computed from sample of size n is a point estimate of the population parameter μ .

1.2 Properties of Good Point Estimator

- The goodness of the estimator is governed by certain properties.
- An estimator possessing the maximum properties is considered good estimator.
- In general, estimator T_n will be rated as good if it differs from $t(\theta)$ by a small quantity ε .
- The merits of an estimator are judged by the properties of the sampling distribution of the estimator.
- The various properties of the estimator whose fulfillment is desired are as follows
 - Unbiasedness.
 - Consistency.
 - Efficiency.
 - Sufficiency.
 - Invariance.

1.2.1 Unbiasedness

“An estimator is defined to be unbiased if the statistic used as an estimator has its expected value equal to the true value of the parameter being estimated”. Let θ be parameter and $\hat{\theta}$ be its estimator. Then $\hat{\theta}$ will be unbiased estimator of θ if

$$E(\hat{\theta}) = \theta$$

Unbiasedness is the property that requires that the probability distribution of $\hat{\theta}$ be necessarily centered at the parameter θ irrespective of value of n

The estimator will be considered **Biased Estimator** if

$$E(\hat{\theta}) \neq \theta$$

The **Bias** of the estimator is given as

$$Bias(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- $Bias(\hat{\theta}) = 0$ if $E(\hat{\theta}) = \theta$. i.e. the estimator is unbiased
- $Bias(\hat{\theta}) < 0$, if $E(\hat{\theta}) < \theta$. i.e. **Negatively Biased Estimator**
- $Bias(\hat{\theta}) > 0$, if $E(\hat{\theta}) > \theta$. i.e. **Positively Biased Estimator**

If the estimator is biased The **Mean Square Error** of the estimator is given as

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + (Bias)^2$$

When we have to compare the performance of the biased estimators of the same parameters, we considered their MSEs. An biased estimator having less MSE will be preferred.

If $Bias = 0$, then $MSE(\hat{\theta}) = Var(\hat{\theta})$

- If the bias of the estimator approaches zero as sample size approaches infinity the estimator is called **Asymptotically Unbiased Estimator**

Examples of Unbiased Estimators

- (1) Sample mean $\bar{X}$ is unbiased estimator of population mean i.e. $E(\bar{X}) = \mu$.
- (2) sample median $\tilde{X}$ is unbiased estimator of population mean when population is normally distributed i.e. $E(\tilde{X}) = \mu$.
- (3) Sample variance s^2 is unbiased estimator of population variance σ^2 . i.e. $E(s^2) = \sigma^2$.
- (4) Sample proportion p is unbiased estimator of population proportion π . i.e. $E(p) = \pi$.

1.2.2 Consistency (Simple Consistency)

- The concept of consistency was introduced by R.A Fisher.

“An estimator is said to be consistent if the statistic to be used as estimator becomes closer and closer to the population parameter being estimated as the sample size n increases”.

Or

1.2.2.1 Strong and Weak Consistency

“An estimator $\hat{\theta}$ is called a consistent estimator of parameter θ if the probability that $\hat{\theta}$ becomes closer and closer to θ , approaches unity with the increasing sample size”. In other words, $\hat{\theta}$ converges to θ in probability terms if sample size approaches infinity. Symbolically we can write as

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$$T_n \xrightarrow{P} t(\theta)$$

$$\lim_{n \rightarrow \infty} p[|\hat{\theta} - \theta| \leq \epsilon] = 1 \dots \dots \dots (1)$$

$$\lim_{n \rightarrow \infty} p[|\hat{\theta} - \theta| > \epsilon] = 0 \dots \dots \dots (2)$$

This consistency is referred as the **strong consistency of the estimator** if the condition (1) is full filled. And the consistency is considered **weak consistency** under condition (2).

- Consistency is a limiting property of estimator and it reveals the behavior of an estimator T_n as n approaches infinity.

1.2.2.2 Mean square Consistency

A biased estimator $\hat{\theta}$ is said to be consistent in mean square of the population parameter θ if and only if its mean square error approaches zero as sample size approaches infinity.

$$MSE(\hat{\theta}) = E[\hat{\theta} - \theta]^2 \rightarrow 0, \text{ As } n \rightarrow \infty$$

- A mean square consistent estimator is also a simple consistent estimator, but not vice versa.

1.2.2.3 Choosing Consistent Estimator

- A consistent estimator may or may not be unbiased.
- Consider a example. $\frac{\sum X_i}{n}$, $\frac{\sum X_i}{n-1}$ and $\frac{\sum X_i}{n-2}$ all are consistent estimators of the population means, but their selection on the basis of consistency is difficult. We have to look for other properties as when, such as unbiasedness.

Consistency may be defined in terms of the variance of the sampling distribution of the estimator as

“ If the variance of the estimator approach to zero as sample size approach to infinity “.

$$var(\hat{\theta}) \rightarrow 0, \text{ as } n \rightarrow \infty.$$

1.2.2.4 Examples of consistent Estimators

(1) Sample mean $\bar{X}$ is unbiased estimator of population mean as

$$\lim_{n \rightarrow \infty} Var(\bar{X}) = \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0$$

(2) Sample proportion $\hat{p}$ is unbiased estimator of population proportion as

$$\lim_{n \rightarrow \infty} Var(\hat{p}) = \lim_{n \rightarrow \infty} \frac{PQ}{n} = 0$$

1.2.2.1 Mean Square Consistency

“A biased estimator $\hat{\theta}$ of parameter θ is said to be mean square consistent estimator if its means square error as well as bias approaches to zero as sample size approaches to infinity”.

$$\lim_{n \rightarrow \infty} [E(\hat{\theta}) - \theta]^2 = 0$$

- Mean square consistency implies that the estimator $\hat{\theta}$ is unbiased as well as variance of $\hat{\theta}$ approach to zero as sample size approach to infinity.

1.2.3 Efficiency

“An unbiased estimator is defined to be efficient if the variance of its sampling distribution is smaller than the variance of any other unbiased estimator of same parameter”.

Suppose we have two unbiased estimators T_1 and T_2 of the same parameter θ . Then T_1 will be efficient estimator of θ if

$$Var(T_1) < Var(T_2)$$

The **Relative Efficiency** of T_1 as compare to T_2 is given as

$$R_{ef} = \frac{Var(T_2)}{Var(T_1)}$$

If T_1 is efficient then $R_{ef} > 1$.

Example: Both the sample mean and median are unbiased estimators of population mean of normal distribution. But the sample mean is efficient estimator as compare to sample median because the variance of sampling distribution of sample mean is less than the variance of sampling distribution of median.

$$R_{ef} = \frac{Var(\tilde{X})}{Var(\bar{X})} = \frac{\frac{\pi\sigma^2}{n}}{\frac{\sigma^2}{n}} = \frac{\pi}{2} = 1.57 > 1$$

1.2.4 Sufficiency

- Concept of sufficiency was given by R.A Fisher

“An estimator is said to be sufficient if the statistic used as an estimator makes use of all the information present in the sample”.

- Any estimator which is not computed by using all the values in the sample is not a sufficient estimator.
- Sufficient estimator provides us maximum information as it is close population parameter, so it also measures variability.

Example: Sample mean is sufficient estimator of population mean but sample median is not sufficient consider the data in ascending order

1,2,3,4,5,6,7

The mean and median of the data is 4. If we add two new values 0 and 8 in the data as

0,1,2,3,4,5,6,7,10

Now, the mean becomes 4.22 and median still remains at 4. It is clear that knowing the median of the data we still need to have more information about the data as changes at both tails of the data will not alter value of median. Whereas, once we know the mean of the data, no more information needed to be extracted from the data because mean is based upon all the values of the data.

Second Definition

“A statistic $S = s(X_1, X_2, \dots, X_n)$ is sufficient for parameter θ of density $f(x; \theta)$ if the conditional distribution of $X_1, X_2, \dots, X_n$ for a given $S = s$ is independent of parameter θ ”.

- From this definition it is clear that one cannot expect to get an information about θ from a distribution that does not contain θ .

1.2.4.1 Neyman Factorization Theorem

Third Definition (Fisher Neyman Factorization Theorem)

“Let $X_1, X_2, \dots, X_n$ be a random sample from a density $f(x; \theta)$. A statistic $S = s(X_1, X_2, \dots, X_n)$ is said to be sufficient statistic for θ if the joint probability density function of $X_1, X_2, \dots, X_n$ can be factorized as

$$f_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = g\{s(x_1, x_2, \dots, x_n); \theta\} h(x_1, x_2, \dots, x_n) = g(s, \theta) h(x_1, x_2, \dots, x_n)$$

Here, $g(x; \theta)$ is function containing the “s” the sufficient statistic and θ the parameter and $h(x_1, x_2, \dots, x_n)$ is the function independent of θ .

1.2.4.2 Minimal Sufficient Statistic

A set of sufficient statistics is minimal if no other set of sufficient statistics condenses the data more than it.

More technically a set of jointly sufficient statistics is defined to be minimal sufficient if and only if it is a function of every other set of sufficient statistics, i.e.

$$T(X_1, X_2, \dots, X_n) = k[t(X_1, X_2, \dots, X_n)]$$

For every statistic $t(\cdot)$.

This definition of minimal sufficient statistic is of little use, that is if the joint p.d.f is properly factorized, it will give a minimal sufficient statistic also.

1.2.5 Invariance

“ If the function of the parameter changes, the estimator also changes with the same functional application , this property is known as invariance”.

- This property is related to the coding of the data.
- Location Invariant: $T = t(X_1, X_2, \dots, X_n)$ will be location invariant if

$$t(x_1 + c, x_2 + c, \dots, x_n + c) = t(x_1, x_2, \dots, x_n) + c$$

Where “c” is constant.

Suppose we have parameter μ and statistic $\bar{x}$. If the parameter changes as $\mu + c$,then the estimator $\bar{x} + c$.

- Scale Invariant: $T = t(X_1, X_2, \dots, X_n)$ will be scale invariant if

$$t(x_1c, x_2c, \dots, x_nc) = ct(x_1, x_2, \dots, x_n)$$

Where “c” is constant.

Suppose we have parameter μ and statistic $\bar{x}$. If the parameter changes as $c\mu$, then the estimator $c\bar{x}$.

- In general, we say that if $\hat{\theta}$ is estimator of θ , then the estimator $T(\hat{\theta})$ is the estimator of $T(\theta)$.
- Any one-one functional relationship between parameter and statistic will ensure the invariant property.

Example : If

$$E(X - \mu)^2 = \sigma^2$$

Then

$$\sqrt{E(x - \mu)^2} = \sigma \quad \text{or} \quad [E(x - \mu)^2]^2 = (\sigma^2)^2$$

1.3 Regularity Conditions

Let Θ be an open interval of the real line and $\{f(\theta, x), \theta \in \Theta\}$ be a family of p.d.f's or p.m.f's.

(i) $\frac{\partial}{\partial \theta} \ln f(x; \theta)$ exist for all x and θ .

(ii) The operation of integration w.r.t x and differentiation w.r.t θ can be interchanged and they yield the same results. Also the operation of integration and differentiation can be interchanged in the expectation of $t(\theta)$.

(a) If $f(x; \theta)$ is p.d.f

$$\frac{\partial}{\partial \theta} \int \dots \int \prod_{i=1}^n f(x_i, \theta) dx_1 dx_2 \dots dx_n = \int \dots \int \frac{\partial}{\partial \theta} \prod_{i=1}^n f(x_i, \theta) dx_1 dx_2 \dots dx_n$$

(b) If $f(x; \theta)$ is p.d.f and $t(x_1, x_2, \dots, x_n)$ is estimator.

$$\frac{\partial}{\partial \theta} \int \dots \int t(x_1, x_2, \dots, x_n) \prod_{i=1}^n f(x_i, \theta) dx_1 dx_2 \dots dx_n = \int \dots \int t(x_1, x_2, \dots, x_n) \frac{\partial}{\partial \theta} \prod_{i=1}^n f(x_i, \theta) dx_1 dx_2 \dots dx_n$$

(c) If $f(x; \theta)$ is p.m.f, for $\mathbf{X}$ being vector

$$\frac{\partial}{\partial \theta} \sum_{\mathbf{x}} f(\mathbf{x}) = \sum_{\mathbf{x}} \frac{\partial}{\partial \theta} f(\mathbf{x})$$

(d) If $f(x; \theta)$ is p.m.f, for $\mathbf{X}$ being vector

$$\frac{\partial}{\partial \theta} \sum_{\mathbf{x}} t(\mathbf{x}) f(\mathbf{x}) = \sum_{\mathbf{x}} t(\mathbf{x}) \frac{\partial}{\partial \theta} f(\mathbf{x})$$

(iii) For all θ in Θ

$$0 \leq E \left[\left\{ \frac{\partial}{\partial \theta} \ln f(x; \theta) \right\} \right]^2 < \infty$$

These regularity conditions are necessary for existence of Cramer Rao lower bound.

1.4 Cramer Rao Lower Bound

Let $X_1, X_2, \dots, X_n$ be independent random samples each with p.m.f or p.d.f $f(x; \theta)$. Let $t_n(X_1, X_2, \dots, X_n)$ be an estimator if

$$L = \prod_{i=1}^n f(x_i; \theta)$$

And if regularity conditions are satisfied

For t_n an unbiased estimator of function θ

$$V(t_n) \geq \frac{[T'(\theta)]^2}{E \left(\frac{\partial \ln L}{\partial \theta} \right)^2}$$

Or

$$V(t_n) \geq \frac{[T'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]}$$

It is interesting to note that Cramer Rao inequality gives the lower bound of $V(t_n)$.

- If we do not use likelihood function then the Cramer Rao inequality will be of the form

$$V(t_n) \geq \frac{[T'(\theta)]^2}{nE\left(\frac{\partial \ln f(x; \theta)}{\partial \theta}\right)^2}$$

Or

$$V(t_n) \geq \frac{[T'(\theta)]^2}{-nE\left[\frac{\partial^2 \ln f(x; \theta)}{\partial \theta^2}\right]}$$

The Cramer Rao inequality (CR inequality) is also called Frechet Cramer and Rao (FCR) inequality.

1.4.1 Uses of Cramer Rao Inequality

- It gives the lower bound for the variance of an unbiased estimator.
- A researcher using an unbiased estimator whose variance is closed to the Cramer Rao lower bound would know that he is using a good estimator.
- It can give the lower bound for the variance of a biased estimator t_n , which is given as

$$V(t_n) \geq \frac{[T'(\theta) + b_n]^2}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2}$$

1.4.2 Uniformly Minimum Variance Bound (UMVB) Estimator

An unbiased estimator whose variance is equal to Cramer Rao lower bound is called Uniformly Minimum Variance Bound (UMVB) estimator. This estimator is also termed as **Most Efficient Estimator**.

Let $X_1, X_2, \dots, X_n$ be random sample from $f(x; \theta)$ and we have two unbiased estimators t_n and g_n of parameter $T(\theta)$. Then the estimator t_n will be UMVB estimator of $T(\theta)$ if and only if

$$V(t_n) < V(g_n)$$

Here g_n may be any other unbiased estimator of $T(\theta)$.

1.4.3 Asymptotically Efficient Estimator

An unbiased estimator whose variance approaches to Cramer Rao lower bound when sample size increases is called Asymptotically Efficient Estimator. The asymptotical efficiency is the limit of its efficiency when sample size approach infinity.

1.4.4 Attainment of Equality in MVB

The inequality sign in the expression of Cramer-Rao inequality arises purely from the use of Cauchy Shewarz inequality and the necessary and sufficient condition is that the Cauchy Shewarz inequality becomes the equality when $[t_n - T(\theta)]$ is proportional to $\frac{\partial \ln L}{\partial \theta}$. For all set of observation

$$\frac{\partial \ln L}{\partial \theta} \propto [t_n - T(\theta)]$$

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) [t_n - T(\theta)]$$

Where $A(\theta)$ is the constant of proportionality is independent of observation but may be function of θ .

Multiple both sides with $[t_n - T(\theta)]$ and taking expectation

$$E[t_n - T(\theta)] \frac{\partial \ln L}{\partial \theta} = A(\theta) E[t_n - T(\theta)]^2$$

$$T'(\theta) = A(\theta) \text{Var}(t_n)$$

Thus $A(\theta)$ will have same sign as $T'(\theta)$.

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)}$$

So, we conclude that if

$$\frac{\partial \ln L}{\partial \theta} = A(\theta) [t_n - T(\theta)]$$

is satisfy then t_n is MVB estimator of $T(\theta)$ with variance $\frac{T'(\theta)}{A(\theta)}$ which is equal to the R.H.S of Cramer Rao lower bound inequality.

1.4.5 Drawbacks of Most Efficient Estimator

1- The UMVUE may exist even if the regularity condition do not satisfy.

2-The definition of efficiency does not cover these cases where the regularity condition are satisfied but the UMVUE is not most efficient, because the variance of the estimator does not achieve the lower bound of CR inequality.

1.5 Best Asymptotically Normal Estimator (BAN)

A sequence of estimators T_n of various sample sizes say $T_n = t_n(x_1, x_2 \dots x_n)$ where n indicates the size of the sample. For sample $T_1 = t_1(X_1), T_2 = t_2(X_1, X_2), \dots, T_n = t_n(X_1, X_2, \dots X_n)$, for various sample sizes we will get different estimators, but out of those estimator a estimator will be Best Asymptotically Normal (BAN) if and only if the following conditions satisfy.

(i) The distribution of $\sqrt{n}(T_n - \tau(\theta))$ approach to normal distribution with mean zero and variance σ^2 .as sample size approach infinity.

$$\lim_{n \rightarrow \infty} \sqrt{n}(T_n - \tau(\theta)) \sim N(0, \sigma^2)$$

(ii) for any small quantity ε

$$\lim_{n \rightarrow \infty} P_\theta[|T_n - \tau(\theta)| > \varepsilon] = 0 \text{ for all } \theta \text{ in } \Theta$$

Same condition as consistency

(iii) if $\{T_n^*\}$ is any sequence of estimators other than $\{T_n\}$ such that

$$\lim_{n \rightarrow \infty} \sqrt{n}(T_n^* - \tau(\theta)) \sim N(0, \sigma^{*2})$$

Then $\sigma^2 \geq \sigma^{*2}$ for all θ in Θ in an open interval.

Since BAN estimators are consistent as well so they are sometimes given another name (CANE) Estimators.

1.6 Comparison Between Sufficiency And Minimum Variance Bound (MVB)

The necessary and sufficient condition for sufficiency is

$$L(x_1, x_2 \dots x_n; \theta) = \prod_{i=1}^n f(x_i; \theta) = k_1(t; \theta) k_2(x_1, x_2 \dots x_n) \dots \dots (1)$$

Where k_2 is independent of θ .

Taking ln both sides

$$\ln L = \ln k_1(t; \theta) + \ln k_2(x_1, x_2 \dots x_n)$$

Diff w.r.t θ

$$\frac{\partial \ln L}{\partial \theta} = \frac{\partial \ln k_1(t; \theta)}{\partial \theta} \dots \dots \dots (2)$$

The condition that MVB estimator exists is given by

$$\frac{\partial \ln L}{\partial \theta} = A(\theta)[t - T(\theta)] \dots \dots \dots (3)$$

On comparing eq (2) and (3) we see that a MVB estimator of $T(\theta)$, can only exist if there is a sufficient statistic.

In fact eq(3) is a special case of eq (2), I.e.

$$\frac{\partial \ln k_1(t; \theta)}{\partial \theta} = A(\theta)[t - T(\theta)] \dots \dots \dots (4)$$

Thus, sufficiency which at first sight seems a more restrictive criterion than attainment of minimum variance bound is in reality a less restrictive one.

For whenever eq(3) holds eq(2) holds.

Also while even of eq(3) does not hold we may still have a sufficient statistic.

2 Question Related to Unbiasedness

2.1 Finding Mean ,Variance and Distribution Function of Continuous Random Variable

Question1; The continuous random variable has the p.d.f $f(x) = nx^{n-1}$, $0 < x < 1$

Find the (i) $E(x)$ (ii) $var(x)$ (iii) The distribution function.

Solution;-

(i) Mean

$$E(x) = \int_0^1 x(nx^{n-1})dx = n \int_0^1 x^n dx = n \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{n}{n+1} [(1)^{n+1} - (0)^{n+1}] = \frac{n}{n+1}$$

$$E(x^2) = \int_0^1 x^2(nx^{n-1})dx = n \int_0^1 x^{n+1} dx = n \frac{x^{n+2}}{n+2} \Big|_0^1 = \frac{n}{n+2} [(1)^{n+2} - (0)^{n+2}] = \frac{n}{n+2}$$

(ii) Variance

$$Var(x) = E(x^2) - [E(x)]^2 = \frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2 = \frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2}$$

$$Var(x) = \frac{n[n^2 + 1 + 2n - n^2 - 2n]}{(n+2)(n+1)^2} = \frac{n}{(n+2)(n+1)^2}$$

(iii) Distribution Function

$$F(x) = \int_0^x nx^{n-1} dx = n \frac{x^n}{n} \Big|_0^x = x^n - (0)^n = x^n$$

2.2 Finding Mean and Variance of Continuous Random Variable

Question1:2 The continuous random variable has the p.d.f $f(x) = \frac{n}{x^{n+1}}$, $x > 1$

Find the (i) $E(x)$ (ii) $var(x)$

(i) Mean

$$E(x) = \int_1^\infty x \left(\frac{n}{x^{n+1}}\right) dx = n \int_1^\infty x^{-n} dx = n \frac{x^{-n+1}}{-n+1} \Big|_1^\infty = \frac{n}{1-n} \left[\frac{1}{(\infty)^{n-1}} - \frac{1}{(1)^{n-1}} \right]$$

$$E(x) = \frac{n}{1-n} [0 - 1] = \frac{n}{n-1}$$

$$E(x^2) = \int_0^{\infty} x^2 \left(\frac{n}{x^{n+1}} \right) dx = n \int_0^{\infty} x^{-n+1} dx = n \frac{x^{-n+2}}{-n+2} \Big|_0^{\infty} = \frac{n}{2-n} \left[\frac{1}{(\infty)^{n-2}} - \frac{1}{(1)^{n-2}} \right]$$

$$E(x^2) = \frac{n}{2-n} [0 - 1] = \frac{n}{n-2}$$

(ii) Variance

$$Var(x) = E(x^2) - [E(x)]^2 = \frac{n}{n-2} - \left(\frac{n}{n-1} \right)^2 = \frac{n[(n-1)^2 - n(n-2)]}{(n-2)(n-1)^2}$$

$$Var(x) = \frac{n[n^2 + 1 - 2n - n^2 + 2n]}{(n-2)(n-1)^2} = \frac{n}{(n-2)(n-1)^2}$$

2.3 Unbiased Estimator of Population Mean

Question 3; Show that the sample mean $\bar{X}$ is unbiased estimator of population mean μ .

Solution-

Let $X_1, X_2, \dots, X_n$ be random sample of size n from a population

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} [X_1 + X_2 + \dots + X_n]$$

Taking Expectation

$$E(\bar{X}) = \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] = \frac{1}{n} [\mu + \mu + \dots + \mu] = \frac{n\mu}{n} = \mu$$

This show that sample mean $\bar{X}$ is unbiased estimator of population Mean μ .

2.4 Biased Estimator of Population Variance

Question 4: Prove that $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is biased estimator of σ^2 . Also find the amount of bias.

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Add and subtract μ

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2 = \frac{1}{n} \sum [(X_i - \mu) - (\bar{X} - \mu)]^2$$

$$S^2 = \frac{1}{n} \sum [(X_i - \mu)^2 + (\bar{X} - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu)]$$

$$S^2 = \frac{1}{n} [\sum (X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2\sum (X_i - \mu)(\bar{X} - \mu)]$$

AS,

$$\sum(X_i - \mu) = \sum X_i - n\mu = n\bar{X} - n\mu = n(\bar{X} - \mu)$$

So,

$$S^2 = \frac{1}{n} [\sum(X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2n(\bar{X} - \mu)^2] = \frac{1}{n} [\sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2]$$

Taking Expectation

$$E(S^2) = \frac{1}{n} [\sum E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2] = \frac{1}{n} [\sum \sigma^2 - n\sigma_{\bar{X}}^2] = \frac{1}{n} \left[n\sigma^2 - \frac{n\sigma^2}{n} \right]$$

$$E(S^2) = \frac{1}{n} [n\sigma^2 - \sigma^2] = \left(\frac{n-1}{n} \right) \sigma^2$$

AS, $E(S^2) \neq \sigma^2$ so it is biased estimator.

$$\text{Bias} = E(S^2) - \sigma^2 = \left(\frac{n-1}{n} \right) \sigma^2 - \sigma^2 = \left(\frac{n-1-n}{n} \right) \sigma^2 = -\frac{1}{n} \sigma^2$$

Hence, S^2 is Negatively Biased.

2.5 Unbiased Estimator of Population Variance

Question 5: Prove that $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is unbiased estimator of σ^2 .

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Add and subtract μ

$$S^2 = \frac{1}{n-1} \sum (X_i - \mu + \mu - \bar{X})^2 = \frac{1}{n-1} \sum [(X_i - \mu) - (\bar{X} - \mu)]^2$$

$$S^2 = \frac{1}{n-1} \sum [(X_i - \mu)^2 + (\bar{X} - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu)]$$

$$S^2 = \frac{1}{n-1} [\sum (X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2\sum (X_i - \mu)(\bar{X} - \mu)]$$

AS,

$$\sum(X_i - \mu) = \sum X_i - n\mu = n\bar{X} - n\mu = n(\bar{X} - \mu)$$

So,

$$S^2 = \frac{1}{n-1} [\sum(X_i - \mu)^2 + n(\bar{X} - \mu)^2 - 2n(\bar{X} - \mu)^2] = \frac{1}{n-1} [\sum(X_i - \mu)^2 - n(\bar{X} - \mu)^2]$$

Taking Expectation

$$E(S^2) = \frac{1}{n-1} [\sum E(X_i - \mu)^2 - nE(\bar{X} - \mu)^2] = \frac{1}{n-1} [\sum \sigma^2 - n\sigma_{\bar{X}}^2] = \frac{1}{n-1} \left[n\sigma^2 - \frac{n\sigma^2}{n} \right]$$

$$E(S^2) = \frac{1}{n-1} [n\sigma^2 - \sigma^2] = \left(\frac{n-1}{n-1}\right) \sigma^2 = \sigma^2$$

2.6 Unbiased Estimator of Population Variance

Question 6: Given a random sample of size n from a population with mean μ and finite variance σ^2 . Show that $\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is unbiased estimator of σ^2 .

Solution

Let

$$T = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

Taking expectation both sides

$$E(T) = \frac{1}{n} \sum_{i=1}^n E(X_i - \mu)^2 = \frac{1}{n} [E(X_1 - \mu)^2 + E(X_2 - \mu)^2 + \dots \dots E(X_n - \mu)^2]$$

$$E(T) = \frac{1}{n} [\sigma^2 + \sigma^2 + \sigma^2 \dots \dots \sigma^2] = \frac{n\sigma^2}{n} = \sigma^2$$

Hence, T unbiased estimator of σ^2 .

Note:

$\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$ is unbiased estimator of σ^2 , but $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is not because here one parameter is estimated to replace μ by $\bar{X}$, so, the estimator will be unbiased if we adjust its degree of freedom therefore, $\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is unbiased estimator of σ^2 .

2.7 Unbiased Estimator of Population Mean (Two Independent Samples).

Question 7: If $\bar{X}_1$ and $\bar{X}_2$ are the means of independent random samples of sizes n_1 and n_2 respectively from $N(\mu, \sigma^2)$. Show that $\theta \bar{X}_1 + (1 - \theta) \bar{X}_2$, $0 < \theta < 1$, is unbiased estimator of μ .

Solution:-

Let

$$T = \theta \bar{X}_1 + (1 - \theta) \bar{X}_2$$

Taking Expectation Both sides

$$E(T) = \theta E(\bar{X}_1) + (1 - \theta) E(\bar{X}_2)$$

$$E(T) = \theta\mu + (1 - \theta)\mu = \theta\mu + \mu - \theta\mu = \mu$$

Hence, T is unbiased estimator of population mean μ .

Note: 1- If $E(T_1) = E(T_2) = \theta$, then the estimator $T = aT_1 + (1 - a)T_2$ will be unbiased for θ

2- If $E(T_1) = E(T_2) = E(T_3) = \theta$, then the estimator $T = aT_1 + bT_2 + (1 - a - b)T_3$ will be unbiased for θ . Similarly, $T = aT_1 + aT_2 + (1 - 2a)T_3$ is unbiased for θ

3- If $E(T_1) = E(T_2) = E(T_3) = E(T_4) = \theta$, then the estimator $T = aT_1 + bT_2 + cT_3 + (1 - a - b - c)T_4$ will be unbiased for θ . Similarly, $T = aT_1 + aT_2 + aT_3 + (1 - 3a)T_4$ is unbiased for θ .

2.8 Pooled Estimator of Population Mean and Variance.

Question 8: Random sample of n observations $X_i (i = 1, 2, \dots, n)$ is drawn from a population with mean μ and variance σ^2 , from sample we compute

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \text{ and } s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Another random sample of “ m ” observations $y_i (i = 1, 2, \dots, m)$ is chosen and from sample we compute

$$\bar{y} = \frac{1}{m} \sum_{i=1}^m y_i \text{ and } s_y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$

Show that $E(\bar{X}_p) = \mu$ and $E(s_p^2) = \sigma^2$

Solution:-

As we Know that

$$\bar{X}_p = \frac{n\bar{x} + m\bar{y}}{n + m}$$

Taking Expectation

$$E(\bar{X}_p) = \frac{nE(\bar{x}) + mE(\bar{y})}{n + m}$$

As $E(\bar{x}) = E(\bar{y}) = \mu$

$$E(\bar{X}_p) = \frac{n\mu + m\mu}{n + m} = \frac{(n + m)\mu}{n + m} = \mu$$

Hence, $\bar{X}_p$ is unbiased Estimator of μ .

Also we have

$$s_p^2 = \frac{(n - 1)s_x^2 + (m - 1)s_y^2}{n + m - 2}$$

Taking expectation

$$E(s_p^2) = \frac{(n - 1)E(s_x^2) + (m - 1)E(s_y^2)}{n + m - 2}$$

As $E(s_x^2) = E(s_y^2) = \sigma^2$

$$E(s_p^2) = \frac{(n - 1)\sigma^2 + (m - 1)\sigma^2}{n + m - 2} = \frac{(n + m - 2)\sigma^2}{n + m - 2} = \sigma^2$$

Hence, s_p^2 is unbiased estimator of σ^2 .

2.9 Standard Deviation is not Unbiased Estimator of σ

Question 9 : if $E(s^2) = \sigma^2$ then show that $E(s) \neq \sigma$ i.e. standard deviation is nt unbiased estimator of σ .

Solution:

By definition of variance

$$V(s) \geq 0$$

$$E(s^2) - [E(s)]^2 \geq 0$$

As $E(s^2) = \sigma^2$

$$\sigma^2 - [E(s)]^2 \geq 0$$

$$\sigma^2 \geq [E(s)]^2$$

Taking Square root

$$\sigma \geq E(s)$$

This implies that $E(s) \neq \sigma$

Hence , S.D is biased estimator of σ .

Note: greater of equal implies two situations, σ is greater than $E(s)$ or equal, for unbiasedness we need equal. Also $\sigma^2 \geq [E(s)]^2$ is not one-one relationship so invariance property will not be valid here.

2.10 Unbiased Estimator for Parameter of Exponential Distribution

Question 10: Show that the sample mean $\bar{X}$ of random sample of size n is unbiased estimator of parameter θ of exponential distribution.

Solution:-

The p.d.f of exponential distribution

$$f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, x > 0$$

$$E(x) = \int_0^{\infty} x \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = \frac{1}{\theta} \int_0^{\infty} x e^{-\frac{x}{\theta}} dx$$

Let

$$t = \frac{x}{\theta} \Rightarrow x = t\theta \Rightarrow dx = \theta dt$$

when $x = 0$ then $t = 0$ and when $x = \infty$ then $t = \infty$

$$E(x) = \frac{1}{\theta} \int_0^{\infty} (t\theta) e^{-t} (\theta dt) = \theta \int_0^{\infty} t e^{-t} dt = \theta \int_0^{\infty} t^{2-1} e^{-t} dt$$

Using Gamma Function

$$E(x) = \theta \Gamma 2 = \theta (2 - 1)! = \theta$$

Now taking

$$\bar{x} = \frac{1}{n} \sum x$$

Taking Expectation

$$E(\bar{x}) = \frac{1}{n} \sum E(x) = \frac{1}{n} n\theta = \theta$$

2.11 Unbiased Estimator for Parameter of Exponential Distribution

Question 10: Show that the sample mean $\bar{X}$ of random sample of size n from the density

$$f(x) = \theta^2 e^{-\theta^2 x}, x > 0$$

is unbiased estimator of its parameter .

Solution:-

The p.d.f of exponential distribution

$$f(x) = \theta^2 e^{-\theta^2 x}, x > 0$$

$$E(x) = \int_0^{\infty} x \theta^2 e^{-\theta^2 x} dx = \theta^2 \int_0^{\infty} x e^{-\theta^2 x} dx$$

Let

$$t = \theta^2 x \Rightarrow x = \frac{t}{\theta^2} \Rightarrow dx = \frac{dt}{\theta^2}$$

when $x = 0$ then $t = 0$ and when $x = \infty$ then $t = \infty$

$$E(x) = \theta^2 \int_0^{\infty} \left(\frac{t}{\theta^2}\right) e^{-t} \left(\frac{dt}{\theta^2}\right) = \frac{1}{\theta^2} \int_0^{\infty} t e^{-t} dt = \frac{1}{\theta^2} \int_0^{\infty} t^{2-1} e^{-t} dt$$

Using Gamma Function

$$E(x) = \frac{1}{\theta^2} \Gamma 2 = \frac{1}{\theta^2} (2-1)! = \frac{1}{\theta^2}$$

Now taking

$$\bar{x} = \frac{1}{n} \sum x$$

Taking Expectation

$$E(\bar{x}) = \frac{1}{n} \sum E(x) = \frac{1}{n} n \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

Note: Some other forms of exponential distribution.

Parameter	p.d.f	$E(\bar{x})$
θ	$\frac{1}{\theta} e^{-\frac{x}{\theta}}$	θ
$\frac{1}{\theta}$	$\theta e^{-x\theta}$	$\frac{1}{\theta}$

$$\theta^2 \quad \frac{1}{\theta^2} e^{-\frac{x}{\theta^2}} \quad \theta^2$$

$$\frac{1}{\theta^2} \quad \theta^2 e^{-x\theta^2} \quad \frac{1}{\theta^2}$$

$$\theta^k \quad \frac{1}{\theta^k} e^{-\frac{x}{\theta^k}} \quad \theta^k$$

$$\frac{1}{\theta^k} \quad \theta^k e^{-x\theta^k} \quad \frac{1}{\theta^k}$$

2.12 Functional Unbiasedness (Poisson Distribution)

Question 12: let X be Poisson random variable having Probability Mass Function

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Suppose $d(\lambda) = e^{-3\lambda}$, then show that $T(x) = (-2)^x$ is unbiased for $d(\lambda)$.

Solution:

$$E[T(x)] = \sum_{i=0}^{\infty} (-2)^x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{(-2\lambda)^x}{x!}$$

$$E[T(x)] = e^{-\lambda} \left[\frac{(-2\lambda)^0}{0!} + \frac{(-2\lambda)^1}{1!} + \frac{(-2\lambda)^2}{2!} + \dots \dots \dots \right]$$

$$E[T(x)] = e^{-\lambda} \left[1 + \frac{(-2\lambda)^1}{1!} + \frac{(-2\lambda)^2}{2!} + \dots \dots \dots \right]$$

Using $e^{-x} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \dots \dots$

$$E[T(x)] = e^{-\lambda} e^{-2\lambda} = e^{-3\lambda}$$

2.13 Functional Unbiasedness (Poisson Distribution)

Question 13: let X be Poisson random variable with parameter θ . Show that unbiased estimator of $T(x)$ is $e^{-(k+1)\theta}$ where $T(x) = (-k)^x$.

Solution:

$$E[T(x)] = \sum_{i=0}^{\infty} (-k)^x \frac{e^{-\theta} \theta^x}{x!} = e^{-\theta} \sum_{i=0}^{\infty} \frac{(-k\theta)^x}{x!}$$

$$E[T(x)] = e^{-\theta} \left[\frac{(-k\theta)^0}{0!} + \frac{(-k\theta)^1}{1!} + \frac{(-k\theta)^2}{2!} + \dots \dots \dots \right]$$

$$E[T(x)] = e^{-\theta} \left[1 + \frac{(-k\theta)^1}{1!} + \frac{(-k\theta)^2}{2!} + \dots \dots \dots \right]$$

Using $e^{-x} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \dots \dots$

$$E[T(x)] = e^{-\theta} e^{-k\theta} = e^{-(k+1)\theta}$$

Note: In Poisson distribution the functional unbiasedness is

Function Unbiased Estimator

$$d(\lambda) = (-2)^x \quad T(x) = e^{-3\lambda}$$

$$d(\lambda) = (-3)^x \quad T(x) = e^{-4\lambda}$$

$$d(\lambda) = (-4)^x \quad T(x) = e^{-5\lambda}$$

$$d(\lambda) = (-k)^x \quad T(x) = e^{-(k+1)\lambda}$$

$$d(\lambda) = (2)^x \quad T(x) = e^{\lambda}$$

$$d(\lambda) = (3)^x \quad T(x) = e^{2\lambda}$$

$$d(\lambda) = (k)^x \quad T(x) = e^{(k-1)\lambda}$$

2.14 sample Mean and Variance Unbiased Estimator of parameter (Poisson Distribution)

Question 14: let X be Poisson random variable having Probability Mass Function $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ Show that sample mean $\bar{X}$ unbiased estimator of λ and also s^2 is unbiased estimator for λ . Hence, show that $\alpha \bar{X} + (1 - \alpha)s^2$ is unbiased estimator of λ

Solution:-

$$E[x] = \sum_{i=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{i=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x(x-1)!} = e^{-\lambda} \lambda \sum_{i=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$E[x] = e^{-\lambda} \lambda \left[\frac{(\lambda)^0}{0!} + \frac{(\lambda)^1}{1!} + \frac{(\lambda)^2}{2!} + \dots \dots \dots \right]$$

$$E[x] = \lambda e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2!} + \dots \dots \dots \right]$$

Using $e^{-x} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \dots \dots$

$$E[x] = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

$$E[x^2] = E[x(x-1) + x] = \sum_{i=0}^{\infty} [x(x-1) + x] \frac{e^{-\lambda} \lambda^x}{x!} = \sum_{i=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{i=1}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$E[x^2] = \sum_{i=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x(x-1)(x-2)!} + \lambda$$

$$E[x^2] = e^{-\lambda} \lambda^2 \sum_{i=0}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$E[x^2] = e^{-\lambda} \lambda^2 [1 + \lambda + \lambda^2 + \dots \dots \dots] + \lambda = e^{-\lambda} \lambda^2 e^{\lambda} + \lambda$$

$$E[x^2] = \lambda^2 + \lambda$$

Now,

$$s^2 = \frac{1}{n-1} \left[\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right] = \frac{1}{n-1} [\sum x_i^2 - n\bar{X}^2] \dots \dots (A)$$

and

$$V(\bar{X}) = E(\bar{X}^2) - (E(\bar{X}))^2 \Rightarrow V(\bar{X}) + (E(\bar{X}))^2 = E(\bar{X}^2)$$

$$E(\bar{X}^2) = \frac{\lambda}{n} + \lambda^2$$

Consider (A), taking expectation both sides

$$E(s^2) = \frac{1}{n-1} [\sum E(x_i^2) - nE(\bar{X}^2)] = \frac{1}{n-1} \left[n(\lambda^2 + \lambda) - n \left(\frac{\lambda}{n} + \lambda^2 \right) \right]$$

$$E(s^2) = \frac{1}{n-1} [n\lambda^2 + n\lambda - \lambda - n\lambda^2] = \frac{1}{n-1} (n-1)\lambda = \lambda$$

This shows that s^2 is unbiased estimator of λ .

Now let

$$T = \alpha \bar{X} + (1 - \alpha)s^2$$

Taking Expectation

$$E(T) = \alpha E(\bar{X}) + (1 - \alpha)E(s^2) = \alpha\lambda + (1 - \alpha)\lambda = \alpha\lambda + \lambda - \alpha\lambda = \lambda$$

Hence T is unbiased estimator of λ .

2.15 Functional Expectation of Estimator (Poisson Distribution).

Question 15: If $X \sim P(m)$ then show that $E(e^{-kx}) = \exp[-m(1 - e^{-k})]$. Hence show that if $\bar{X}$ is arithmetic mean of n independent variables $X_1, X_2, \dots, X_n$ each having Poisson distribution with mean “ m ” then show that $E(e^{-\bar{X}}) = \exp[-nm(1 - e^{-\frac{1}{n}})]$.

Solution:

$$E[e^{-kx}] = \sum_{x=0}^{\infty} e^{-kx} \frac{e^{-m} m^x}{x!} = e^{-m} \sum_{x=0}^{\infty} \frac{(e^{-k} m)^x}{x!}$$

$$E[e^{-kx}] = e^{-m} \left[\frac{(e^{-k} m)^0}{0!} + \frac{(e^{-k} m)^1}{1!} + \frac{(e^{-k} m)^2}{2!} + \dots \dots \dots \right]$$

Using $e^{-x} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \dots \dots$

$$E[x] = e^{-m} e^{me^{-k}} = e^{-m(1-e^{-k})} = \exp\{-m(1 - e^{-k})\}$$

Now,

$$E[e^{-\bar{X}}] = E\left[e^{-\frac{\sum X_i}{n}}\right] = E\left[e^{-\frac{1}{n}(X_1 + X_2 + \dots + X_n)}\right] = E\left[e^{-\frac{X_1}{n}} \cdot e^{-\frac{X_2}{n}} \dots \dots e^{-\frac{X_n}{n}}\right]$$

Since $X_1, X_2, \dots, X_n$ are independent so

$$E[e^{-\bar{X}}] = E\left(e^{-\frac{X_1}{n}}\right) \cdot E\left(e^{-\frac{X_2}{n}}\right) \dots \dots E\left(e^{-\frac{X_n}{n}}\right)$$

Let $-\frac{1}{n} = t$

$$E\left(e^{-\frac{X_1}{n}}\right) = E(e^{tX_1}) = E(e^{tX})$$

$$E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-m} m^x}{x!} = e^{-m} \sum_{x=0}^{\infty} \frac{(e^t m)^x}{x!} = e^{-m} e^{me^t}$$

$$E[e^{tX}] = e^{-m(1-e^t)} = \exp\{-m(1-e^t)\} = \exp\left\{-m\left(1-e^{-\frac{1}{n}}\right)\right\}$$

Therefore,

$$E[e^{-\bar{X}}] = \exp\left\{-m\left(1-e^{-\frac{1}{n}}\right)\right\} \exp\left\{-m\left(1-e^{-\frac{1}{n}}\right)\right\} \dots \dots \exp\left\{-m\left(1-e^{-\frac{1}{n}}\right)\right\}$$

$$E[e^{-\bar{X}}] = \exp\left\{-mn\left(1-e^{-\frac{1}{n}}\right)\right\}$$

2.16 Unbiased Estimator and its variance in Chi-Square Distribution

Question 16: let $X_1, X_2, \dots, X_n$ denotes a random sample from $N(0, \theta)$, $0 < \theta < \infty$. Show that $\frac{\sum X_i^2}{n}$ is an unbiased estimator of θ and has the variance $\frac{2\theta^2}{n}$.

Solution:

Given that $X_1, X_2, \dots, X_n$ denotes the random sample from a $N(0, \theta)$.

Let

$$\frac{\sum X_i^2}{n} = S^2 \Rightarrow \sum X_i^2 = nS^2 \dots \dots (1)$$

As by definition of chi-square

$$\frac{nS^2}{\theta} \sim \chi^2(n)$$

As expectation of chi-square is equal to degree of freedom so

$$E\left[\frac{nS^2}{\theta}\right] = n$$

From (1)

$$E\left[\frac{\sum X_i^2}{\theta}\right] = n \Rightarrow E\left[\frac{\sum X_i^2}{n}\right] = \theta$$

Also, the variance of chi-square distribution is two times degree of freedom

$$V\left[\frac{\sum X_i^2}{\theta}\right] = 2n$$

$$\frac{1}{\theta^2} V[\sum X_i^2] = 2n$$

Divide both sides with n^2

$$\frac{1}{n^2 \theta^2} V[\sum X_i^2] = \frac{2n}{n^2}$$

$$V \left[\frac{\sum X_i^2}{n} \right] = \frac{2\theta^2}{n}$$

2.17 Biased Estimator of Population Standard Deviation and Amount of Bias.

Question 17 : Let $X_1, X_2, \dots, X_n$ be a random sample from some population with finite mean μ . prove that s is not in general unbiased for σ . Hence, prove that the amount of bias is

$$\text{Bias}(\sigma, s) = \sigma \left[\sqrt{\frac{2}{n-1}} \Gamma\left(\frac{n}{2}\right) \left\{ \Gamma\left(\frac{n-1}{2}\right) \right\}^{-1} - 1 \right]$$

Solution:-

As we know that

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

Let $\chi^2 = w$, the p.d.f of chi-square is

$$f(w) = \frac{e^{-\frac{w}{2}} (w)^{\left(\frac{n-1}{2}\right)-1}}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \quad 0 < w < \infty$$

$$\frac{(n-1)s^2}{\sigma^2} = w \Rightarrow \frac{\sqrt{n-1}s}{\sigma} = w^{\frac{1}{2}}$$

Taking Expectation

$$E \left[\frac{\sqrt{n-1}s}{\sigma} \right] = E \left(w^{\frac{1}{2}} \right)$$

So,

$$E \left[\frac{\sqrt{n-1}s}{\sigma} \right] = \int_0^\infty w^{\frac{1}{2}} \frac{e^{-\frac{w}{2}} (w)^{\left(\frac{n-1}{2}\right)-1}}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} dw = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^\infty e^{-\frac{w}{2}} (w)^{\left(\frac{n-2}{2}\right)} dw$$

Let

$$z = \frac{w}{2} \Rightarrow 2z = w \Rightarrow 2dz = dw$$

when $w = 0, z = 0$, when $w = \infty, z = \infty$

$$E \left[\frac{\sqrt{n-1}s}{\sigma} \right] = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^{\infty} e^{-z} (2z)^{\left(\frac{n-1}{2}\right)} (2dz)$$

$$E \left[\frac{\sqrt{n-1}s}{\sigma} \right] = \frac{2^{\frac{n}{2}-1+1}}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^{\infty} (z)^{\frac{n}{2}-1} e^{-z} (dz) = \frac{2^{\frac{n}{2}-\frac{n}{2}+\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right)$$

$$E \left[\frac{\sqrt{n-1}s}{\sigma} \right] = \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right)$$

$$E[s] = \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right) \sqrt{n-1}} \Gamma\left(\frac{n}{2}\right) \sigma = \frac{1}{\Gamma\left(\frac{n-1}{2}\right) \sqrt{\frac{n-1}{2}}} \Gamma\left(\frac{n}{2}\right) \sigma$$

$$E[s] = \sqrt{\frac{2}{n-1}} \Gamma\left(\frac{n}{2}\right) \left\{ \Gamma\left(\frac{n-1}{2}\right) \right\}^{-1} \sigma \dots \dots (A)$$

As $E(s) \neq \sigma$ so s is biased estimator of σ

For Amount of Bias

$$\text{Bias}(\sigma, s) = E(s) - \sigma = \sqrt{\frac{2}{n-1}} \Gamma\left(\frac{n}{2}\right) \left\{ \Gamma\left(\frac{n-1}{2}\right) \right\}^{-1} \sigma - \sigma$$

$$\text{Bias}(\sigma, s) = \sigma \left[\sqrt{\frac{2}{n-1}} \Gamma\left(\frac{n}{2}\right) \left\{ \Gamma\left(\frac{n-1}{2}\right) \right\}^{-1} - 1 \right]$$

2.18 Finding Constant “C” from Expression of Normally Distributed Variates

Question 18: Let $X_1, X_2, \dots, X_6$ be independent random samples from normal distribution with mean μ and variance σ^2 , and let $M = C\{(X_1 - X_2)^2 + (X_3 - X_4)^2 + (X_5 - X_6)^2\}$, for what value of “C”, M is an unbiased estimator of σ^2 .

Solution:-

As the variates are distributed normally so

$$E(X) = \mu, \text{Var}(X) = \sigma^2$$

And

$$Var(X) = E(X^2) - [E(X)]^2 \Rightarrow E(X^2) = Var(X) + [E(X)]^2 \Rightarrow E(X^2) = \sigma^2 + \mu^2$$

Also

$$E(X_i X_j) = E(X_i)E(X_j) = \mu \cdot \mu = \mu^2, \text{ for } i \neq j$$

As

$$M = C\{(X_1 - X_2)^2 + (X_3 - X_4)^2 + (X_5 - X_6)^2\}$$

Taking Expectations

$$E(M) = CE [X_1^2 + X_2^2 - 2X_1X_2 + X_3^2 + X_4^2 - 2X_3X_4 + X_5^2 + X_6^2 - 2X_5X_6]$$

$$\sigma^2 = CE [\sigma^2 + \mu^2 + \sigma^2 + \mu^2 - 2\mu^2 + \sigma^2 + \mu^2 + \sigma^2 + \mu^2 - 2\mu^2 + \sigma^2 + \mu^2 + \sigma^2 + \mu^2 - 2\mu^2]$$

$$\sigma^2 = 6C\sigma^2$$

$$C = \frac{1}{6}$$

2.19 Finding Constant “s” from Expression of Poisson Variates

Question 19: Let $X_1, X_2, \dots, X_n$ be independent random samples from Poisson distribution with parameter μ , and let $M = d\{X_1(X_1 - X_2) + X_2(X_2 - X_3) + \dots + X_{n-1}(X_{n-1} - X_n)^2\}$, for what value of “d”, M is an unbiased estimator of μ .

Solution:

As the variates are distributed normally so

$$E(X) = Var(X) = \mu$$

And

$$Var(X) = E(X^2) - [E(X)]^2 \Rightarrow E(X^2) = Var(X) + [E(X)]^2 \Rightarrow E(X^2) = \mu + \mu^2$$

Also

$$E(X_i X_j) = E(X_i)E(X_j) = \mu \cdot \mu = \mu^2, \text{ for } i \neq j$$

As

$$M = d\{X_1(X_1 - X_2) + X_2(X_2 - X_3) + \dots + X_{n-1}(X_{n-1} - X_n)\}$$

$$M = d[X_1^2 - X_1X_2 + X_2^2 - X_2X_3 + \dots + X_{n-1}^2 - X_{n-1}X_n]$$

Taking Expectation

$$E(M) = d[E(X_1^2) - E(X_1X_2) + E(X_2^2) - E(X_2X_3) + \dots + E(X_{n-1}^2) - E(X_{n-1}X_n)]$$

$$\mu = d[\mu + \mu^2 - \mu^2 + \mu + \mu^2 - \mu^2 + \dots + \mu + \mu^2 - \mu^2]$$

$$\mu = d[(n-1)\mu]$$

$$d = \frac{1}{n-1}$$

2.20 Finding Constants k_1 and k_2 By Minimization of Variance

Question 20: Let Y_1 and Y_2 be two stochastically independent unbiased estimators of θ and of $V(Y_1)$ is twice than $V(Y_2)$. Find the constants k_1 and k_2 so that $k_1Y_1 + k_2Y_2$ is unbiased estimator of θ with smallest possible variance.

Solution:-

Given that Y_1 and Y_2 are unbiased Estimator of θ .

$$E(Y_1) = E(Y_2) = \theta$$

Also given $k_1Y_1 + k_2Y_2$ is unbiased estimator of θ . i.e.

Let $T = k_1Y_1 + k_2Y_2$

$$E(T) = E[k_1Y_1 + k_2Y_2] = \theta$$

$$k_1E(Y_1) + k_2E(Y_2) = \theta$$

$$k_1\theta + k_2\theta = \theta$$

$$k_1 + k_2 = 1 \dots \dots \dots (A)$$

$$k_1 = 1 - k_2 \text{ or } k_2 = 1 - k_1$$

As given

$$V(Y_1) = 2V(Y_2)$$

Let

$$V(Y_2) = V, \text{ therefore } V(Y_1) = 2V$$

$$V(T) = V(k_1Y_1 + k_2Y_2) = k_1^2V(Y_1) + k_2^2V(Y_2)$$

$$V(T) = 2k_1^2V + (1 - k_1)^2V \dots \dots (1)$$

Now, for Minimize variance of T, we need to minimize T. Diff (1) w.r.t k_1

$$\frac{\partial V(T)}{\partial k_1} = 4k_1V - 2(1 - k_1)V = 6k_1V - 2V \dots \dots (2)$$

$$\text{Put } \frac{\partial V(T)}{\partial k_1} = 0$$

$$6k_1V - 2V = 0 \Rightarrow 6k_1V = 2V$$

$$k_1 = \frac{1}{3}$$

Diff (2) w.r.t k_1

$$\frac{\partial^2 V(T)}{\partial k_1^2} = 6V > 0, \text{ As } V > 0$$

Put k_1 in (A)

$$k_2 = 1 - \frac{1}{3} = \frac{2}{3}$$

2.21 Minimizing the Variance for Given Expression

Question 21: Let $X_1, X_2, \dots, X_n$ be an independent random sample from some distribution having mean μ and variance σ^2 .

(a) Show that $\sum_{i=1}^n a_i x_i$ is an unbiased estimator of μ for any set of unknown constants $a_1, a_2, \dots, a_n$ satisfying $\sum_{i=1}^n a_i = 1$.

(b) If $\sum_{i=1}^n a_i = 1$ show that $V(\sum a_i x_i)$ is minimized for $a_i = \frac{1}{n}, i = 1, 2, \dots, n$

Solution:-

Let

$$T = \sum_{i=1}^n a_i x_i$$

Taking Expectation

$$E(T) = \sum_{i=1}^n a_i E(x_i) = \sum_{i=1}^n a_i \mu$$

As

$$\sum_{i=1}^n a_i = 1$$

$$E(T) = \mu$$

Hence, T is unbiased estimator of μ .

$$V(T) = E[T - E(T)]^2$$

$$V(T) = E \left[\sum_{i=1}^n a_i x_i - \sum_{i=1}^n a_i \mu \right]^2 = E \left[\sum_{i=1}^n a_i (x_i - \mu) \right]^2$$

$$V(T) = \sum_{i=1}^n a_i^2 E(x_i - \mu)^2 + \sum_{i \neq j} a_i a_j (x_i - \mu)(x_j - \mu)$$

As samples are independent so, the covariance term will be zero and

$$E(x_i - \mu)^2 = \sigma^2$$

$$V(T) = \sum_{i=1}^n a_i^2 \sigma^2 \dots \dots (1)$$

We can minimize $V(T)$ by minimizing $\sum a_i^2$, using LaGrange Multiplier

Let

$$\phi = \sum_{i=1}^n a_i^2 - \lambda \left(\sum_{i=1}^n a_i - 1 \right)$$

Diff w.r.t a_i

$$\frac{\partial \phi}{\partial a_i} = 2 \sum_{i=1}^n a_i - \sum \lambda = 2(1) - n\lambda = 2 - n\lambda$$

Put $\frac{\partial \phi}{\partial a_i} = 0$

$$2 - n\lambda = 0 \Rightarrow \lambda = \frac{2}{n}$$

Consider

$$2 \sum_{i=1}^n a_i - n\lambda = 0$$

Dropping Σ

$$2a_i - \lambda = 0$$

$$a_i = \frac{\lambda}{2}$$

Putting $\lambda = \frac{2}{n}$

$$a_i = \frac{\frac{2}{n}}{2} = \frac{1}{n}$$

Putting a_i in (1)

$$V(T) = \sum_{i=1}^n \left(\frac{1}{n} \right)^2 \sigma^2 = \frac{n}{n^2} \sigma^2 = \frac{\sigma^2}{n}$$

2.22 Unbiased Estimator of Probability of Success in Negative Binomial Distribution

Question 22: The probability of $(m-1)$ success in the first $(n-1)$ trials with the a success at n th trial is

$$\binom{n-1}{m-1} \omega^m x^{n-m}, \quad \text{where } x = 1 - \omega \text{ and } n = m, m+1, m+2, \dots$$

Show that $P^* = \frac{m-1}{n-1}$ is unbiased estimator of ω .

Solution:

$$\begin{aligned} E(P^*) &= \sum_{n=m}^{\infty} \left(\frac{m-1}{n-1} \right) \binom{n-1}{m-1} \omega^m x^{n-m} \\ E(P^*) &= \sum_{n=m}^{\infty} \left(\frac{m-1}{n-1} \right) \frac{(n-1)!}{(m-1)!(n-1-m+1)!} \omega^m x^{n-m} \\ E(P^*) &= \sum_{n=m}^{\infty} \left(\frac{m-1}{n-1} \right) \frac{(n-1)(n-2)!}{(m-1)(m-2)!(n-m)!} \omega^m x^{n-m} \\ E(P^*) &= \sum_{n=m}^{\infty} \frac{(n-2)!}{(m-2)!(n-m)!} \omega^m x^{n-m} = \omega \sum_{n=m}^{\infty} \binom{n-2}{m-2} \omega^{m-1} x^{n-m} \\ E(P^*) &= \omega \left[\binom{m-2}{m-2} \omega^{m-1} x^{m-m} + \binom{m+1-2}{m-2} \omega^{m-1} x^{m+1-m} + \binom{m+2-2}{m-2} \omega^{m-1} x^{m+2-m} + \dots \right] \\ E(P^*) &= \omega \left[\omega^{m-1} + (m-1) \omega^{m-1} x + \frac{m(m-1)}{2!} \omega^{m-1} x^2 + \dots \right] \\ E(P^*) &= \omega \omega^{m-1} \left[1 + (m-1)x + \frac{m(m-1)}{2!} x^2 + \dots \right] \end{aligned}$$

Using $(1-x)^{-n} = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$

$$E(P^*) = \omega \omega^{m-1} (1-x)^{-m+1} = \omega \omega^{m-1} \omega^{-m+1} = \omega$$

Thus

$$E(P^*) = E \left[\frac{m-1}{n-1} \right] = \omega$$

2.23 Unbiased Estimator of Sample Proportion in Binomial and Negative Binomial

Question 23: A sample of n objects is taken from an infinite population in which proportion $p = 1 - q$ of object processing a certain attribute (A).

(a) Prove that $\frac{\bar{p}\bar{q}}{n-1}$ is an unbiased estimator of $\frac{pq}{n}$ where, $\bar{p} = 1 - \bar{q}$ is the proportion of objects possessing an attribute (A) in the sample.

(b) Instead of taking a sample in the above manner, the objects of the sample are selected at random in succession until a fixed number “ m ” of objects possessing (A) is obtained in the sample. Prove that $\frac{m-1}{n-1}$ is an unbiased estimator of p , but $\frac{m}{n}$ is not unbiased estimation of p .

Solution:-

(a) let

$$T = \frac{\bar{p}\bar{q}}{n-1}$$

Taking Expectation

$$E(T) = E\left(\frac{\bar{p}\bar{q}}{n-1}\right) = \frac{1}{n-1}E(\bar{p}\bar{q}) = \frac{1}{n-1}[E\{\bar{p}(1-\bar{p})\}] = \frac{1}{n-1}[E(\bar{p}) - E(\bar{p}^2)]$$

As we have $E(\bar{p}) = p$ and $var(\bar{p}) = \frac{pq}{n}$

$$Var(\bar{p}) = E(\bar{p}^2) - [E(\bar{p})]^2$$

$$E(\bar{p}^2) = Var(\bar{p}) + [E(\bar{p})]^2 = \frac{pq}{n} + p^2$$

So,

$$E(T) = \frac{1}{n-1}\left[p - \frac{pq}{n} - p^2\right] = \frac{1}{n-1}\left[p\left(\frac{n-q-np}{n}\right)\right] = \frac{p}{n(n-1)}[n(1-p) - q]$$

$$E(T) = \frac{p}{n(n-1)}[nq - q] = \frac{pq}{n(n-1)}(n-1) = \frac{pq}{n}$$

Hence, $\frac{\bar{p}\bar{q}}{n-1}$ is unbiased estimator of $\frac{pq}{n}$

(b) we have the probability function

$$\binom{n-1}{m-1} p^m q^{n-m}, \quad \text{where } q = 1 - p \text{ and } n = m, m+1, m+2, \dots$$

It is the probability function in which first (n-1) items there are (m-1) A's and that the nth item is an A.

Let

$$P^* = \binom{m-1}{n-1}$$

$$E(P^*) = \sum_{n=m}^{\infty} \binom{m-1}{n-1} \binom{n-1}{m-1} p^m q^{n-m}$$

$$E(P^*) = \sum_{n=m}^{\infty} \binom{m-1}{n-1} \frac{(n-1)!}{(m-1)!(n-1-m+1)!} p^m q^{n-m}$$

$$E(P^*) = \sum_{n=m}^{\infty} \binom{m-1}{n-1} \frac{(n-1)(n-2)!}{(m-1)(m-2)!(n-m)!} p^m q^{n-m}$$

$$E(P^*) = \sum_{n=m}^{\infty} \frac{(n-2)!}{(m-2)!(n-m)!} p^m q^{n-m} = p \sum_{n=m}^{\infty} \binom{n-2}{m-2} p^{m-1} q^{n-m}$$

$$E(P^*) = p \left[\binom{m-2}{m-2} p^{m-1} q^{m-m} + \binom{m+1-2}{m-2} p^{m-1} q^{m+1-m} + \binom{m+2-2}{m-2} p^{m-1} q^{m+2-m} + \dots \right]$$

$$E(P^*) = p \left[p^{m-1} + (m-1) p^{m-1} q + \frac{m(m-1)}{2!} p^{m-1} q^2 + \dots \right]$$

$$E(P^*) = p p^{m-1} \left[1 + (m-1)q + \frac{m(m-1)}{2!} q^2 + \dots \right]$$

Using $(1-x)^{-n} = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$

$$E(P^*) = p p^{m-1} (1-q)^{-m+1} = p p^{m-1} p^{-m+1} = p$$

Thus

$$E(P^*) = E \left[\frac{m-1}{n-1} \right] = p$$

Now, for

$$P^* = \binom{m}{n}$$

$$E(P^*) = \sum_{n=m}^{\infty} \binom{m}{n} \binom{n-1}{m-1} p^m q^{n-m}$$

$$\begin{aligned}
E(P^*) &= \sum_{n=m}^{\infty} \binom{m}{n} \frac{(n-1)!}{(m-1)!(n-1-m+1)!} p^m q^{n-m} \\
E(P^*) &= \sum_{n=m}^{\infty} \binom{m}{n} \frac{(n-1)!}{(m-1)!(n-m)!} p^m q^{n-m} \\
E(P^*) &= \frac{m}{(m-1)!} \frac{p^m}{q^m} \sum_{n=m}^{\infty} \frac{(n-1)!}{n(n-m)!} q^n \\
E(P^*) &= \frac{m}{(m-1)!} \frac{p^m}{q^m} \left[\frac{(m-1)!}{m(m-m)!} q^m + \frac{(m+1-1)!}{(m+1)(m+1-m)!} q^{m+1} + \dots \right] \\
E(P^*) &= \frac{m}{(m-1)!} \frac{p^m}{q^m} \left[\frac{(m-1)!}{m} q^m + \frac{(m)!}{(m+1)} q^{m+1} + \frac{(m+1)!}{(m+2)2!} q^{m+2} \dots \right] \\
E(P^*) &= \frac{m}{(m-1)!} \frac{p^m q^m}{q^m} \left[\frac{(m-1)!}{m} + \frac{(m)!}{(m+1)} q^1 + \frac{(m+1)!}{(m+2)2!} q^2 \dots \right] \\
E(P^*) &= \frac{mp^m}{(m-1)!} \left[\frac{(m-1)!}{m} + \frac{(m)!}{(m+1)} q^1 + \frac{(m+1)!}{(m+2)2!} q^2 \dots \right]
\end{aligned}$$

Hence

$$E(P^*) \neq p$$

So $P^* = \left(\frac{m}{n}\right)$ is biased estimator of p.

2.24 Unbiased Estimator of Probability of Success in Negative Binomial Distribution

Question 24: A lepidopterist wishes to estimate the frequency with which an unusual form of certain species of butterfly occur in a particular distribution. He catches individual species of the species until he has obtained exactly B of this form and lies total catch is then T. Find the probability distribution of T. Is $\frac{B-1}{T-1}$, $B > 1$ and unbiased estimate of the relative frequency of the series under consideration.

Solution :-

Let the relative frequency of the series under consideration be p. consider last number to be caught which must be of type B with probability p. The probability of there being (B-1) members out of (T-1) is

$$\binom{T-1}{B-1} p^{B-1} q^{T-B}, \quad \text{where } q = 1 - p \text{ and } T = B, B+1, B+2, \dots$$

Thus the required probability distribution will be

$$\binom{T-1}{B-1} p^B q^{T-B}$$

Which is negative binomial distribution

Let

$$P^* = \left(\frac{B-1}{T-1} \right)$$

$$E(P^*) = \sum_{T=B}^{\infty} \left(\frac{B-1}{T-1} \right) \binom{T-1}{B-1} p^B q^{T-B}$$

$$E(P^*) = \sum_{T=B}^{\infty} \left(\frac{B-1}{T-1} \right) \frac{(T-1)!}{(B-1)!(T-1-B+1)!} p^B q^{T-B}$$

$$E(P^*) = \sum_{T=B}^{\infty} \left(\frac{B-1}{T-1} \right) \frac{(T-1)(T-2)!}{(B-1)(B-2)!(T-B)!} p^B q^{T-B}$$

$$E(P^*) = \sum_{T=B}^{\infty} \frac{(T-2)!}{(B-2)!(T-B)!} p^B q^{T-B} = p \sum_{T=B}^{\infty} \binom{T-2}{B-2} p^{B-1} q^{T-B}$$

$$E(P^*) = p \left[\binom{B-2}{B-2} p^{B-1} q^{B-B} + \binom{B+1-2}{B-2} p^{B-1} q^{B+1-B} + \binom{B+2-2}{B-2} p^{B-1} q^{B+2-B} + \dots \right]$$

$$E(P^*) = p \left[p^{B-1} + (B-1)p^{B-1}q + \frac{B(B-1)}{2!} p^{B-1}q^2 + \dots \right]$$

$$E(P^*) = p p^{B-1} \left[1 + (B-1)q + \frac{B(B-1)}{2!} q^2 + \dots \right]$$

Using $(1-x)^{-n} = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$

$$E(P^*) = p p^{B-1} (1-q)^{-B+1} = p p^{B-1} p^{-B+1} = p$$

Thus

$$E(P^*) = E \left[\frac{B-1}{T-1} \right] = p$$

2.25 Asymptotical Unbiased Estimator in Bernoulli Distribution

Question 25: In a sequence of “n” Bernoulli trials with the probability of success “p”, “X” successes were observed. Show that $\hat{p}(1 - \hat{p})^2$ is not unbiased estimator of $p(1 - p)$, but the bias approaches zero as sample size approaches infinity.

Solution:-

Let

$$T = \hat{p}(1 - \hat{p})^2 = \hat{p}(1 + \hat{p}^2 - 2\hat{p}) = \hat{p} + \hat{p}^3 - 2\hat{p}^2$$

Taking Expectation

$$E(T) = E(\hat{p}) + E(\hat{p}^3) - 2E(\hat{p}^2) = E\left(\frac{X}{n}\right) + E\left(\frac{X}{n}\right)^3 - 2E\left(\frac{X}{n}\right)^2$$

$$E(T) = \frac{1}{n} E(X) + \frac{1}{n^3} E(X^3) - \frac{2}{n^2} E(X^2)$$

As in the sequence of n Bernoulli trials

$$E(X) = np, \quad E(X^2) = n(n-1)p^2 + np, \quad E(X^3) = n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np$$

$$E(T) = \frac{1}{n}(np) + \frac{1}{n^3}[n(n-1)(n-2)p^3 + 3n(n-1)p^2 + np] - \frac{2}{n^2}[n(n-1)p^2 + np]$$

$$E(T) = p + \frac{(n-1)(n-2)}{n^2}p^3 + \frac{3(n-1)}{n^2}p^2 + \frac{1}{n^2}p - \frac{2(n-1)}{n}p^2 - \frac{2}{n}p$$

$$E(T) = p + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)p^3 + 3\left(\frac{1}{n} - \frac{1}{n^2}\right)p^2 + \frac{1}{n^2}p - 2\left(1 - \frac{1}{n}\right)p^2 - \frac{2}{n}p$$

Taking Limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} E(T) = p + (1-0)(1-0)p^3 + 3(0-0)p^2 + (0)p - 2(1-0)p^2 - (0)p$$

$$\lim_{n \rightarrow \infty} E(T) = p + p^3 - 2p^2 = p(1 + p^2 - 2p) = p(1 - p)^2 = pq^2$$

Here, we see that

$$E(T) = E[\hat{p}(1 - \hat{p})^2] \rightarrow p(1 - p)^2 \text{ As } n \rightarrow \infty$$

Now,

$$\lim_{n \rightarrow \infty} \text{Bias} = \lim_{n \rightarrow \infty} [E(T) - p(1 - p)] = p(1 - p)^2 - p(1 - p)^2 = 0$$

As Bias $\rightarrow 0$ when $n \rightarrow \infty$, So, T is asymptotically unbiased estimator of $p(1 - p)^2$

2.26 Order Statistic of Random Sample of Size 3 From Uniform

Question 26: Let $Y_1 < Y_2 < Y_3$ be a set of order Statistic corresponding to a random sample of size 3 from uniform distribution on the interval $(\theta, 2\theta)$, $\theta > 0$. Show that both y_1 and y_3 are biased estimators of θ . Also show that y_1 is asymptotically unbiased, but y_3 is not. Also find the linear combination of y_1 and y_3 which is unbiased estimator of θ .

Solution:-

The p.d.f of uniform distribution is

$$f(y) = \frac{1}{2\theta - \theta} = \frac{1}{\theta}, \text{ for } (\theta, 2\theta)$$

The distribution function

$$F(y) = \int_{\theta}^y \frac{1}{\theta} dy = \frac{y}{\theta} \Big|_{\theta}^y = \frac{y - \theta}{\theta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_1 of random sample of 3 $n = 3, r = 1$

$$g(y_1) = \frac{3!}{(1-1)!(3-1)!} [F(y_1^3)]^{1-1} [1 - F(y_1^3)]^{3-1} f(y_1^3)$$

$$g(y_1) = 3 \left[1 - \frac{y_1 - \theta}{\theta} \right]^2 \frac{1}{\theta}$$

$$E(y_1) = \int_{\theta}^{2\theta} y_1 g(y_1) dy_1 = \int_{\theta}^{2\theta} y_1 3 \left[1 - \frac{y_1 - \theta}{\theta} \right]^2 \frac{1}{\theta} dy_1 = \frac{3}{\theta} \int_{\theta}^{2\theta} y_1 \left[1 - \frac{y_1 - \theta}{\theta} \right]^2 dy_1$$

Let

$$z = 1 - \frac{y_1 - \theta}{\theta} \Rightarrow \frac{y_1 - \theta}{\theta} = 1 - z \Rightarrow y_1 = \theta + \theta(1 - z) \Rightarrow y_1 = \theta(2 - z)$$

$$dy_1 = -\theta dz$$

$$\text{When } y_1 = \theta \text{ Then } z = 1 - \frac{\theta - \theta}{\theta} = 1 - 0 = 1$$

$$\text{When } y_1 = 2\theta \text{ Then } z = 1 - \frac{2\theta - \theta}{\theta} = 1 - \frac{\theta}{\theta} = 1 - 1 = 0$$

$$E(y_1) = \frac{3}{\theta} \int_1^0 \theta(2-z)z^2(-\theta dz) = \theta \int_0^1 (6z^2 - 3z^3) dz = \theta \left[6 \int_0^1 z^2 dz - 3 \int_0^1 z^3 dz \right]$$

$$E(y_1) = \theta \left[6 \left(\frac{z^3}{3} \right) \Big|_0^1 - 3 \left(\frac{z^4}{4} \right) \Big|_0^1 \right] = \theta \left[2(1-0) - \frac{3}{4}(1-0) \right] = \theta \left[2 - \frac{3}{4} \right] = \frac{5}{4} \theta$$

Hence,

$$E(y_1) = \frac{5}{4} \theta \neq \theta$$

So, it is biased estimator of θ .

$$\text{Bias}(y_1) = E(y_1) - \theta = \frac{5}{4} \theta - \theta = \frac{1}{4} \theta$$

Since we have random sample of size 3 i.e. $n=3$

$$\text{Bias}(y_1) = \frac{1}{3+1} \theta = \frac{1}{n+1} \theta$$

Applying limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \text{Bias}(y_1) = \lim_{n \rightarrow \infty} \frac{1}{n+1} \theta = 0$$

Hence y_1 is asymptotically unbiased estimator of θ .

Now Distribution of y_3 of random sample of 3 $n = 3, r = 3$

$$g(y_3) = \frac{3!}{(3-1)!(3-3)!} [F(y_3^3)]^{3-1} [1 - F(y_3^3)]^{3-3} f(y_3^3)$$

$$g(y_3) = 3 \left[\frac{y_3 - \theta}{\theta} \right]^2 \frac{1}{\theta}$$

$$E(y_3) = \int_{\theta}^{2\theta} y_3 g(y_3) dy_3 = \int_{\theta}^{2\theta} y_3 3 \left[\frac{y_3 - \theta}{\theta} \right]^2 \frac{1}{\theta} dy_3 = \frac{3}{\theta} \int_{\theta}^{2\theta} y_3 \left[\frac{y_3 - \theta}{\theta} \right]^2 dy_3$$

Let

$$z = \frac{y_3 - \theta}{\theta} \Rightarrow y_3 - \theta = \theta z \Rightarrow y_3 = \theta(z + 1)$$

$$dy_3 = \theta dz$$

$$\text{When } y_3 = \theta \text{ Then } z = \frac{\theta - \theta}{\theta} = 0$$

$$\text{When } y_3 = 2\theta \text{ Then } z = \frac{2\theta - \theta}{\theta} = \frac{\theta}{\theta} = 1$$

$$E(y_3) = \frac{3}{\theta} \int_0^1 \theta(z+1)z^2(\theta dz) = \theta \int_0^1 (3z^3 + 3z^2)dz = 3\theta \left[\int_0^1 z^3 dz + \int_0^1 z^2 dz \right]$$

$$E(y_3) = 3\theta \left[\left(\frac{z^4}{4} \right) \Big|_0^1 + \left(\frac{z^3}{3} \right) \Big|_0^1 \right] = 3\theta \left[\frac{1}{4}(1-0) + \frac{1}{3}(1-0) \right] = 3\theta \left[\frac{7}{12} \right] = \frac{7}{4}\theta$$

Hence,

$$E(y_3) = \frac{7}{4}\theta \neq \theta$$

So, it is biased estimator of θ .

$$\text{Bias}(y_3) = E(y_3) - \theta = \frac{7}{4}\theta - \theta = \frac{3}{4}\theta$$

Since we have random sample of size 3 i.e. $n=3$

$$\text{Bias}(y_3) = \frac{3}{3+1}\theta = \frac{n}{n+1}\theta$$

Applying limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \text{Bias}(y_3) = \lim_{n \rightarrow \infty} \frac{n}{n+1}\theta = \theta$$

Hence y_3 is not asymptotically unbiased estimator of θ . As bias does not approach to zero when sample size approach to infinity.

Now, to obtain the linear combination of y_1 and y_3 which will be unbiased for θ . Consider

$$E(y_1) = \frac{5}{4}\theta \Rightarrow E\left(\frac{4y_1}{5}\right) = \theta$$

Hence let

$$T_1 = \frac{4y_1}{5}$$

Unbiased estimator of θ

Also

$$E(y_3) = \frac{7}{4}\theta \Rightarrow E\left(\frac{4y_3}{7}\right) = \theta$$

Hence let

$$T_2 = \frac{4y_3}{7}$$

Unbiased estimator of θ

Now Linear combination of y_1 and y_2 , Let $T = \frac{1}{2}[T_1 + T_2]$ will be unbiased estimator of θ .

$$E(T) = \frac{1}{2}[E(T_1) + E(T_2)] = \frac{1}{2}(\theta + \theta) = \frac{2\theta}{2} = \theta$$

Hence required linear combination of y_1 and y_n which is unbiased for θ

$$T = \frac{1}{2} \left[\left(\frac{4}{5} \right) y_1 + \left(\frac{4}{7} \right) y_n \right]$$

2.27 Order Statistic of Random Sample of Size n From Uniform

Question 27: Let $Y_1 < Y_2 \dots < Y_n$ be a set of order Statistic corresponding to a random sample of size n from uniform distribution on the interval $(\theta, 2\theta)$, $\theta > 0$. Show that both y_1 and y_n are biased estimators of θ . Also show that y_1 is asymptotically unbiased, but y_n is not. Also find the linear combination of y_1 and y_3 which is unbiased estimator of θ .

Solution:-

The p.d.f of uniform distribution is

$$f(y) = \frac{1}{2\theta - \theta} = \frac{1}{\theta}, \text{ for } (\theta, 2\theta)$$

The distribution function

$$F(y) = \int_{\theta}^y \frac{1}{\theta} dy = \frac{y}{\theta} \Big|_{\theta}^y = \frac{y - \theta}{\theta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_1 of random sample of n $n = n, r = 1$

$$g(y_1) = \frac{n!}{(1-1)!(n-1)!} [F(y_1^n)]^{1-1} [1 - F(y_1^n)]^{n-1} f(y_1^n)$$

$$g(y_1) = n \left[1 - \frac{y_1 - \theta}{\theta} \right]^{n-1} \frac{1}{\theta}$$

$$E(y_1) = \int_{\theta}^{2\theta} y_1 g(y_1) dy_1 = \int_{\theta}^{2\theta} y_1 n \left[1 - \frac{y_1 - \theta}{\theta} \right]^{n-1} \frac{1}{\theta} dy_1 = \frac{n}{\theta} \int_{\theta}^{2\theta} y_1 \left[1 - \frac{y_1 - \theta}{\theta} \right]^{n-1} dy_1$$

Let

$$z = 1 - \frac{y_1 - \theta}{\theta} \Rightarrow \frac{y_1 - \theta}{\theta} = 1 - z \Rightarrow y_1 = \theta + \theta(1 - z) \Rightarrow y_1 = \theta(2 - z)$$

$$dy_1 = -\theta dz$$

$$\text{When } y_1 = \theta \text{ Then } z = 1 - \frac{\theta - \theta}{\theta} = 1 - 0 = 1$$

$$\text{When } y_1 = 2\theta \text{ Then } z = 1 - \frac{2\theta - \theta}{\theta} = 1 - \frac{\theta}{\theta} = 1 - 1 = 0$$

$$E(y_1) = \frac{n}{\theta} \int_1^0 \theta(2-z)z^{n-1}(-\theta dz) = n\theta \int_0^1 (2z^{n-1} - z^n)dz = n\theta \left[2 \int_0^1 z^{n-1}dz - \int_0^1 z^n dz \right]$$

$$E(y_1) = n\theta \left[2 \left(\frac{z^n}{n} \right) \Big|_0^1 - \left(\frac{z^{n+1}}{n+1} \right) \Big|_0^1 \right] = n\theta \left[\frac{2}{n}(1-0) - \frac{1}{n+1}(1-0) \right] = n\theta \left[\frac{2n+2-n}{n(n+1)} \right] = \frac{n+2}{n+1} \theta$$

Hence,

$$E(y_1) = \frac{n+2}{n+1} \theta \neq \theta$$

So, it is biased estimator of θ .

$$Bias(y_1) = E(y_1) - \theta = \frac{n+2}{n+1} \theta - \theta = \left(\frac{n+2-n-1}{n+1} \right) \theta = \frac{1}{n+1} \theta$$

Applying limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} Bias(y_1) = \lim_{n \rightarrow \infty} \frac{1}{n+1} \theta = 0$$

Hence y_1 is asymptotically unbiased estimator of θ .

Now Distribution of y_n of random sample of $n, r = n$

$$g(y_n) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1-F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n \left[\frac{y_n - \theta}{\theta} \right]^{n-1} \frac{1}{\theta}$$

$$E(y_n) = \int_{\theta}^{2\theta} y_n g(y_n) dy_n = \int_{\theta}^{2\theta} y_n n \left[\frac{y_n - \theta}{\theta} \right]^{n-1} \frac{1}{\theta} dy_n = \frac{n}{\theta} \int_{\theta}^{2\theta} y_n \left[\frac{y_n - \theta}{\theta} \right]^{n-1} dy_n$$

Let

$$z = \frac{y_n - \theta}{\theta} \Rightarrow y_n - \theta = \theta z \Rightarrow y_n = \theta(z + 1)$$

$$dy_n = \theta dz$$

$$\text{When } y_n = \theta \text{ Then } z = \frac{\theta - \theta}{\theta} = 0$$

$$\text{When } y_n = 2\theta \text{ Then } z = \frac{2\theta - \theta}{\theta} = \frac{\theta}{\theta} = 1$$

$$E(y_n) = \frac{n}{\theta} \int_0^1 \theta(z+1)z^{n-1}(\theta dz) = n\theta \int_0^1 (z^n + z^{n-1})dz = n\theta \left[\int_0^1 z^n dz + \int_0^1 z^{n-1} dz \right]$$

$$E(y_n) = n\theta \left[\left(\frac{z^{n+1}}{n+1} \right) \Big|_0^1 + \left(\frac{z^n}{n} \right) \Big|_0^1 \right] = n\theta \left[\frac{1}{n+1}(1-0) + \frac{1}{n}(1-0) \right] = n\theta \left[\frac{n+n+1}{n(n+1)} \right] = \frac{2n+1}{n+1}\theta$$

Hence,

$$E(y_n) = \frac{2n+1}{n+1}\theta \neq \theta$$

So, it is biased estimator of θ .

$$\text{Bias}(y_n) = E(y_n) - \theta = \frac{2n+1}{n+1}\theta - \theta = \left(\frac{2n+1-n-1}{n+1} \right)\theta = \frac{n}{n+1}\theta$$

$$\text{Bias}(y_n) = \frac{3}{3+1}\theta = \frac{n}{n+1}\theta$$

Applying limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \text{Bias}(y_n) = \lim_{n \rightarrow \infty} \frac{n}{n+1}\theta = \theta$$

Hence y_n is not asymptotically unbiased estimator of θ . As bias does not approach to zero when sample size approach to infinity.

Now, to obtain the linear combination of y_1 and y_n which will be unbiased for θ . Consider

$$E(y_1) = \frac{n+2}{n+1}\theta \Rightarrow E\left[\left(\frac{n+1}{n+2}\right)y_1\right] = \theta$$

Hence let

$$T_1 = \left(\frac{n+1}{n+2}\right)y_1$$

Unbiased estimator of θ

Also

$$E(y_n) = \frac{2n+1}{n+1}\theta \Rightarrow E\left[\left(\frac{n+1}{2n+1}\right)y_n\right] = \theta$$

Hence let

$$T_2 = \left(\frac{n+1}{2n+1}\right)y_n$$

Unbiased estimator of θ

Now Linear combination of y_1 and y_2 , Let $T = \frac{1}{2}[T_1 + T_2]$ will be unbiased estimator of θ .

$$E(T) = \frac{1}{2}[E(T_1) + E(T_2)] = \frac{1}{2}(\theta + \theta) = \frac{2\theta}{2} = \theta$$

Hence required linear combination of y_1 and y_n which is unbiased for θ

$$T = \frac{1}{2}\left[\left(\frac{n+1}{n+2}\right)y_1 + \left(\frac{n+1}{2n+1}\right)y_n\right]$$

2.28 Order Statistic of Random Sample of Size 3 From Uniform

Question 28: Let $X_{(1)}^{(3)} < X_{(2)}^{(3)} < X_{(3)}^{(3)}$ be a set of order Statistic corresponding to a random sample of size 3 from uniform distribution on the interval $(0, \lambda)$, $0 < x < \lambda$. Show that both $4X_{(1)}^{(3)}$, $2X_{(2)}^{(3)}$ and $\frac{4}{3}X_{(3)}^{(3)}$ are all unbiased estimators of λ .

Solution:-

The p.d.f of uniform distribution is

$$f(x) = \frac{1}{\lambda - 0} = \frac{1}{\lambda}, \quad \text{for } 0 < x < \lambda$$

The distribution function

$$F(x) = \int_0^x \frac{1}{\lambda} dx = \frac{x}{\lambda} \Big|_0^x = \frac{x - 0}{\lambda} = \frac{x}{\lambda}$$

As Distribution of order statistic $x_{(r)}^{(n)}$ is

$$g \left[x_{(r)}^{(n)} \right] = \frac{n!}{(r-1)!(n-r)!} \left[F \left(x_{(r)}^{(n)} \right) \right]^{r-1} \left[1 - F \left(x_{(r)}^{(n)} \right) \right]^{n-r} dF \left(F \left(x_{(r)}^{(n)} \right) \right)$$

Let

$$y_1 = x_{(1)}^{(3)}$$

Now Distribution of y_1 of random sample of 3 $n = 3, r = 1$

$$g(y_1) = \frac{3!}{(1-1)!(3-1)!} \left[F(y_1) \right]^{1-1} \left[1 - F(y_1) \right]^{3-1} f(y_1)$$

$$g(y_1) = 3 \left[1 - \frac{y_1}{\lambda} \right]^2 \frac{1}{\lambda}$$

$$E(4y_1) = \int_0^\lambda 4y_1 g(y_1) dy_1 = \int_0^\lambda 4y_1 3 \left[1 - \frac{y_1}{\lambda} \right]^2 \frac{1}{\lambda} dy_1 = \frac{12}{\lambda} \int_0^\lambda y_1 \left[1 - \frac{y_1}{\lambda} \right]^2 dy_1$$

Let

$$z = 1 - \frac{y_1}{\lambda} \Rightarrow \frac{y_1}{\lambda} = 1 - z \Rightarrow y_1 = \lambda(1 - z)$$

$$dy_1 = -\lambda dz$$

$$\text{When } y_1 = 0 \text{ Then } z = 1 - \frac{0}{\lambda} = 1 - 0 = 1$$

$$\text{When } y_1 = \lambda \text{ Then } z = 1 - \frac{\lambda}{\lambda} = 1 - 1 = 0$$

$$E(4y_1) = \frac{12}{\lambda} \int_1^0 \lambda(1-z)z^2(-\lambda dz) = 12\lambda \int_0^1 (z^2 - z^3)dz = 12\lambda \left[\int_0^1 z^2 dz - \int_0^1 z^3 dz \right]$$

$$E(4y_1) = 12\lambda \left[\left(\frac{z^3}{3} \right) \Big|_0^1 - \left(\frac{z^4}{4} \right) \Big|_0^1 \right] = 12\lambda \left[\frac{1}{3}(1-0) - \frac{1}{4}(1-0) \right] = 12\lambda \left[\frac{1}{3} - \frac{1}{4} \right] = 12\lambda \left(\frac{4-3}{12} \right) = \lambda$$

Hence,

$$E(4y_1) = \lambda$$

So, it is unbiased estimator of λ .

Let

$$y_2 = x_{(2)}^{(3)}$$

Now Distribution of y_2 of random sample of 3 $n = 3, r = 2$

$$g(y_2) = \frac{3!}{(2-1)!(3-2)!} [F(y_2^3)]^{2-1} [1-F(y_2^3)]^{3-2} f(y_2^3)$$

$$g(y_2) = 6 \frac{y_2}{\lambda} \left[1 - \frac{y_2}{\lambda} \right] \frac{1}{\lambda} = \frac{6y_2}{\lambda^2} \left[1 - \frac{y_2}{\lambda} \right]$$

$$E(2y_2) = \int_0^\lambda 2y_2 g(y_2) dy_2 = \int_0^\lambda 2y_2 \frac{6y_2}{\lambda^2} \left[1 - \frac{y_2}{\lambda} \right] dy_2 = \frac{12}{\lambda^2} \int_0^\lambda (y_2)^2 \left[1 - \frac{y_2}{\lambda} \right] dy_2$$

Let

$$z = 1 - \frac{y_2}{\lambda} \Rightarrow \frac{y_2}{\lambda} = 1 - z \Rightarrow y_2 = \lambda(1 - z)$$

$$dy_1 = -\lambda dz$$

$$\text{When } y_1 = 0 \text{ Then } z = 1 - \frac{0}{\lambda} = 1 - 0 = 1$$

$$\text{When } y_1 = \lambda \text{ Then } z = 1 - \frac{\lambda}{\lambda} = 1 - 1 = 0$$

$$E(2y_2) = \frac{12}{\lambda^2} \int_1^0 \lambda^2(1-z)^2 z (-\lambda dz) = 12\lambda \int_0^1 (z + z^3 - 2z^2) dz = 12\lambda \left[\int_0^1 z dz + \int_0^1 z^3 dz - 2 \int_0^1 z^2 dz \right]$$

$$E(2y_2) = 12\lambda \left[\left(\frac{z^2}{2} \right) \Big|_0^1 + \left(\frac{z^4}{4} \right) \Big|_0^1 - 2 \left(\frac{z^3}{3} \right) \Big|_0^1 \right] = 12\lambda \left[\frac{1}{2}(1-0) + \frac{1}{4}(1-0) - \frac{2}{3}(1-0) \right]$$

$$E(2y_2) = 12\lambda \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right] = 12\lambda \left[\frac{6+3-8}{12} \right] = 12\lambda \left[\frac{1}{12} \right] = \lambda$$

Hence,

$$E(2y_2) = \lambda$$

So, it is unbiased estimator of λ

Let

$$y_3 = x_{(3)}^{(3)}$$

Now Distribution of y_2 of random sample of 3 $n = 3, r = 3$

$$g(y_2) = \frac{3!}{(3-1)!(3-3)!} [F(y_3^3)]^{3-1} [1 - F(y_3^3)]^{3-3} f(y_3^3)$$

$$g(y_3) = 3 \left(\frac{y_3}{\lambda} \right)^2 \frac{1}{\lambda}$$

$$E\left(\frac{4}{3}y_3\right) = \int_0^{\lambda} \frac{4}{3}y_3 g(y_3) dy_3 = \int_0^{\lambda} \frac{4}{3}y_3 3 \left(\frac{y_3}{\lambda} \right)^2 \frac{1}{\lambda} dy_3 = \frac{4}{\lambda^3} \int_0^{\lambda} (y_3)^3 dy_3$$

$$E\left(\frac{4}{3}y_3\right) = \frac{4}{\lambda^3} \left(\frac{y_3^4}{4} \right) \Big|_0^{\lambda} = \frac{1}{\lambda^3} (\lambda^4 - 0) = \lambda$$

Hence,

$$E\left(\frac{4}{3}y_3\right) = \lambda$$

So, it is unbiased estimator of λ .

2.29 Unbiased Estimator Using Chi-Square Distribution

Question 29: Show that in estimating σ from $N(0, \sigma^2)$, the estimates

$$(1) \quad t_1 = \left(\frac{1}{2} \sum X_i\right)^{\frac{1}{2}} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n+1}{2}\right)} \quad (2) \quad t_2 = \left[\frac{1}{2} \sum (X_i - \bar{X})^2\right]^{\frac{1}{2}} \frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}$$

Both are unbiased estimator of σ .

Solution:-

(1) Consider

$$t_1 = \left(\frac{1}{2} \sum X_i\right)^{\frac{1}{2}} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n+1}{2}\right)} = \frac{\Gamma\left(\frac{n}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)} (\sum X_i)^{\frac{1}{2}} = K_1 (\sum X_i)^{\frac{1}{2}}$$

Where

$$K_1 = \frac{\Gamma\left(\frac{n}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n+1}{2}\right)}$$

Taking Expectation

$$E(t_1) = K_1 E(\sum X_i)^{\frac{1}{2}} \dots \dots \dots (A)$$

As $X_i \sim N(0, \sigma^2)$, So,

$$u = \frac{\sum X_i^2}{\sigma^2} \sim \chi_{(n)}^2 \Rightarrow \sum X_i^2 = \sigma^2 u \Rightarrow (\sum X_i^2)^{\frac{1}{2}} = \sigma u^{\frac{1}{2}}$$

Taking Expectation

$$E(\sum X_i^2)^{\frac{1}{2}} = \sigma E(u^{\frac{1}{2}}) \dots \dots \dots (B)$$

Now we need to find $E(u^{\frac{1}{2}})$ and $u \sim \chi_{(n)}^2$, the p.d.f of u .

$$f(u) = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} u^{\frac{n}{2}-1} e^{-\frac{u}{2}}, \quad 0 < u < \infty$$

$$E(u^{\frac{1}{2}}) = \int_0^{\infty} u^{\frac{1}{2}} \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} u^{\frac{n}{2}-1} e^{-\frac{u}{2}} du = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} u^{\frac{n-1}{2}} e^{-\frac{u}{2}} du$$

Let

$$t = \frac{u}{2} \Rightarrow 2t = u \Rightarrow 2dt = du$$

when $u = 0$ then $t = 0$, when $u = \infty$ then $t = \infty$

$$E\left(u^{\frac{1}{2}}\right) = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} (2t)^{\frac{n-1}{2}} e^{-t} (2dt) = \frac{2^{\frac{n-1}{2}} \cdot 2}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} t^{\frac{n-1}{2}} e^{-t} dt$$

$$E\left(u^{\frac{1}{2}}\right) = \frac{2^{\frac{n}{2} - \frac{n}{2} - \frac{1}{2} + 1}}{\Gamma\left(\frac{n}{2}\right)} \int_0^{\infty} t^{\left(\frac{n+1}{2}\right) - 1} e^{-t} dt = \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)} \Gamma\left(\frac{n+1}{2}\right)$$

Put in Equation (B)

$$E(\sum X_i^2)^{\frac{1}{2}} = \sigma \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)} \Gamma\left(\frac{n+1}{2}\right)$$

Put in Equation (A)

$$E(t_1) = K_1 \sigma \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)} \Gamma\left(\frac{n+1}{2}\right) = \frac{\Gamma\left(\frac{n}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n+1}{2}\right) \Gamma\left(\frac{n}{2}\right)} \frac{1}{2^{\frac{1}{2}}} \Gamma\left(\frac{n+1}{2}\right) \sigma = \sigma$$

Hence t_1 is unbiased estimator of σ .

(2) Consider

$$t_2 = \left[\frac{1}{2} \sum (X - \bar{X})^2 \right]^{\frac{1}{2}} \frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} = \frac{\Gamma\left(\frac{n-1}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)} \left[\sum (X - \bar{X})^2 \right]^{\frac{1}{2}} = K_2 \left[\sum (X - \bar{X})^2 \right]^{\frac{1}{2}}$$

Where

$$K_2 = \frac{\Gamma\left(\frac{n-1}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)}$$

Taking Expectation

$$E(t_2) = K_2 E\left[\sum (X - \bar{X})^2\right]^{\frac{1}{2}} \dots \dots \dots (C)$$

As $X_i \sim N(0, \sigma^2)$, So,

$$u = \frac{\sum (X - \bar{X})^2}{\sigma^2} \sim \chi_{(n-1)}^2 \Rightarrow \sum (X - \bar{X})^2 = \sigma^2 u \Rightarrow \left[\sum (X - \bar{X})^2 \right]^{\frac{1}{2}} = \sigma u^{\frac{1}{2}}$$

Taking Expectation

$$E\left[\sum (X - \bar{X})^2\right]^{\frac{1}{2}} = \sigma E\left(u^{\frac{1}{2}}\right) \dots \dots \dots (D)$$

Now we need to find $E\left(u^{\frac{1}{2}}\right)$ and $u \sim \chi^2_{(n-1)}$, the p.d.f of u .

$$f(u) = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} u^{\frac{n-1}{2}-1} e^{-\frac{u}{2}}, \quad 0 < u < \infty$$

$$E\left(u^{\frac{1}{2}}\right) = \int_0^{\infty} u^{\frac{1}{2}} \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} u^{\frac{n-1}{2}-1} e^{-\frac{u}{2}} du = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^{\infty} u^{\frac{n}{2}-1} e^{-\frac{u}{2}} du$$

Let

$$t = \frac{u}{2} \Rightarrow 2t = u \Rightarrow 2dt = du$$

when $u = 0$ then $t = 0$, when $u = \infty$ then $t = \infty$

$$E\left(u^{\frac{1}{2}}\right) = \frac{1}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^{\infty} (2t)^{\frac{n}{2}-1} e^{-t} (2dt) = \frac{2^{\frac{n}{2}-1} \cdot 2}{2^{\frac{n-1}{2}} \Gamma\left(\frac{n-1}{2}\right)} \int_0^{\infty} t^{\frac{n}{2}-1} e^{-t} dt$$

$$E\left(u^{\frac{1}{2}}\right) = \frac{2^{\frac{n}{2}-1-\frac{n-1}{2}+1}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right) = \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right)$$

Put in Equation (D)

$$E[\sum(X - \bar{X})^2]^{\frac{1}{2}} = \sigma \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right)$$

Put in Equation (C)

$$E(t_2) = K_2 \sigma \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right) = \frac{\Gamma\left(\frac{n-1}{2}\right) \left(\frac{1}{2}\right)^{\frac{1}{2}}}{\Gamma\left(\frac{n}{2}\right)} \frac{2^{\frac{1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)} \Gamma\left(\frac{n}{2}\right) \sigma = \sigma$$

Hence, t_2 is unbiased estimator of σ .

2.30 Unbiased Estimators Sum of Bernoulli R.V of Random Sample size n

Question 30: Let X be a random sample of size n from Bernoulli distribution and $T = \sum X_i$, show that

$$(a) \quad \frac{T}{n} \quad (b) \quad \frac{T(T-1)}{n(n-1)} \quad (c) \quad \frac{T(T-1)(T-2)}{n(n-1)(n-2)}$$

Are respectively unbiased estimators of θ , θ^2 and θ^3 .

Solution:-

The probability Mass Function of Bernoulli Distribution with parameter θ

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1$$

(a) Consider

$$\frac{T}{n} = \frac{\sum X_i}{n}$$

Taking Expectation

$$E\left(\frac{T}{n}\right) = E\left(\frac{\sum X_i}{n}\right) = \frac{1}{n} \sum E(X_i) \dots \dots (A)$$

Now,

$$E(X_i) = \sum_{x=0}^1 x f(x; \theta) = 0 \cdot \theta^0 (1 - \theta)^1 + 1 \cdot \theta^1 (1 - \theta)^0 = 0 + \theta = \theta$$

Put in (A)

$$E\left(\frac{T}{n}\right) = \frac{1}{n} \sum \theta = \frac{n}{n} \theta = \theta$$

Hence, $\frac{T}{n}$ is unbiased estimator of θ .

(b) Consider

$$\frac{T(T-1)}{n(n-1)} = \frac{1}{n(n-1)} [\sum X_i (\sum X_i - 1)] = \frac{1}{n(n-1)} [(\sum X_i)^2 - \sum X_i]$$

Taking Expectation

$$E \left[\frac{T(T-1)}{n(n-1)} \right] = \frac{1}{n(n-1)} [E(T^2) - E(T)] = \frac{1}{n(n-1)} [E(\sum X_i)^2 - E(\sum X_i)] \dots \dots \dots (B)$$

Now

$$E(\sum X_i) = \sum E(X_i) = n\theta$$

As

$$Var(\sum X_i) = E(\sum X_i)^2 - [E(\sum X_i)]^2$$

$$E(\sum X_i)^2 = Var(\sum X_i) + [E(\sum X_i)]^2$$

As X is Bernoulli random variable with parameter θ so $\sum X_i$ will have Binomial distribution with parameter n and θ . Having mean $n\theta$ and variance $n\theta(1-\theta)$.

$$E(\sum X_i)^2 = n\theta(1-\theta) + (n\theta)^2 = n\theta - n\theta^2 + n^2\theta^2$$

Put $E(\sum X_i)$ and $E(\sum X_i)^2$ in (B)

$$E \left[\frac{T(T-1)}{n(n-1)} \right] = \frac{1}{n(n-1)} [n\theta - n\theta^2 + n^2\theta^2 - n\theta] = \frac{1}{n(n-1)} [-n\theta^2 + n^2\theta^2]$$

$$E \left[\frac{T(T-1)}{n(n-1)} \right] = \frac{1}{n(n-1)} n\theta^2(n-1) = \theta^2$$

Hence $E \left[\frac{T(T-1)}{n(n-1)} \right]$ is unbiased estimator of θ^2 .

(c) Consider

$$\left[\frac{T(T-1)(T-2)}{n(n-1)(n-2)} \right] = \frac{1}{n(n-1)(n-2)} [T^3 - 3T^2 + 2T]$$

Taking Expectation

$$E \left[\frac{T(T-1)(T-2)}{n(n-1)(n-2)} \right] = \frac{1}{n(n-1)(n-2)} E[T(T-1)(T-2)] \dots \dots (c)$$

In Binomial

$$E(T) = n\theta, \quad E[T(T-1)] = n(n-1)\theta^2, \quad E[T(T-1)(T-2)] = n(n-1)(n-2)\theta^3$$

Put in (c)

$$E\left[\frac{T(T-1)(T-2)}{n(n-1)(n-2)}\right] = \frac{1}{n(n-1)(n-2)} n(n-1)(n-2)\theta^3 = \theta^3$$

2.31 Unbiased Estimator of Sum of Bernoulli R.V of Random Sample size n

Question 30.A: Let X has mean p and variance pq and $S_n = X_1 + X_2 + \dots + X_n \sim B(n, p)$

- (a) Prove that $\frac{S_n}{n}$ is unbiased estimator of p.
- (b) Show that $\left(1 - \frac{S_n}{n}\right)$ is unbiased estimator of (1-p) or q, but $\frac{S_n}{n} \left(1 - \frac{S_n}{n}\right)$ is not unbiased estimator of pq. Find the bias.
- (c) Prove that $\frac{S_n}{n} \left(1 - \frac{S_n}{n}\right)$ is asymptotically unbiased estimator of pq.
- (d) Prove that $\frac{S_n}{n}$ is consistent in mean square error as an estimate of p.
- (e) Prove that $\frac{S_n}{n} \left(1 - \frac{S_n}{n}\right)$ is consistent estimator of pq.

Solution

(a)

$$\frac{S_n}{n} = \frac{1}{n} [X_1 + X_2 + \dots + X_n]$$

Taking Expectation

$$E\left(\frac{S_n}{n}\right) = \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] = \frac{1}{n} [p + p + \dots + p] = \frac{np}{n} = p$$

Hence, $\frac{S_n}{n}$ is unbiased estimator of p.

(b)

$$E\left[1 - \frac{S_n}{n}\right] = E(1) - E\left(\frac{S_n}{n}\right) = 1 - p = q$$

So $\left(1 - \frac{S_n}{n}\right)$ is unbiased estimator of 1-p or q.

$$E\left[\frac{S_n}{n} \left(1 - \frac{S_n}{n}\right)\right] = E\left(\frac{S_n}{n}\right) - E\left(\frac{S_n}{n}\right)^2 = p - E\left(\frac{S_n}{n}\right)^2$$

As

$$\text{Var}\left(\frac{S_n}{n}\right) = E\left(\frac{S_n}{n}\right)^2 - \left[E\left(\frac{S_n}{n}\right)\right]^2 = E\left(\frac{S_n}{n}\right)^2 - p^2$$

$$E\left(\frac{S_n}{n}\right)^2 = \text{Var}\left(\frac{S_n}{n}\right) + p^2 = \frac{1}{n^2}\text{Var}(S_n) + p^2 = \frac{npq}{n^2} + p^2 = \frac{pq}{n} + p^2$$

So,

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = p - \frac{pq}{n} - p^2 = \frac{np - pq - np^2}{n} = \frac{np(1 - p) - pq}{n} = \frac{npq - pq}{n}$$

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = \left(\frac{n-1}{n}\right)pq \neq pq$$

Hence, $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is not unbiased estimator of pq .

Now, for Bias

$$\text{Bias} = E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] - pq = \left(\frac{n-1}{n}\right)pq - pq = \left(\frac{n-1-n}{n}\right)pq = -\frac{pq}{n}$$

(c)

Taking limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \text{Bias} = \lim_{n \rightarrow \infty} -\frac{pq}{n} = 0$$

Hence, $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is asymptotically unbiased estimator of pq .

(d)

$$E\left(\frac{S_n}{n} - p\right)^2 = \text{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2}\text{Var}(S_n) = \frac{npq}{n^2} = \frac{pq}{n}$$

Taking limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} E\left(\frac{S_n}{n} - p\right)^2 = \lim_{n \rightarrow \infty} \frac{pq}{n} = 0$$

Hence $\frac{S_n}{n}$ is consistent in mean square for the parameter p .

(e) As

$$E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = \left(\frac{n-1}{n}\right)pq = pq - \frac{pq}{n}$$

Taking limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} E\left[\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)\right] = \lim_{n \rightarrow \infty} pq - \frac{pq}{n} = pq$$

Hence $\frac{S_n}{n}\left(1 - \frac{S_n}{n}\right)$ is consistent in mean square for the parameter pq .

3 Question Related to Consistency

3.1 Sample Mean of Exponential Distribution Consistent Estimator of Population Mean

Question 31: Show that the sample mean of exponential distribution from a random sample of size n is consistent estimator of population mean.

Solution:-

The p.d.f of exponential distribution is

$$f(x, \lambda) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, \quad x > 0$$

Similarly,

$$E(x^r) = \int_0^{\infty} x^r \frac{1}{\lambda} e^{-\frac{x}{\lambda}} dx = \frac{1}{\lambda} \int_0^{\infty} x^r e^{-\frac{x}{\lambda}} dx$$

Let

$$t = \frac{x}{\lambda} \Rightarrow t\lambda = x \Rightarrow \lambda dt = dx$$

$$E(x^r) = \frac{1}{\lambda} \int_0^{\infty} (t\lambda)^r e^{-t} \lambda dt = \lambda^r \int_0^{\infty} t^{r+1-1} e^{-t} dt = \lambda^r \Gamma(r+1)$$

$$E(x) = \lambda \Gamma(1+1) = \lambda \Gamma(2) = \lambda 1! = \lambda$$

$$E(x^2) = \lambda^2 \Gamma(2+1) = \lambda^2 \Gamma(3) = \lambda^2 2! = 2\lambda^2$$

$$Var(x) = 2\lambda^2 - (\lambda)^2 = \lambda^2$$

Now, Variance of $\bar{x}$

$$Var(\bar{x}) = \frac{\lambda^2}{n} \quad \text{and} \quad S.D(\bar{x}) = \frac{\lambda}{\sqrt{n}}$$

For Simple consistency apply limit n tends to infinity on $Var(\bar{x})$

$$\lim_{n \rightarrow \infty} \text{Var}(\bar{x}) = \lim_{n \rightarrow \infty} \frac{\lambda^2}{n} = 0$$

As the variance approach to zero when sample size approach to infinity so sample mean is consistent estimator of population mean.

For strong consistency

$$p[|\bar{x} - \mu| < \varepsilon] = p\left[\left|\frac{\bar{x} - \mu}{\frac{\lambda}{\sqrt{n}}}\right| < \frac{\varepsilon}{\frac{\lambda}{\sqrt{n}}}\right] = p\left[|Z| < \frac{\sqrt{n}}{\lambda} \varepsilon\right] = p\left(-\frac{\sqrt{n}}{\lambda} \varepsilon < Z < \frac{\sqrt{n}}{\lambda} \varepsilon\right)$$

Taking limit n tends to infinity

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = p(-\infty < Z < \infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}} dZ = 1$$

This shows that sample mean is strong consistent estimator of population mean as

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = 1$$

3.2 Sample Mean of Normal Distribution Consistent Estimator of Population Mean

Question 32: Show that the sample mean of exponential distribution from a random sample of size n is consistent estimator of population mean.

Solution:-

$$p[|\bar{x} - \mu| < \varepsilon] = p\left[\left|\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}\right| < \frac{\varepsilon}{\frac{\sigma}{\sqrt{n}}}\right] = p\left[|Z| < \frac{\sqrt{n}}{\sigma} \varepsilon\right] = p\left(-\frac{\sqrt{n}}{\sigma} \varepsilon < Z < \frac{\sqrt{n}}{\sigma} \varepsilon\right)$$

Taking limit n tends to infinity

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = p(-\infty < Z < \infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}} dZ = 1$$

This shows that sample mean is strong consistent estimator of population mean as

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = 1$$

3.3 Consistent Estimator in Cauchy Distribution

Question 33: Show that for random sample from the distribution

$$f(x) = \frac{1}{\pi[1 + (x - \theta)^2]} \quad -\infty < x < \infty$$

The sample is not consistent estimator for θ . but the sample median is consistent estimator of θ .

Solution:

In Cauchy distribution the distribution of mean and single observation is same.

$$f(x) = \frac{1}{\pi[1 + (x - \theta)^2]}, \quad f(\bar{x}) = \frac{1}{\pi[1 + (\bar{x} - \theta)^2]}$$

$$p[|\bar{x} - \mu| < \varepsilon] = p[|z| < \varepsilon] = p(-\varepsilon < z < \varepsilon), \text{ where } z = \bar{x} - \mu$$

Taking limit n tends to infinity

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = p(-\varepsilon < z < \varepsilon) = \int_{-\varepsilon}^{\varepsilon} \frac{1}{\pi[1 + z^2]} dz$$

As it is an even function so

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] = \int_0^{\varepsilon} \frac{1}{\pi[1 + z^2]} dz = \frac{1}{\pi} \tan^{-1} z \Big|_0^{\varepsilon} = \frac{1}{\pi} [\tan^{-1} \varepsilon - \tan^{-1} 0] = \frac{\tan^{-1} \varepsilon}{\pi} \neq 1$$

As $\varepsilon > 0$, and $\tan^{-1} \varepsilon \neq \pi$

Hence, sample mean is not the consistent estimator of θ . Because

$$\lim_{n \rightarrow \infty} p[|\bar{x} - \mu| < \varepsilon] \neq 1$$

Now for sample median

$$p[|x_{med} - \mu| < \varepsilon] = p\left[\left|\frac{x_{med} - \mu}{\frac{\pi}{2\sqrt{n}}}\right| < \frac{\varepsilon}{\frac{\pi}{2\sqrt{n}}}\right] = p\left[|Z| < \frac{2\sqrt{n}}{\pi} \varepsilon\right] = p\left(-\frac{2\sqrt{n}}{\pi} \varepsilon < Z < \frac{2\sqrt{n}}{\pi} \varepsilon\right)$$

Taking limit n tends to infinity

$$\lim_{n \rightarrow \infty} p[|x_{med} - \mu| < \varepsilon] = p(-\infty < Z < \infty) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = 1$$

This shows that sample median is strong consistent estimator of population mean as

$$\lim_{n \rightarrow \infty} p[|x_{med} - \mu| < \varepsilon] = 1$$

3.4 Consistent Estimator in Uniform distribution (0,a).

Question 34: let X is a random sample of size n from $U(0,a)$

(i) Check $\hat{M} = \text{Max}(y_1, y_2, \dots, y_n)$ is biased or unbiased for “a”. (ii) If biased, find the amount of bias. (iii) Check for asymptotically unbiasedness. (iv) Show that $\hat{M}$ is consistent estimator for “a”.

Solution:

(i) The p.d.f of uniform distribution is

$$f(x) = \frac{1}{a} \Rightarrow \frac{1}{a}, \text{ for } (0, a)$$

The distribution function

$$F(x) = \int_0^x \frac{1}{a} dx = \frac{x}{a} \Big|_0^x = \frac{x}{a}$$

$$\hat{M} = \text{Max}(y_1, y_2, \dots, y_n) = y_n$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_n of random sample of $r = n$

$$g(\hat{M}) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1 - F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n \left[\frac{y_n}{a} \right]^{n-1} \frac{1}{a}$$

$$E(\hat{M}) = \int_0^a y_n g(y_n) dy_n = \int_0^a y_n n \left[\frac{y_n}{a} \right]^{n-1} \frac{1}{a} dy_n = \frac{n}{a^n} \int_0^a (y_n)^n dy_n$$

$$E(\hat{M}) = \frac{n}{a^n} \left[\left(\frac{y_n^{n+1}}{n+1} \right) \Big|_0^a \right] = \frac{n}{a^n} \left[\frac{1}{n+1} (a^{n+1} - 0) \right] = \frac{na^{n+1}}{a^n(n+1)} = \frac{na}{n+1}$$

Hence,

$$E(\hat{M}) = \frac{na}{n+1} \neq a$$

So, it is biased estimator of a .

(ii) Amount of Bias

$$Bias(\hat{M}) = E(\hat{M}) - a = \frac{na}{n+1} - a = \left(\frac{na - na - a}{n+1} \right) a = \frac{-a}{n+1}$$

(iii) Asymptotically Unbiasedness

Applying limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} Bias(\hat{M}) = \lim_{n \rightarrow \infty} \frac{-a}{n+1} = 0$$

Hence $\hat{M}$ is asymptotically unbiased estimator of a . As bias approaches to zero when sample size approach to infinity

(iv) Consistency

$$E(\hat{M}^2) = \int_0^a y_n^2 g(y_n) dy_n = \int_0^a y_n^2 n \left[\frac{y_n}{a} \right]^{n-1} \frac{1}{a} dy_n = \frac{n}{a^n} \int_0^a (y_n)^{n+1} dy_n$$

$$E(\hat{M}^2) = \frac{n}{a^n} \left[\left(\frac{y_n^{n+2}}{n+2} \right) \Big|_0^a \right] = \frac{n}{a^n} \left[\frac{1}{n+2} (a^{n+2} - 0) \right] = \frac{na^{n+2}}{a^n(n+2)} = \frac{na^2}{n+2}$$

$$var(\hat{M}) = E(\hat{M}^2) - [E(\hat{M})]^2 = \frac{na^2}{n+2} - \left(\frac{na}{n+1} \right)^2 = na^2 \left[\frac{1}{n+2} - \frac{n}{(n+1)^2} \right]$$

$$var(\hat{M}) = na^2 \left[\frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)^2} \right] = na^2 \left[\frac{n^2 + 1 + 2n - n^2 - 2n}{(n+2)(n+1)^2} \right]$$

$$var(\hat{M}) = \frac{na^2}{(n+2)(n+1)^2}$$

Taking limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} var(\hat{M}) = \lim_{n \rightarrow \infty} \frac{na^2}{(n+2)(n+1)^2} = \lim_{n \rightarrow \infty} \frac{\frac{a^2}{n^2}}{\left(1 + \frac{2}{n}\right) \left(1 + \frac{1}{n}\right)^2} = 0$$

Hence $\hat{M}$ is consistent estimator of a . As variance approach to zero as sample size approach to infinity

Note: Mid -Range: It is function of the order statistic and is defined as if $y_1 < y_2 \dots < y_n$. be order statistic then $\frac{y_1 + y_n}{2}$, is called mid-range.

3.5 Mean Square Consistency of Distribution of Mid-range

Question 35 The distribution of mid-range of a random sample of size n from a uniform distribution under the interval $(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ have the density function

$$f(t) = n2^{n-1} \left[\frac{1}{2} - |t - \theta| \right]^{n-1}$$

Show that the mid-range is consistent in mean square error and unbiased estimator of θ .

Solution:-

(1) Unbiasedness

$$\begin{aligned}
 E(t - \theta) &= \int_{\theta - \frac{1}{2}}^{\theta + \frac{1}{2}} (t - \theta) n2^{n-1} \left[\frac{1}{2} - |t - \theta| \right]^{n-1} dt \\
 E(t - \theta) &= \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta) n2^{n-1} \left[\frac{1}{2} - \{-(t - \theta)\} \right]^{n-1} dt + \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta) n2^{n-1} \left[\frac{1}{2} - \{(t - \theta)\} \right]^{n-1} dt \\
 E(t - \theta) &= n2^{n-1} \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta) \left[\frac{1}{2} + t - \theta \right]^{n-1} dt + n2^{n-1} \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta) \left[\frac{1}{2} - t + \theta \right]^{n-1} dt \\
 E(t - \theta) &= n2^{n-1} [I_1 + I_2] \dots \dots \dots (A)
 \end{aligned}$$

Take,

$$I_1 = \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta) \left[\frac{1}{2} + t - \theta \right]^{n-1} dt$$

Let

$$u = \frac{1}{2} + t - \theta \Rightarrow u - \frac{1}{2} = t - \theta \Rightarrow du = dt$$

$$\text{when } t = \theta - \frac{1}{2} \text{ then } u = \theta - \frac{1}{2} - \theta + \frac{1}{2} = 0$$

$$\text{when } t = \theta \text{ then } u = \theta - \theta + \frac{1}{2} = \frac{1}{2}$$

$$I_1 = \int_0^{\frac{1}{2}} \left(u - \frac{1}{2}\right) u^{n-1} du = \int_0^{\frac{1}{2}} u^n du - \frac{1}{2} \int_0^{\frac{1}{2}} u^{n-1} du = \frac{u^{n+1}}{n+1} \Big|_0^{\frac{1}{2}} - \frac{1}{2} \frac{u^n}{n} \Big|_0^{\frac{1}{2}}$$

$$I_1 = \frac{1}{n+1} \left[\left(\frac{1}{2}\right)^{n+1} - 0 \right] - \frac{1}{2n} \left[\left(\frac{1}{2}\right)^n - 0 \right] = \left(\frac{1}{2}\right)^{n+1} \left[\frac{1}{n+1} - \frac{1}{n} \right] = \left(\frac{1}{2}\right)^{n+1} \left[\frac{n - n - 1}{n(n+1)} \right]$$

$$I_1 = -\frac{\left(\frac{1}{2}\right)^{n+1}}{n(n+1)}$$

Take,

$$I_2 = \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta) \left[\frac{1}{2} - t + \theta \right]^{n-1} dt$$

Let

$$u = \frac{1}{2} - t + \theta \Rightarrow \frac{1}{2} - u = t - \theta \Rightarrow -du = dt$$

$$\text{when } t = \theta \text{ then } u = \frac{1}{2} - \theta + \theta = \frac{1}{2}$$

$$\text{when } t = \theta + \frac{1}{2} \text{ then } u = \frac{1}{2} - \theta - \frac{1}{2} + \theta = 0$$

$$I_2 = \int_{\frac{1}{2}}^0 \left(\frac{1}{2} - u\right) u^{n-1} (-du) = \frac{1}{2} \int_0^{\frac{1}{2}} u^{n-1} du - \int_0^{\frac{1}{2}} u^n du = \frac{1}{2} \frac{u^n}{n} \Big|_0^{\frac{1}{2}} - \frac{u^{n+1}}{n+1} \Big|_0^{\frac{1}{2}}$$

$$I_2 = \frac{1}{2n} \left[\left(\frac{1}{2} \right)^n - 0 \right] - \frac{1}{n+1} \left[\left(\frac{1}{2} \right)^{n+1} - 0 \right] = \left(\frac{1}{2} \right)^{n+1} \left[\frac{1}{n} - \frac{1}{n+1} \right] = \left(\frac{1}{2} \right)^{n+1} \left[\frac{n+1-n}{n(n+1)} \right]$$

$$I_2 = \frac{\left(\frac{1}{2} \right)^{n+1}}{n(n+1)}$$

Putting I_1 and I_2 in (A)

$$E(t - \theta) = n2^{n-1} \left[-\frac{\left(\frac{1}{2} \right)^{n+1}}{n(n+1)} + \frac{\left(\frac{1}{2} \right)^{n+1}}{n(n+1)} \right] = 0$$

$$E(t) = \theta$$

This shows that t is unbiased estimator of θ .

(ii) Mean Square Consistency

$$Var(t) = E(t - \theta)^2 = \int_{\theta - \frac{1}{2}}^{\theta + \frac{1}{2}} (t - \theta)^2 n2^{n-1} \left[\frac{1}{2} - |t - \theta| \right]^{n-1} dt$$

$$E(t - \theta)^2 = \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta)^2 n2^{n-1} \left[\frac{1}{2} - \{-(t - \theta)\} \right]^{n-1} dt + \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta)^2 n2^{n-1} \left[\frac{1}{2} - \{(t - \theta)\} \right]^{n-1} dt$$

$$E(t - \theta)^2 = n2^{n-1} \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta)^2 \left[\frac{1}{2} + t - \theta \right]^{n-1} dt + n2^{n-1} \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta)^2 \left[\frac{1}{2} - t + \theta \right]^{n-1} dt$$

$$E(t - \theta)^2 = n2^{n-1} [I_3 + I_4] \dots \dots (B)$$

Take,

$$I_3 = \int_{\theta - \frac{1}{2}}^{\theta} (t - \theta)^2 \left[\frac{1}{2} + t - \theta \right]^{n-1} dt$$

Let

$$u = \frac{1}{2} + t - \theta \Rightarrow u - \frac{1}{2} = t - \theta \Rightarrow du = dt$$

$$\text{when } t = \theta - \frac{1}{2} \text{ then } u = \theta - \frac{1}{2} - \theta + \frac{1}{2} = 0$$

$$\text{when } t = \theta \text{ then } u = \theta - \theta + \frac{1}{2} = \frac{1}{2}$$

$$I_3 = \int_0^{\frac{1}{2}} \left(u - \frac{1}{2}\right)^2 u^{n-1} du = \int_0^{\frac{1}{2}} u^{n+1} du + \frac{1}{4} \int_0^{\frac{1}{2}} u^{n-1} du - \int_0^{\frac{1}{2}} u^n du = \frac{u^{n+2}}{n+2} \Big|_0^{\frac{1}{2}} + \frac{1}{4} \frac{u^n}{n} \Big|_0^{\frac{1}{2}} - \frac{u^{n+1}}{n+1} \Big|_0^{\frac{1}{2}}$$

$$I_3 = \frac{1}{n+2} \left[\left(\frac{1}{2}\right)^{n+2} - 0 \right] + \frac{1}{4n} \left[\left(\frac{1}{2}\right)^n - 0 \right] - \frac{1}{n+1} \left[\left(\frac{1}{2}\right)^{n+1} - 0 \right] = \frac{\left(\frac{1}{2}\right)^{n+2}}{n+2} + \frac{\left(\frac{1}{2}\right)^{n+2}}{n} - \frac{\left(\frac{1}{2}\right)^{n+1}}{n+1}$$

$$I_3 = \left(\frac{1}{2}\right)^{n+2} \left[\frac{1}{n+2} + \frac{1}{n} - \frac{2}{(n+1)} \right]$$

Take,

$$I_4 = \int_{\theta}^{\theta + \frac{1}{2}} (t - \theta)^2 \left[\frac{1}{2} - t + \theta \right]^{n-1} dt$$

Let

$$u = \frac{1}{2} - t + \theta \Rightarrow \frac{1}{2} - u = t - \theta \Rightarrow -du = dt$$

$$\text{when } t = \theta \text{ then } u = \frac{1}{2} - \theta + \theta = \frac{1}{2}$$

$$\text{when } t = \theta + \frac{1}{2} \text{ then } u = \frac{1}{2} - \theta - \frac{1}{2} + \theta = 0$$

$$I_3 = \int_{\frac{1}{2}}^0 \left(\frac{1}{2} - u\right)^2 u^{n-1} (-du) = \frac{1}{4} \int_0^{\frac{1}{2}} u^{n-1} du + \int_0^{\frac{1}{2}} u^{n+1} du + - \int_0^{\frac{1}{2}} u^n du = \frac{1}{4} \frac{u^n}{n} \Big|_0^{\frac{1}{2}} + \frac{u^{n+2}}{n+2} \Big|_0^{\frac{1}{2}} - \frac{u^{n+1}}{n+1} \Big|_0^{\frac{1}{2}}$$

$$I_4 = \frac{1}{4n} \left[\left(\frac{1}{2} \right)^n - 0 \right] + \frac{1}{n+2} \left[\left(\frac{1}{2} \right)^{n+2} - 0 \right] - \frac{1}{n+1} \left[\left(\frac{1}{2} \right)^{n+1} - 0 \right] = \frac{\left(\frac{1}{2} \right)^{n+2}}{n} + \frac{\left(\frac{1}{2} \right)^{n+2}}{n+2} - \frac{\left(\frac{1}{2} \right)^{n+1}}{n+1}$$

$$I_4 = \left(\frac{1}{2} \right)^{n+2} \left[\frac{1}{n} + \frac{1}{n+2} - \frac{2}{(n+1)} \right]$$

Put I_3 and I_4 in (B)

$$E(t - \theta)^2 = n2^{n-1} \left[\left(\frac{1}{2} \right)^{n+2} \left[\frac{1}{n+2} + \frac{1}{n} - \frac{2}{(n+1)} \right] + \left(\frac{1}{2} \right)^{n+2} \left[\frac{1}{n} + \frac{1}{n+2} - \frac{2}{(n+1)} \right] \right]$$

$$E(t - \theta)^2 = n2^{n-1} 2 \left(\frac{1}{2} \right)^{n+2} \left[\frac{1}{n} + \frac{1}{n+2} - \frac{2}{(n+1)} \right]$$

$$E(t - \theta)^2 = \frac{n}{4} \left[\frac{1}{n} + \frac{1}{n+2} - \frac{2}{(n+1)} \right] = \frac{n}{4} \left[\frac{(n+1)(n+2) + n(n+1) - 2n(n+2)}{n(n+1)(n+2)} \right]$$

$$E(t - \theta)^2 = \frac{1}{4} \left[\frac{n^2 + 3n + 2 + n^2 + n - 2n^2 - 4n}{(n+1)(n+2)} \right] = \frac{1}{4} \left[\frac{2}{(n+1)(n+2)} \right]$$

$$Var(t) = E(t - \theta)^2 = \frac{1}{2(n+1)(n+2)}$$

Taking limit $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} Var(t) = \lim_{n \rightarrow \infty} \frac{1}{2(n+1)(n+2)} = 0$$

Hence, t is consistent estimator of θ . As its variance approach to zero when sample size approach infinity.

4. Question Related to CRLB and MVB

4.1 MVB Estimator in Exponential Distribution

Question 36: Show that $\bar{X} = \frac{1}{n} \sum X_i$ in exponential distribution with mean θ is MVB estimator has variance Also Obtain the CRLB for variance.

Solution:-

The p.d.f of exponential distribution with mean θ

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad x > 0$$

The likelihood function

$$L = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) = \frac{1}{\theta} e^{-\frac{x_1}{\theta}} \frac{1}{\theta} e^{-\frac{x_2}{\theta}} \dots \frac{1}{\theta} e^{-\frac{x_n}{\theta}}$$

$$L = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum x_i}$$

Taking ln both sides

$$\ln L = n \ln 1 - n \ln \theta - \frac{1}{\theta} \sum x_i$$

Diff w.r.t to θ

$$\frac{\partial \ln L}{\partial \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum x_i = \frac{n}{\theta^2} \left[\frac{\sum x_i}{n} - \theta \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n}, \quad T(\theta) = \theta, \quad A(\theta) = \frac{n}{\theta^2}$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{1}{\frac{n}{\theta^2}} = \frac{\theta^2}{n}$$

This shows the $\frac{1}{n} \sum x_i$ is MVB estimator with variance $\frac{\theta^2}{n}$.

For Crammer Rao Lower bound of Variance

Consider

$$\frac{\partial \ln L}{\partial \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum x_i$$

Diff again w.r.t θ

$$\frac{\partial^2 \ln L}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum x_i$$

Taking Expectation

$$E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum E(x_i) \text{ --- (A)}$$

$$E(x) = \int_0^{\infty} x \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$$

Let

$$t = \frac{x}{\theta} \Rightarrow t\theta = x \Rightarrow \theta dt = dx$$

$$E(x) = \int_0^{\infty} t\theta \frac{1}{\theta} e^{-\theta t} dt = \theta \int_0^{\infty} t^{2-1} e^{-\theta t} dt = \theta \Gamma 2 = \theta.1! = \theta$$

Put in A

$$E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = \frac{n}{\theta^2} - \frac{2}{\theta^3} n\theta = \frac{n}{\theta^2} - \frac{2n}{\theta^2} = -\frac{n}{\theta^2}$$

Using CRLB

$$\text{Var}(t_n) \geq \frac{[T'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]} \geq \frac{1}{-\left(-\frac{n}{\theta^2}\right)} \geq \frac{\theta^2}{n}$$

Which is require lower bound, it can be observed that $\frac{1}{n} \sum x_i$ is MVB Estimator of θ

4.2 CRLB E in Exponential Distribution

Question 37: Find the Crammer Rao Lower bound for unbiased estimator of

$$(a) \quad \phi(\theta) = \frac{1}{\theta} \quad (b) \quad \phi(\theta) = P(X > k), k \text{ is know}$$

For $f(x) = \theta e^{-x\theta}, 0 < x < \infty$, Does there exist a MVBUE of any function of θ .

Solution;-

(i)

$$f(x; \theta) = \theta e^{-x\theta} \quad x > 0$$

The likelihood function

$$L = f(x_1; \theta) f(x_2; \theta) \dots \dots f(x_n; \theta) = \theta e^{-x_1 \theta} \theta e^{-x_2 \theta} \dots \dots \dots \theta e^{-x_n \theta}$$

$$L = (\theta)^n e^{-\theta \sum x_i}$$

Taking ln both sides

$$\ln L = n \ln \theta - \theta \sum x_i$$

Diff w.r.t to θ

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} - \sum x_i$$

Diff w.r.t to θ

$$\frac{\partial^2 \ln L}{\partial \theta^2} = \frac{-n}{\theta^2}$$

Taking Expectation

$$E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = \frac{-n}{\theta^2}$$

$$\phi(\theta) = \frac{1}{\theta}, \quad \phi'(\theta) = \frac{-1}{\theta^2}$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]} \geq \frac{\left(\frac{-1}{\theta^2} \right)^2}{-\left(\frac{-n}{\theta^2} \right)} \geq \frac{1}{n\theta^2}$$

(ii)

$$\phi(\theta) = P(X \geq k) = \int_k^{\infty} \theta e^{-x\theta} dx = -e^{-x\theta} \Big|_k^{\infty} = -[e^{-\infty} - e^{-k\theta}] = -[0 - e^{-k\theta}] = e^{-k\theta}$$

$$\phi(\theta) = e^{-k\theta}, \quad \phi'(\theta) = -ke^{-k\theta}$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E\left[\frac{\partial^2 \ln L}{\partial \theta^2}\right]} \geq \frac{(-ke^{-k\theta})^2}{-\left(\frac{-n}{\theta^2}\right)} \geq \frac{k^2 \theta^2 e^{-2k\theta}}{n} \geq \frac{(k\theta e^{-k\theta})^2}{n}$$

(b) MVBUE

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta} - \sum x_i = -n \left[\frac{\sum x_i}{n} - \frac{1}{\theta} \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n}, \quad T(\theta) = \frac{1}{\theta}, \quad A(\theta) = -n$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{-\frac{1}{\theta^2}}{-n} = \frac{1}{n\theta^2}$$

This shows the $\frac{1}{n} \sum x_i$ is MVB estimator with variance. $\frac{1}{n\theta^2}$.

4.3 CRLB E in Exponential Distribution

Question 38: Find the Crammer Rao Lower bound for unbiased estimator of

$$(a) \phi(\theta) = \frac{1}{\theta} \quad (b) \phi(\theta) = P(X > k), k \text{ is know}$$

For $f(x) = \frac{1}{\theta} e^{-\frac{x}{\theta}}, 0 < x < \infty$, Does there exist a MVBUE of any function of θ .

Solution;-

(i)

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \quad x > 0$$

The likelihood function

$$L = f(x_1; \theta) f(x_2; \theta) \dots \dots f(x_n; \theta) = \frac{1}{\theta} e^{-\frac{x_1}{\theta}} \frac{1}{\theta} e^{-\frac{x_2}{\theta}} \dots \dots \frac{1}{\theta} e^{-\frac{x_n}{\theta}}$$

$$L = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum x_i}$$

Taking ln both sides

$$\ln L = n \ln 1 - n \ln \theta - \frac{1}{\theta} \sum x_i$$

Diff w.r.t to θ

$$\frac{\partial \ln L}{\partial \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum x_i$$

Diff again w.r.t θ

$$\frac{\partial^2 \ln L}{\partial \theta^2} = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum x_i$$

Taking Expectation

$$E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum E(x_i) \text{ --- (A)}$$

$$E(x) = \int_0^{\infty} x \frac{1}{\theta} e^{-\frac{x}{\theta}} dx$$

Let

$$t = \frac{x}{\theta} \Rightarrow t\theta = x \Rightarrow \theta dt = dx$$

$$E(x) = \int_0^{\infty} t\theta \frac{1}{\theta} e^{-t} \theta dt = \theta \int_0^{\infty} t^{2-1} e^{-t} dt = \theta \Gamma 2 = \theta \cdot 1! = \theta$$

Put in A

$$E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right] = \frac{n}{\theta^2} - \frac{2}{\theta^3} n\theta = \frac{n}{\theta^2} - \frac{2n}{\theta^2} = -\frac{n}{\theta^2}$$

$$\phi(\theta) = \frac{1}{\theta}, \quad \phi'(\theta) = -\frac{1}{\theta^2}$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L}{\partial \theta^2} \right]} \geq \frac{\left(\frac{-1}{\theta^2} \right)^2}{-\left(\frac{-n}{\theta^2} \right)} \geq \frac{1}{n\theta^2}$$

(ii)

$$\phi(\theta) = P(X \geq k) = \int_k^{\infty} \frac{1}{\theta} e^{-\frac{x}{\theta}} d\theta = -e^{-\frac{x}{\theta}} \Big|_k^{\infty} = -\left[e^{-\infty} - e^{-\frac{k}{\theta}} \right] = -\left[0 - e^{-\frac{k}{\theta}} \right] = e^{-\frac{k}{\theta}}$$

$$\phi(\theta) = e^{-\frac{k}{\theta}}, \quad \phi'(\theta) = -\frac{k}{\theta^2} e^{-\frac{k}{\theta}}$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E\left[\frac{\partial^2 \ln L}{\partial \theta^2}\right]} \geq \frac{\left(-\frac{k}{\theta^2} e^{-\frac{k}{\theta}}\right)^2}{-\left(\frac{-n}{\theta^2}\right)} \geq \frac{\left(ke^{-\frac{k}{\theta}}\right)^2}{\theta^4} \times \frac{\theta^2}{n} \geq \frac{\left(ke^{-\frac{k}{\theta}}\right)^2}{n\theta^2}$$

(b) MVBUE

$$\frac{\partial \ln L}{\partial \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum x_i = \frac{n}{\theta^2} \left[\frac{\sum x_i}{n} - \theta \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n}, \quad T(\theta) = \theta, \quad A(\theta) = \frac{n}{\theta^2}$$

Here, $T(\theta) = \theta$, but it should be equal to $\frac{1}{\theta}$, so there does not exist MVBUE.

4.5 CRLB and MVB for Normal Distribution

Question 39: Find Crammer Rao Lower bound for normal distribution for

(1) $\phi(\theta) = \mu$, when σ^2 is known

(2) $\phi(\theta) = \sigma^2$, when is known

(3) $\phi(\theta) = \sigma^2$, when $\mu = 0$

Solution:-

(1)

The pdf of normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$L(\mu) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum (x-\mu)^2}$$

Taking ln both sides

$$\ln L(\mu) = \ln \left[\frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2} \right]$$

$$\ln L(\mu) = \ln 1 - \frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2 \ln e$$

$$\ln L(\mu) = -\frac{n}{2} \ln(\sigma^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x-\mu)^2$$

Differentiate partially w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{2}{2\sigma^2} \sum_{i=1}^n (x-\mu) (-1)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x-\mu)$$

Differentiate, (1) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (0-1) = -\frac{n}{\sigma^2}$$

$$E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu) \right] = -\frac{n}{\sigma^2}$$

$$\phi(\theta) = \mu, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\mu)}{\partial \mu^2} \right]} \geq \frac{(1)^2}{-\left(-\frac{n}{\sigma^2}\right)} \geq \frac{\sigma^2}{n}$$

(2)

Now, to find C.R.L.B for σ^2 when μ known

$$\frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{\partial}{\partial \sigma^2} [\ln(\sigma^2)] - \frac{\partial}{\partial \sigma^2} \left[\frac{n}{2} \ln(2\pi) \right] - \frac{1}{2} \sum_{i=1}^n (x-\mu)^2 \frac{\partial}{\partial \sigma^2}$$

$$\frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{1}{\sigma^2} (1) - \frac{1}{2} \sum_{i=1}^n (x-\mu)^2 (-1) (\sigma^2)^{-2} (1)$$

Again differentiate (2) w,r,t σ^2

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-1} + \frac{1}{2} \sum_{i=1}^n (x - \mu)^2 \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-2}$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{n}{2} (\sigma^2)^{-2} - \frac{2}{2} \sum_{i=1}^n (x - \mu)^2 (\sigma^2)^{-3}$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (x - \mu)^2$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{1}{2\sigma^6} \left[n\sigma^2 - 2 \sum_{i=1}^n (x - \mu)^2 \right]$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{1}{2\sigma^6} \left[\sum_{i=1}^n (x - \mu)^2 - 2 \sum_{i=1}^n (x - \mu)^2 \right]$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n (x - \mu)^2 \right]$$

$$E \left[\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) \right] = \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n E(x - \mu)^2 \right] = \frac{-n\sigma^2}{2\sigma^6} = -\frac{n}{2\sigma^4}$$

$$\phi(\theta) = \sigma^2, \quad \phi'(\theta) = 1$$

PUACP

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\sigma^2)}{\partial \sigma^2} \right]} \geq \frac{(1)^2}{-\left(-\frac{n}{2\sigma^4}\right)} \geq \frac{2\sigma^4}{n}$$

(3)

Now, to find C.R.L.B for σ^2 when $\mu = 0$ known

$$\frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{\partial}{\partial \sigma^2} [\ln(\sigma^2)] - \frac{\partial}{\partial \sigma^2} \left[\frac{n}{2} \ln(2\pi) \right] - \frac{1}{2} \sum_{i=1}^n x_i^2 \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-1}$$

$$\frac{\partial}{\partial \sigma^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{1}{\sigma^2} (1) - \frac{1}{2} \sum_{i=1}^n (x)^2 (-1)(\sigma^2)^{-2} (1)$$

Again differentiate (2) w,r,t σ^2

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = -\frac{n}{2} \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-1} + \frac{1}{2} \sum_{i=1}^n (x)^2 \frac{\partial}{\partial \sigma^2} (\sigma^2)^{-2}$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{n}{2} (\sigma^2)^{-2} - \frac{2}{2} \sum_{i=1}^n (x)^2 (\sigma^2)^{-3}$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (x)^2$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{1}{2\sigma^6} \left[n\sigma^2 - 2 \sum_{i=1}^n (x)^2 \right]$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{1}{2\sigma^6} \left[\sum_{i=1}^n (x)^2 - 2 \sum_{i=1}^n (x)^2 \right]$$

$$\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) = \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n (x)^2 \right]$$

$$E \left[\frac{\partial^2}{\partial(\sigma^2)^2} \ln L(\sigma^2) \right] = \frac{-1}{2\sigma^6} \left[\sum_{i=1}^n E(x^2) \right] = \frac{-n\sigma^2}{2\sigma^6} = -\frac{n}{2\sigma^4}$$

$$\phi(\theta) = \sigma^2, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$Var(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\sigma^2)}{\partial \sigma^2} \right]} \geq \frac{(1)^2}{-\left(-\frac{n}{2\sigma^4}\right)} \geq \frac{2\sigma^4}{n}$$

4.5 CRLB and MVB for Normal Distribution

Question 40: (a) Obtain the MVB estimates for an unknown population parameter μ when θ^2 is known and for θ^2 when $\mu = 0$ respectively, in normal distribution with mean μ and variance θ^2 . Also find their variances

(b) Find C.R.L.B for μ .,

(c) Find MVB estimator of θ . Comment in either case.

(d) Find MVB for θ by using C.R.L.B

Solution

(a) MVB for μ when θ^2 known

The pdf of normal distribution $N(0, \theta^2)$

$$f(x) = \frac{1}{\theta\sqrt{2\pi}} e^{-\frac{1}{2\theta^2}(x-\mu)^2}$$

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\theta\sqrt{2\pi}} e^{-\frac{1}{2\theta^2}(x-\mu)^2}$$

$$L(\mu) = \frac{1}{\theta^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\theta^2} \sum (x-\mu)^2}$$

Taking ln both sides

$$\ln L(\mu) = \ln \left[\frac{1}{\theta^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\theta^2} \sum_{i=1}^n (x-\mu)^2} \right]$$

$$\ln L(\mu) = \ln 1 - \frac{n}{2} \ln(\theta^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\theta^2} \sum_{i=1}^n (x-\mu)^2 \ln e$$

$$\ln L(\mu) = -\frac{n}{2} \ln(\theta^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\theta^2} \sum_{i=1}^n (x-\mu)^2$$

Differentiate partially w.r.t μ

$$\frac{\partial}{\partial \mu} \ln L(\mu) = -\frac{2}{2\theta^2} \sum_{i=1}^n (x-\mu) (-1)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\theta^2} \sum_{i=1}^n (x-\mu) \dots \dots \dots (A)$$

$$\frac{\partial}{\partial \mu} \ln L(\mu) = \frac{1}{\theta^2} (\sum x_i - n\mu) = \frac{n}{\theta^2} \left(\frac{\sum x_i}{n} - \mu \right)$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n}, \quad T(\theta) = \mu, \quad A(\theta) = \frac{n}{\theta^2}$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{1}{\frac{n}{\theta^2}} = \frac{\theta^2}{n}$$

(b) C.R.L.B for μ θ^2 known

Diff, (A) w.r.t μ

$$\frac{\partial^2}{\partial \mu^2} \ln L(\mu) = \frac{1}{\theta^2} \sum_{i=1}^n (0 - 1) = -\frac{n}{\theta^2}$$

$$E \left[\frac{\partial^2}{\partial \mu^2} \ln L(\mu) \right] = -\frac{n}{\theta^2}$$

$$\phi(\theta) = \mu, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\mu)}{\partial \mu^2} \right]} \geq \frac{(1)^2}{-\left(-\frac{n}{\theta^2}\right)} \geq \frac{\theta^2}{n}$$

(a) MVB for θ^2 when $\mu = 0$

The pdf of normal distribution $N(0, \theta^2)$

$$f(x) = \frac{1}{\theta\sqrt{2\pi}} e^{-\frac{1}{2\theta^2}(x)^2}$$

The likelihood function

$$L(\theta^2) = \prod_{i=1}^n \frac{1}{\theta\sqrt{2\pi}} e^{-\frac{1}{2\theta^2}(x)^2}$$

$$L(\theta^2) = \frac{1}{\theta^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\theta^2} \sum_{i=1}^n (x)^2}$$

Taking ln both sides

$$\ln L(\theta^2) = \ln \left[\frac{1}{\theta^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\theta^2} \sum_{i=1}^n (x)^2} \right]$$

$$\ln L(\theta^2) = \ln 1 - \frac{n}{2} \ln(\theta^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\theta^2} \sum_{i=1}^n (x_i)^2$$

$$\ln L(\theta^2) = -\frac{n}{2} \ln(\theta^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\theta^2} \sum_{i=1}^n (x_i)^2$$

$$\frac{\partial}{\partial \theta^2} \ln L(\theta^2) = -\frac{n}{2} \frac{\partial}{\partial \theta^2} [\ln(\theta^2)] - \frac{\partial}{\partial \theta^2} \left[\frac{n}{2} \ln(2\pi) \right] - \frac{1}{2} \sum_{i=1}^n x_i^2 \frac{\partial}{\partial \theta^2} (\theta^2)^{-1}$$

$$\frac{\partial}{\partial \theta^2} \ln L(\theta^2) = -\frac{n}{2} \frac{1}{\theta^2} (1) - \frac{1}{2} \sum_{i=1}^n (x_i)^2 (-1)(\theta^2)^{-2} (1)$$

$$\frac{\partial}{\partial \theta^2} \ln L(\theta^2) = -\frac{n}{2} \frac{1}{\theta^2} + \frac{1}{2\theta^4} \sum_{i=1}^n (x_i)^2 \dots \dots \dots (B)$$

$$\frac{\partial}{\partial \theta^2} \ln L(\theta^2) = \frac{n}{2\theta^4} \left[\frac{1}{n} \sum_{i=1}^n (x_i)^2 - \theta^2 \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i^2}{n}, \quad T(\theta) = \theta^2, \quad A(\theta) = \frac{n}{2\theta^4}$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{1}{\frac{n}{2\theta^4}} = \frac{2\theta^4}{n}$$

(c) MVB for θ when $\mu = 0$

Take

$$\ln L(\theta) = -\frac{n}{2} \ln(\theta^2) - \frac{n}{2} \ln(2\pi) - \frac{1}{2\theta^2} \sum_{i=1}^n (x_i)^2$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{n}{2} \frac{\partial}{\partial \theta} [\ln(\theta^2)] - \frac{\partial}{\partial \theta} \left[\frac{n}{2} \ln(2\pi) \right] - \frac{1}{2} \sum_{i=1}^n x_i^2 \frac{\partial}{\partial \theta} (\theta^{-2})$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{n}{2} \frac{1}{\theta^2} (2\theta) - \frac{1}{2} \sum_{i=1}^n (x_i)^2 (-2)(\theta)^{-3} (1)$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^n (x_i)^2 \dots \dots \dots (C)$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \frac{n}{\theta^3} \left[\frac{1}{n} \sum_{i=1}^n (x_i)^2 - \theta^2 \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i^2}{n}, \quad T(\theta) = \theta^2, \quad A(\theta) = \frac{n}{\theta^3}$$

Here $T(\theta)$ should be equal to θ but $T(\theta) = \theta^2$, so MVB of θ does not exist. So, MVB estimator of θ^2 is not invariant.

(d) C.R.L.B for θ

Diff (C) w.r.t θ

$$\frac{\partial^2}{\partial \theta^2} \ln L(\theta) = \frac{n}{\theta^2} - \frac{3}{\theta^4} \sum_{i=1}^n (x_i)^2$$

$$E \left[\frac{\partial^2}{\partial \theta^2} \ln L(\theta) \right] = \frac{n}{\theta^2} - \frac{3}{\theta^4} E \left[\sum_{i=1}^n (x_i)^2 \right] = \frac{n}{\theta^2} - \frac{3}{\theta^4} (n\theta^2) = \frac{n}{\theta^2} - \frac{3n}{\theta^2} = -\frac{2n}{\theta^2}$$

$$\phi(\theta) = \theta, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} \right]} = \frac{(1)^2}{-\left(-\frac{2n}{\theta^2}\right)} = \frac{\theta^2}{2n}$$

4.6 MVB for Cauchy Distribution (Non-existence)

Question 41 : Prove that MVB estimator of θ in Cauchy distribution does not exist.

Solution:-

The likelihood function

$$L(\theta) = \prod_{i=1}^n \frac{1}{\pi \{1 + (x_i - \theta)^2\}}$$

$$L(\theta) = \pi^{-n} \prod_{i=1}^n \{1 + (x_i - \theta)^2\}^{-1}$$

Taking ln both sides

$$\ln L(\theta) = -n \ln \pi - \ln \prod_{i=1}^n [1 + (x_i - \theta)^2]$$

$$\ln L(\theta) = -n \ln \pi - \sum \ln [1 + (x_i - \theta)^2]$$

Diff w.r.t θ

$$\frac{\partial}{\partial \theta} \ln L(\theta) = -\sum \frac{1}{[1 + (x_i - \theta)^2]} [0 + 2(x_i - \theta)(0 - 1)]$$

$$\frac{\partial}{\partial \theta} \ln L(\theta) = \sum \frac{2(x_i - \theta)}{[1 + (x_i - \theta)^2]}$$

Here $\frac{\partial}{\partial \theta} \ln L(\theta)$ cannot be expressed in the form $A(\theta)[t_n - T(\theta)]$, so MVB estimator does not exist.

4.7 MVB for Binomial Distribution

Question 42; Find MVB estimator for “np” of binomial distribution , where n is known.

Solution

The likelihood function

$$L(p) = \prod_{i=1}^n \binom{n}{x_i} p^{x_i} (1-p)^{n-x_i}$$

$$L(p) = \prod_{i=1}^n \binom{n}{x_i} p^{\sum x_i} (1-p)^{n^2 - \sum x_i}$$

$$L(p) = k p^{\sum x_i} (1-p)^{n^2 - \sum x_i}$$

Taking “ln” both sides

$$\ln L(p) = \ln k + \sum x_i \ln p + (n^2 - \sum x_i) \ln (1-p)$$

Differentiate w.r.t p

$$\frac{\partial}{\partial p} \ln L(p) = 0 + \sum x_i \frac{\partial}{\partial p} \ln p + (n^2 - \sum x_i) \frac{\partial}{\partial p} \ln (1-p)$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{\sum x_i}{p} - \frac{n^2 - \sum x_i}{1-p} = \frac{(1-p)\sum x_i - p(n^2 - \sum x_i)}{p(1-p)} = \frac{\sum x_i - p\sum x_i - pn^2 + p\sum x_i}{p(1-p)}$$

$$\frac{\partial}{\partial p} \ln L(p) = \frac{\sum x_i - pn^2}{p(1-p)} = \frac{n}{p(1-p)} \left[\frac{1}{n} \sum x_i - np \right] = \frac{n}{pq} \left[\frac{1}{n} \sum x_i - np \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n} \quad , \quad T(\theta) = np, \quad A(\theta) = \frac{n}{pq}$$

$$Var(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{n}{\frac{n}{pq}} = pq$$

4.8 MVB and CRLB for Poisson Distribution

Question 43: For $X_1, X_2, \dots, X_n$ independent random variables each with Poisson with mean λ , where λ is unknown. Find the MVB and Crammer Rao Lower Bound on the variance of any unbiased estimator t_n for λ .

Solution:

(a) MVB for λ .

The probability Mass Function of Poisson Distribution is

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$L(\lambda) = \frac{e^{-\lambda n} \lambda^{\sum x_i}}{x_1! \cdot x_2! \cdot \dots \cdot x_n!}$$

Taking “ln” Both sides

$$\ln L(\lambda) = -\lambda n \ln e + \sum x_i \ln \lambda - \ln (x_1! \cdot x_2! \cdot \dots \cdot x_n!)$$

$$\ln L(\lambda) = -\lambda n + \sum x_i \ln \lambda - \sum \ln (x_i!)$$

Diff w.r.t λ

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = -n \frac{\partial}{\partial \lambda} \lambda + \sum x_i \frac{\partial}{\partial \lambda} \ln \lambda - \frac{\partial}{\partial \lambda} \sum \ln (x_i!)$$

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = -n + \frac{\sum x_i}{\lambda} \dots \dots \dots (1)$$

$$\frac{\partial}{\partial \lambda} \ln L(\lambda) = \frac{n}{\lambda} \left[\frac{\sum x_i}{n} - \lambda \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\sum x_i}{n}, \quad T(\theta) = \lambda, \quad A(\theta) = \frac{n}{\lambda}$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{1}{\frac{n}{\lambda}} = \frac{\lambda}{n}$$

(b) MVB for λ .

Diff (1) w.r.t λ

$$\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) = -\frac{\sum x_i}{\lambda^2}$$

$$E \left[\frac{\partial^2}{\partial \lambda^2} \ln L(\lambda) \right] = - \frac{\sum E(x_i)}{\lambda^2} = - \frac{n\lambda}{\lambda^2} = - \frac{n}{\lambda}$$

$$\phi(\theta) = \lambda, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E \left[\frac{\partial^2 \ln L(\lambda)}{\partial \lambda^2} \right]} \geq \frac{(1)^2}{-\left(-\frac{n}{\lambda}\right)} \geq \frac{\lambda}{n}$$

4.9 MVB Estimator for parameter of Gamma Distribution

Question 44: For two parameter Gamma distribution with p.d.f

$$f(x; \alpha, \beta) = \frac{1}{\Gamma \alpha \beta^\alpha} e^{-\frac{x}{\beta}} x^{\alpha-1}, x > 0$$

If α is known then show that $\bar{x} = \frac{\sum x_i}{n}$ is MVB estimator of β or Find the MVB estimator of the unknown parameter β , where, α is known.

Solution:

The likelihood function

$$\begin{aligned} L(\beta) &= \prod_{i=1}^n \frac{1}{\Gamma \alpha \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\beta) &= \prod_{i=1}^n \frac{1}{(\alpha-1)! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\beta) &= \prod_{i=1}^n \frac{\alpha}{\alpha(\alpha-1)! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\beta) &= \prod_{i=1}^n \frac{\alpha}{\alpha! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^{\alpha-1} \\ L(\beta) &= \prod_{i=1}^n x_i^{\alpha-1} \frac{\alpha^n}{(\alpha! \beta^\alpha)^n} e^{-\frac{\sum x_i}{\beta}} \end{aligned}$$

Taking ln both sides

$$\ln L(\beta) = (\alpha-1) \ln \prod_{i=1}^n x_i + n \ln \alpha - n \ln \alpha! - n \alpha \ln \beta - \frac{\sum x_i}{\beta}$$

Diff (1) w.r.t β

$$\frac{\partial}{\partial \beta} \ln L(\beta) = -\frac{n\alpha}{\beta} + \frac{\sum x_i}{\beta^2} \dots \dots \dots (1)$$

$$\frac{\partial}{\partial \beta} \ln L(\beta) = \frac{n\alpha}{\beta^2} \left[\frac{1}{\alpha} \frac{\sum x_i}{n} - \beta \right] = \frac{n\alpha}{\beta^2} \left[\frac{\bar{X}}{\alpha} - \beta \right]$$

Compare with $A(\theta)[t_n - T(\theta)]$

$$t_n = \frac{\bar{X}}{\alpha}, \quad T(\theta) = \beta, \quad A(\theta) = \frac{n\alpha}{\beta^2}$$

$$\text{Var}(t_n) = \frac{T'(\theta)}{A(\theta)} = \frac{1}{\frac{n\alpha}{\beta^2}} = \frac{\beta^2}{n\alpha}$$

Hence the MVB estimator of parameter β is $\frac{\bar{X}}{\alpha}$ with variance $\frac{\beta^2}{n\alpha}$

(b) C.R.L.B for β .

Diff (1) w.r.t β

$$\frac{\partial^2}{\partial \beta^2} \ln L(\beta) = \frac{n\alpha}{\beta^2} - \frac{2\sum x_i}{\beta^3}$$

$$E \left[\frac{\partial^2}{\partial \beta^2} \ln L(\beta) \right] = \frac{n\alpha}{\beta^2} - \frac{2\sum E(x_i)}{\beta^3} \dots \dots \dots (2)$$

$$E(x) = \int_0^\infty x \frac{1}{\Gamma\alpha \beta^\alpha} e^{-\frac{x}{\beta}} x^{\alpha-1} dx = \int_0^\infty x^\alpha \frac{1}{\Gamma\alpha \beta^\alpha} e^{-\frac{x}{\beta}} dx$$

Let

$$t = \frac{x}{\beta} \Rightarrow x = t\beta \Rightarrow dx = \beta dt$$

$$E(x) = \frac{1}{\Gamma\alpha \beta^\alpha} \int_0^\infty (t\beta)^\alpha e^{-t} \beta dt = \frac{\beta^{\alpha+1}}{\Gamma\alpha \beta^\alpha} \int_0^\infty t^{\alpha+1-1} e^{-t} dt$$

$$E(x) = \frac{\beta}{\Gamma\alpha} \Gamma(\alpha + 1) = \frac{\beta}{(\alpha - 1)!} \alpha! = \frac{\beta}{(\alpha - 1)!} \alpha(\alpha - 1)! = \alpha\beta$$

Put in (2)

$$E \left[\frac{\partial^2}{\partial \beta^2} \ln L(\beta) \right] = \frac{n\alpha}{\beta^2} - \frac{2n\alpha\beta}{\beta^3} = \frac{n\alpha}{\beta^2} - \frac{2n\alpha}{\beta^2} = -\frac{n\alpha}{\beta^2}$$

$$\phi(\theta) = \beta, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{-E\left[\frac{\partial^2 \ln L(\beta)}{\partial \beta^2}\right]} \geq \frac{(1)^2}{-\left(-\frac{n\alpha}{\beta^2}\right)} \geq \frac{\beta^2}{n\alpha}$$

4.10 CRLB for Given P.d.f

Question 45: Given the p.d.f

$$f(x; \theta) = [\pi\{1 + (x_i - \theta)^2\}]^{-1}$$

Show that the Crammer Rao Lower Bound of an unbiased estimator of θ is $\frac{2}{n}$. Where n is the size of random sample from this distribution.

Solution:-

Given p.d.f

$$f(x; \theta) = [\pi\{1 + (x_i - \theta)^2\}]^{-1}$$

Taking ln both sides

$$\ln f = \ln[\pi\{1 + (x_i - \theta)^2\}]^{-1} = -\ln \pi - \ln\{1 + (x_i - \theta)^2\}$$

Diff w.r.t θ

$$\frac{\partial \ln f}{\partial \theta} = -\frac{1}{1 + (x_i - \theta)^2} [0 + 2(x_i - \theta)(-1)] = \frac{2(x_i - \theta)}{1 + (x_i - \theta)^2}$$

$$E\left(\frac{\partial \ln f}{\partial \theta}\right)^2 = \int_{-\infty}^{\infty} \left[\frac{2(x_i - \theta)}{1 + (x_i - \theta)^2}\right]^2 \frac{1}{\pi\{1 + (x_i - \theta)^2\}} dx_i$$

$$E\left(\frac{\partial \ln f}{\partial \theta}\right)^2 = \frac{4}{\pi} \int_{-\infty}^{\infty} \frac{(x_i - \theta)^2}{[1 + (x_i - \theta)^2]^3} dx_i$$

Put

$$x_i - \theta = \tan \phi \quad \Rightarrow dx_i = \sec^2 \phi d\phi$$

$$\text{When } x = -\infty, \text{ then } \tan \phi = -\infty \Rightarrow \phi = \tan^{-1}(-\infty) = -\frac{\pi}{2}$$

$$\text{When } x = \infty, \text{ then } \tan \phi = \infty \Rightarrow \phi = \tan^{-1}(\infty) = \frac{\pi}{2}$$

$$E\left(\frac{\partial \ln f}{\partial \theta}\right)^2 = \frac{4}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tan^2 \phi}{[1 + \tan^2 \phi]^3} \sec^2 \phi d\phi$$

$$E \left(\frac{\partial \ln f}{\partial \theta} \right)^2 = \frac{4}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tan^2 \phi}{[\sec^2 \phi]^3} \sec^2 \phi d\phi = \frac{4}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tan^2 \phi}{\sec^6 \phi} \sec^2 \phi d\phi = \frac{4}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\tan^2 \phi}{\sec^4 \phi} d\phi$$

$$E \left(\frac{\partial \ln f}{\partial \theta} \right)^2 = \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \frac{\tan^2 \phi}{\sec^4 \phi} d\phi = \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \frac{\sin^2 \phi}{\cos^2 \phi} \cos^4 \phi d\phi = \frac{8}{\pi} \int_0^{\frac{\pi}{2}} \sin^2 \phi \cos^2 \phi d\phi$$

Using Wallis Formula

$$\int_0^{\frac{\pi}{2}} \sin^p x \cos^q x dx = \frac{(p-1)(p-3) \dots (q-1)(q-3) \dots \pi}{(p+q)(p+q-2) \dots}, \quad p \text{ is even}$$

$$E \left(\frac{\partial \ln f}{\partial \theta} \right)^2 = \frac{8}{\pi} \cdot \frac{(2-1)(2-1)}{(2+2)(2+2-2)} \pi = \frac{8}{\pi} \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{1}{2}$$

$$\phi(\theta) = \theta, \quad \phi'(\theta) = 1$$

Crammer Rao Lower Bound

$$\text{Var}(t_n) \geq \frac{[\phi'(\theta)]^2}{n E \left[\frac{\partial \ln f}{\partial \theta} \right]^2} \geq \frac{(1)^2}{n \left(\frac{1}{2} \right)} \geq \frac{2}{n}$$

4.11 MVB of Order Statistic From Uniform (Variance Less than MVB)

Question 46: Given

$$f(x; \theta) = \frac{1}{\theta}, \quad 0 < x < \theta, \quad \theta > 0$$

Compute the reciprocal of $n E \left[\frac{\partial \ln f}{\partial \theta} \right]^2$ and compare this with the variance of $\frac{(n+1)}{n} y_n$ where y_n is the largest item of a random sample of size n from this distribution. Show that the least attainable variance may be less than C.R.L.B

Solution:-

$$f(x; \theta) = \frac{1}{\theta}$$

Taking ln both sides

$$\ln f = \ln \left(\frac{1}{\theta} \right) = \ln 1 - \ln \theta$$

Diff w.r.t θ

$$\frac{\partial \ln f}{\partial \theta} = -\frac{1}{\theta}$$

$$nE \left(\frac{\partial \ln f}{\partial \theta} \right)^2 = \frac{n}{\theta^2}$$

The C.R.L.B is obtained through the reciprocal of $nE \left(\frac{\partial \ln f}{\partial \theta} \right)^2$ is $\frac{\theta^2}{n}$

The distribution function

$$F(x) = \int_0^x \frac{1}{\theta} dx = \frac{x}{\theta} \Big|_0^x = \frac{x}{\theta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_n of random sample of $r = n$

$$g(y_n) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1 - F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n \left[\frac{y_n}{\theta} \right]^{n-1} \frac{1}{\theta}$$

$$E(y_n) = \int_0^\theta y_n g(y_n) dy_n = \int_0^\theta y_n n \left[\frac{y_n}{\theta} \right]^{n-1} \frac{1}{\theta} dy_n = \frac{n}{\theta^n} \int_0^\theta (y_n)^n dy_n$$

$$E(y_n) = \frac{n}{\theta^n} \left[\left(\frac{y_n^{n+1}}{n+1} \right) \Big|_0^\theta \right] = \frac{n}{\theta^n} \left[\frac{1}{n+1} (\theta^{n+1} - 0) \right] = \frac{n\theta^{n+1}}{\theta^n(n+1)} = \frac{n\theta}{n+1}$$

$$E \left(\frac{n+1}{n} y_n \right) = \left(\frac{n}{n+1} \right) \left(\frac{n\theta}{n+1} \right) = \theta$$

Hence, $\frac{n+1}{n}y_n$ is unbiased estimator of θ .

$$E(y_n^2) = \int_0^\theta y_n^2 g(y_n) dy_n = \int_0^\theta y_n^2 n \left[\frac{y_n}{\theta} \right]^{n-1} \frac{1}{\theta} dy_n = \frac{n}{\theta^n} \int_0^\theta (y_n)^{n+1} dy_n$$

$$E(y_n^2) = \frac{n}{\theta^n} \left[\left(\frac{y_n^{n+2}}{n+2} \right) \Big|_0^\theta \right] = \frac{n}{\theta^n} \left[\frac{1}{n+2} (\theta^{n+2} - 0) \right] = \frac{n\theta^{n+2}}{\theta^n(n+2)} = \frac{n\theta^2}{n+2}$$

$$var(y_n) = E(y_n) - [E(y_n)]^2 = \frac{n\theta^2}{n+2} - \left(\frac{n\theta}{n+1} \right)^2 = n\theta^2 \left[\frac{1}{n+2} - \frac{n}{(n+1)^2} \right]$$

$$var(y_n) = n\theta^2 \left[\frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)^2} \right] = n\theta^2 \left[\frac{n^2 + 1 + 2n - n^2 - 2n}{(n+2)(n+1)^2} \right]$$

$$var(y_n) = \frac{n\theta^2}{(n+2)(n+1)^2}$$

Now

$$var\left(\frac{n+1}{n}y_n\right) = \left(\frac{n+1}{n}\right)^2 var(y_n) = \left(\frac{n+1}{n}\right)^2 \frac{n\theta^2}{(n+2)(n+1)^2} = \frac{\theta^2}{n(n+2)}$$

Now, comparing $var\left(\frac{n+1}{n}y_n\right)$ with the Crammer Rao Lower bound

$$\frac{\theta^2}{n(n+2)} < \frac{\theta^2}{n}$$

$$\frac{1}{n(n+2)} < \frac{1}{n}$$

$$\frac{1}{n+2} < 1$$

Which is true so, it is clear that the variance of $\frac{n+1}{n}y_n$ is less than the CRLB

Since, the variance of the $\frac{n+1}{n}y_n$ falls below the CRLB, so, to investigate the possible reason for this we will check the validity of Regularity conditions.

$$\frac{\partial}{\partial \theta} \int f(x) dx = \frac{\partial}{\partial \theta} \left[\int_0^{\theta} \frac{1}{\theta} dx \right] = \frac{\partial}{\partial \theta} \left[\frac{1}{\theta} (\theta - 0) \right] = \frac{\partial}{\partial \theta} (1) = 0$$

And

$$\int \frac{\partial f(x)}{\partial \theta} dx = \int_0^{\theta} \frac{\partial}{\partial \theta} \left(\frac{1}{\theta} \right) dx = \int_0^{\theta} \left(\frac{-1}{\theta^2} \right) dx = \frac{-1}{\theta^2} (\theta - 0) = \frac{-1}{\theta}$$

Here,

$$\frac{\partial}{\partial \theta} \int f(x) dx \neq \int \frac{\partial f(x)}{\partial \theta} dx$$

So, the variance of estimator will be less than the CRLB when the Regularity condition does not fulfill.

4.12 Uniqueness Theorem

Question 47: State and prove Uniqueness theorem. Or show that if MV estimator exists it is essentially unique irrespective of any bound is attained.

Statement: A Minimum Variance Unbiased Estimator (MVU) estimator is unique in the sense that T_1 and T_2 are MVU estimators for $T(\theta)$, then $T_1 = T_2$, almost surely.

Proof:

As T_1 and T_2 both are unbiased estimators of $T(\theta)$. So,

$$E(T_1) = E(T_2) = T(\theta)$$

Also, T_1 and T_2 both are minimum variance estimators so

$$V(T_1) = V(T_2) = V, \quad \forall \theta \in \Theta \dots \dots \dots (1)$$

Consider another unbiased estimator of $T(\theta)$

$$T = \frac{1}{2} [T_1 + T_2]$$

i.e.

$$E(T) = \frac{1}{2} [E(T_1) + E(T_2)] = \frac{1}{2} [T(\theta) + T(\theta)] = \frac{2T(\theta)}{2} = T(\theta)$$

Now variance of T.

$$V(T) = \frac{1}{4}V[T_1 + T_2] = \frac{1}{4}[V(T_1) + V(T_2) + 2COV(T_1, T_2)]$$

Using (1)

$$V(T) = \frac{1}{4}[V + V + 2COV(T_1, T_2)] = \frac{2}{4}[V + COV(T_1, T_2)] = \frac{1}{2}[V + COV(T_1, T_2)]$$

As

$$\rho = \frac{COV(T_1, T_2)}{\sqrt{V(T_1)V(T_2)}} = \frac{COV(T_1, T_2)}{\sqrt{V \cdot V}} = \frac{COV(T_1, T_2)}{V}$$

So,

$$\rho V = COV(T_1, T_2)$$

Hence,

$$V(T) = \frac{1}{2}[V + \rho V] = \frac{V}{2}[1 + \rho]$$

Since, T_1 and T_2 are MVU estimators so

$$V(T) \geq V(T_1) = V(T_2)$$

$$\frac{V}{2}[1 + \rho] \geq V$$

$$1 + \rho \geq 2 \Rightarrow \rho \geq 1$$

But as we know that $|\rho| \leq 1$, or maximumly $\rho = 1$, when we consider $\rho = 1$, then T_1 and T_2 must have a linear relationship of the form.

$$T_1 = \alpha + \beta T_2 \dots \dots \dots (2)$$

Taking expectation

$$E(T_1) = \alpha + \beta E(T_2)$$

$$T(\theta) = \alpha + \beta T(\theta) \dots \dots \dots (3)$$

Also

$$V(T_1) = V(\alpha + \beta T_2) = \beta^2 V(T_2)$$

From (1)

$$V = \beta^2 V \Rightarrow \beta^2 = 1 \Rightarrow \beta = \pm 1$$

Since $\rho = 1$, so $\beta = 1$, Put $\beta = 1$ in (3)

$$T(\theta) = \alpha + T(\theta) \Rightarrow \alpha = 0$$

Put $\alpha = 0, \beta = 1$ in (2)

$$T_1 = T_2, \text{ As desired}$$

4.13 Cramer Rao Inequality

Question 48 : State and prove Cramer Rao Inequality

Statement: Let $X_1, X_2, \dots, X_n$ be independent random samples each with p.m.f or p.d.f $f(x; \theta)$.

Let $t_n(x_1, x_2, \dots, x_n)$ be an estimator. If $L = \prod_{i=1}^n f(x; \theta)$ and if regularity conditions are satisfied.

(1) For t_n an unbiased estimator of function of θ

$$V(t_n) \geq \frac{[T'(\theta)]^2}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2} \quad \text{or} \quad V(t_n) \geq \frac{[T'(\theta)]^2}{-E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)}$$

(2) For t_n being an unbiased estimator of θ .

$$V(t_n) \geq \frac{1}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2} \quad \text{or} \quad V(t_n) \geq \frac{1}{-E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)}$$

(3) For t_n an estimator of function of θ , say $T(\theta)$ and let $b_n(\theta)$ be the bias of t_n i.e.

$$b_n(\theta) = E(t_n) - T(\theta)$$

Then

$$E[t_n - T(\theta)]^2 \geq \frac{[T'(\theta) + b_n'(\theta)]^2}{E\left(\frac{\partial \ln L}{\partial \theta}\right)^2} \quad \text{or} \quad E[t_n - T(\theta)]^2 \geq \frac{[T'(\theta) + b_n'(\theta)]^2}{-E\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)}$$

Proof:

(1) Consider an unbiased estimator t_n of some function of θ say $T(\theta)$.

$$E(t_n) = \int \int \dots \int t_n f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta) dx_1 dx_2 \dots dx_n = T(\theta)$$

$$\int \int \dots \int t_n L dx_1 dx_2 \dots dx_n = T(\theta) \dots \dots \dots (A)$$

Diff (A) w.r.t to θ

$$\int \int \dots \int t_n \frac{\partial L}{\partial \theta} dx_1 dx_2 \dots dx_n = T'(\theta)$$

$$\int \int \dots \int t_n \left(\frac{1}{L} \frac{\partial L}{\partial \theta}\right) L dx_1 dx_2 \dots dx_n = T'(\theta)$$

$$\int \int \dots \int t_n \left(\frac{\partial \ln L}{\partial \theta}\right) L dx_1 dx_2 \dots dx_n = T'(\theta)$$

It can be written as

$$E \left[t_n \left(\frac{\partial \ln L}{\partial \theta} \right) \right] = T'(\theta) \dots \dots \dots (B)$$

$$E(t_n)E \left(\frac{\partial \ln L}{\partial \theta} \right) = T'(\theta) \dots \dots \dots (1)$$

Since,

$$\int \int \dots \dots \dots \int f(x_1, \theta) f(x_2, \theta) \dots \dots f(x_n, \theta) dx_1 dx_2 \dots dx_n = 1$$

$$\int \int \dots \dots \dots \int L dx_1 dx_2 \dots dx_n = 1$$

Diff w.r.t θ

$$\int \int \dots \dots \dots \int \frac{\partial L}{\partial \theta} dx_1 dx_2 \dots dx_n = 0$$

$$\int \int \dots \dots \dots \int \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) L dx_1 dx_2 \dots dx_n = 0$$

$$\int \int \dots \dots \dots \int \left(\frac{\partial \ln L}{\partial \theta} \right) L dx_1 dx_2 \dots dx_n = 0$$

$$E \left(\frac{\partial \ln L}{\partial \theta} \right) = 0$$

Put in (1)

$$E(t_n)(0) = T'(\theta)$$

$$T'(\theta) = 0$$

Now, equation (B) can be written as

$$E \left[\{t_n - T(\theta)\} \left(\frac{\partial \ln L}{\partial \theta} \right) \right] = T'(\theta)$$

$$E[t_n - T(\theta)]E \left[\left(\frac{\partial \ln L}{\partial \theta} \right) \right] = T'(\theta)$$

Squaring and using Cauchy Shewarz Inequality

$$[T'(\theta)]^2 \leq E[t_n - T(\theta)]^2 E \left[\left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right]$$

$$[T'(\theta)]^2 \leq V(t_n) E \left[\left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right]$$

$$\Rightarrow V(t_n) \geq \frac{[T'(\theta)]^2}{E \left[\left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right]} \dots \dots \dots (C) \text{ or } V(t_n) \geq \frac{[T'(\theta)]^2}{-E \left[\left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right]}$$

$$\therefore E \left[\left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right] = -E \left[\left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right]$$

(2) Since t_n is unbiased estimator of θ not for $T(\theta)$

$$T(\theta) = 1$$

$$T'(\theta) = 0$$

Putting in (C)

$$V(t_n) = \frac{1}{E \left[\left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right]} \quad \text{or} \quad V(t_n) \geq \frac{1}{-E \left[\left(\frac{\partial^2 \ln L}{\partial \theta^2} \right) \right]}$$

(3) Since

$$b_n(\theta) = E(t_n) - T(\theta)$$

$$E(t_n) = b_n(\theta) + T(\theta)$$

As

$$E(t_n) = \int \int \dots \int t_n L dx_1 dx_2 \dots dx_n$$

So,

$$\int \int \dots \int t_n L dx_1 dx_2 \dots dx_n = b_n(\theta) + T(\theta)$$

Diff w.r.t θ

$$\int \int \dots \int t_n \frac{\partial L}{\partial \theta} dx_1 dx_2 \dots dx_n = b'_n(\theta) + T'(\theta)$$

$$\int \int \dots \int t_n \left(\frac{1}{L} \frac{\partial L}{\partial \theta} \right) L dx_1 dx_2 \dots dx_n = b'_n(\theta) + T'(\theta)$$

$$\int \int \dots \int t_n \left(\frac{\partial \ln L}{\partial \theta} \right) L dx_1 dx_2 \dots dx_n = b'_n(\theta) + T'(\theta)$$

$$E \left[t_n \left(\frac{\partial \ln L}{\partial \theta} \right) \right] = b'_n(\theta) + T'(\theta)$$

Similarly,

$$E \left[\{t_n - T(\theta)\} \left(\frac{\partial \ln L}{\partial \theta} \right) \right] = b'_n(\theta) + T'(\theta)$$

$$E \{t_n - T(\theta)\} E \left[\left(\frac{\partial \ln L}{\partial \theta} \right) \right] = b'_n(\theta) + T'(\theta)$$

Squaring and using Cauchy Shewarz Inequality

$$[b'_n(\theta) + T'(\theta)]^2 \leq E[t_n - T(\theta)]^2 E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right]$$

$$[b'_n(\theta) + T'(\theta)]^2 \leq E[t_n - T(\theta)]^2 E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right]$$

$$\Rightarrow E[t_n - T(\theta)]^2 \geq \frac{[b'_n(\theta) + T'(\theta)]^2}{E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right]} \text{ or } E[t_n - T(\theta)]^2 \geq \frac{[b'_n(\theta) + T'(\theta)]^2}{-E\left[\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)\right]}$$

$$\therefore E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right] = -E\left[\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)\right]$$

Note: $E\left[\left(\frac{\partial \ln L}{\partial \theta}\right)^2\right] = -E\left[\left(\frac{\partial^2 \ln L}{\partial \theta^2}\right)\right]$ is called amount of information in the sample.



5.0 Questions Related to Efficiency

5.1 Relative Efficiency of sample Mean and Median of Normal Distribution

Question 49: For normal distribution compare the efficiency of mean and median.

Solution:-

As we known that both sample mean and median are unbiased estimators of population mean.

Also we have

$$Var(\bar{X}) = \frac{\sigma^2}{n} \quad \text{and} \quad Var(\tilde{X}) = \frac{\pi\sigma^2}{2n}$$

The relative efficiency of sample mean against sample median is

$$R_{ef} = \frac{Var(\tilde{X})}{Var(\bar{X})} = \frac{\frac{\pi\sigma^2}{2n}}{\frac{\sigma^2}{n}} = \frac{\pi}{2} = 1.57 > 1$$

Since $R_{ef} > 1$, so sample mean is more efficient than sample median.

Note: $R_{ef} = 1.57$, indicates that sample mean is 57% more efficient than sample median, the efficiency which sample mean can produce from sample of 100 observation, the sample median will need to have sample of 157 observations to produce the same.

The relative efficiency of sample median against sample mean is

$$R_{ef} = \frac{Var(\bar{X})}{Var(\tilde{X})} = \frac{\frac{\sigma^2}{n}}{\frac{\pi\sigma^2}{2n}} = \frac{2}{\pi} = 0.63 < 1$$

Since $R_{ef} = 0.63$, The efficiency of sample median is only 63% against sample mean.

Note: this means that sample mean obtained from a random sample of 63 observation is equally as efficient as sample median obtained from sample of 100 observations.

5.2 Relative Efficiency in Uniform Distribution (0, β)

Question 50: (a) If $X_1, X_2, \dots, X_n$ constitutes a random sample from the Uniform population with $\alpha = 0$, over the interval (α, β) show that the largest sample value (i.e. n th order statistic y_n) is a biased estimator of the parameter β . Also modify the parameter of β to make it unbiased.

(b) Also from the some information $X_1, X_2, \dots, X_n$ constitute a random sample from uniform population with $\alpha = 0$, over the interval (α, β) show that $\frac{n+1}{n}y_n$ is an unbiased estimator of β .

(i) Show that $2\bar{X}$ is also unbiased estimator of β .

(ii) Compare the Efficiency of these two estimators of β .

Solution:

Let $X_1, X_2, \dots, X_n$ constitute a random sample from uniform population with $\alpha = 0$, over the interval (α, β)

$$f(x; \theta) = \frac{1}{\beta - \alpha} = \frac{1}{\beta}, \quad 0 \leq X \leq \beta$$

The distribution function

$$F(x) = \int_0^x \frac{1}{\beta} dx = \frac{x}{\beta} \Big|_0^x = \frac{x}{\beta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_n of random sample of $r = n$

$$g(y_n) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1 - F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n \left[\frac{y_n}{\beta} \right]^{n-1} \frac{1}{\beta}$$

$$E(y_n) = \int_0^{\beta} y_n g(y_n) dy_n = \int_0^{\beta} y_n n \left[\frac{y_n}{\beta} \right]^{n-1} \frac{1}{\beta} dy_n = \frac{n}{\beta^n} \int_0^{\beta} (y_n)^n dy_n$$

$$E(y_n) = \frac{n}{\beta^n} \left[\left(\frac{y_n^{n+1}}{n+1} \right) \Big|_0^\beta \right] = \frac{n}{\beta^n} \left[\frac{1}{n+1} (\beta^{n+1} - 0) \right] = \frac{n\beta^{n+1}}{\beta^n(n+1)} = \frac{n\beta}{n+1}$$

Hence,

$$E(y_n) = \frac{n\beta}{n+1} \neq \beta$$

Hence y_n is biased estimator of β .

In order to modify the parameter to make y_n its unbiased estimator the parameter should be $\frac{n\beta}{n+1}$

So that

$$E(y_n) = \frac{n\beta}{n+1}$$

(b)

$$E\left(\frac{n+1}{n} y_n\right) = \left(\frac{n+1}{n}\right) \left(\frac{n\beta}{n+1}\right) = \beta$$

Hence, $\frac{n+1}{n} y_n$ is unbiased estimator of β .

For variance of $\frac{n+1}{n} y_n$

$$E(y_n^2) = \int_0^\beta y_n^2 g(y_n) dy_n = \int_0^\beta y_n^2 n \left[\frac{y_n}{\beta} \right]^{n-1} \frac{1}{\beta} dy_n = \frac{n}{\beta^n} \int_0^\beta (y_n)^{n+1} dy_n$$

$$E(y_n^2) = \frac{n}{\beta^n} \left[\left(\frac{y_n^{n+2}}{n+2} \right) \Big|_0^\beta \right] = \frac{n}{\beta^n} \left[\frac{1}{n+2} (\beta^{n+2} - 0) \right] = \frac{n\beta^{n+2}}{\beta^n(n+2)} = \frac{n\beta^2}{n+2}$$

$$var(y_n) = E(y_n) - [E(y_n)]^2 = \frac{n\beta^2}{n+2} - \left(\frac{n\beta}{n+1} \right)^2 = n\beta^2 \left[\frac{1}{n+2} - \frac{n}{(n+1)^2} \right]$$

$$var(y_n) = n\beta^2 \left[\frac{(n+1)^2 - n(n+2)}{(n+2)(n+1)^2} \right] = n\beta^2 \left[\frac{n^2 + 1 + 2n - n^2 - 2n}{(n+2)(n+1)^2} \right]$$

$$var(y_n) = \frac{n\beta^2}{(n+2)(n+1)^2}$$

Now

$$var\left(\frac{n+1}{n} y_n\right) = \left(\frac{n+1}{n}\right)^2 var(y_n) = \left(\frac{n+1}{n}\right)^2 \frac{n\beta^2}{(n+2)(n+1)^2} = \frac{\beta^2}{n(n+2)}$$

(i) To show $2\bar{X}$ is unbiased estimator of β .

$$E(x) = \int_0^{\beta} x \frac{1}{\beta} dx = \frac{1}{2\beta} x^2 \Big|_0^{\beta} = \frac{1}{2\beta} (\beta^2 - 0) = \frac{\beta}{2}$$

$$E(2\bar{X}) = 2E\left(\frac{\sum X_i}{n}\right) = 2 \sum E(x) = \frac{2n}{n} \left(\frac{\beta}{2}\right) = \beta$$

Hence $2\bar{X}$ is unbiased estimator of β .

$$E(x^2) = \int_0^{\beta} x^2 \frac{1}{\beta} dx = \frac{1}{3\beta} x^3 \Big|_0^{\beta} = \frac{1}{3\beta} (\beta^3 - 0) = \frac{\beta^2}{3}$$

$$Var(2\bar{X}) = 4Var(\bar{X}) = 4 \frac{var(x)}{n} \dots \dots (A)$$

$$Var(x) = E(x^2) - [E(x)]^2 = \frac{\beta^2}{3} - \left(\frac{\beta}{2}\right)^2 = \frac{\beta^2}{3} - \frac{\beta^2}{4} = \frac{4\beta^2 - 3\beta^2}{12} = \frac{\beta^2}{12}$$

Put in (A)

$$Var(2\bar{X}) = 4 \frac{\frac{\beta^2}{12}}{n} = \frac{\beta^2}{3n}$$

(ii) Relative Efficiency of $\frac{n+1}{n} y_n$ and $2\bar{X}$

$$R_{ef} = \frac{Var\left(\frac{n+1}{n} y_n\right)}{Var(2\bar{X})} = \frac{\frac{\beta^2}{n(n+2)}}{\frac{\beta^2}{3n}} = \frac{3}{n+2}$$

1- If $n=1$ (unit sample size), $R_{ef} = 1$, both estimators are equally efficient.

2- If $n > 1$, $R_{ef} < 1$, $\frac{n+1}{n} y_n$ is more efficient than $2\bar{X}$.

3-2- If $n < 1$, (which is not possible practically) $R_{ef} > 1$, $2\bar{X}$ is more efficient than $\frac{n+1}{n} y_n$.

5.3 Expression Related to Efficiency in Terms of Correlation Co-efficient

Question 51: if t^* is most efficient estimator of θ and t is any other estimator with efficiency E . Show that

$$V(t^* - t) = \left(\frac{1}{E} - 1\right) V(t^*), \text{ such that } \rho = \sqrt{E}$$

Solution:

Since t^* is most efficient estimator, given that

$$E(t^*) = \theta \quad \text{and} \quad \text{Var}(t^*) = V \text{ (Minimum Variance)}$$

$$V(t^* - t) = V(t^*) + V(t) - 2\text{Cov}(t^*, t) \dots \dots \dots (1)$$

The efficiency of t is

$$\frac{V(t^*)}{V(t)} = E \Rightarrow V(t) = \frac{V(t^*)}{E} = \frac{V}{E}$$

As,

$$\text{Cov}(t^*, t) = \rho \sqrt{V(t^*)V(t)} = \rho \sqrt{V \cdot \frac{V}{E}} = \frac{\rho V}{\sqrt{E}}$$

Put in (1)

$$V(t^* - t) = V + \frac{V}{E} - 2 \frac{\rho V}{\sqrt{E}}$$

Put $\rho = \sqrt{E}$

$$V(t^* - t) = V + \frac{V}{E} - 2 \frac{V\sqrt{E}}{\sqrt{E}} = V + \frac{V}{E} - 2V = \frac{V}{E} - V = V \left(\frac{1}{E} - 1 \right)$$

Hence,

$$V(t^* - t) = V(t^*) \left(\frac{1}{E} - 1 \right)$$

5.4 Finding Coefficient of Correlation in terms of Efficiency.

Question 52: If T is a MVUE of $T(\theta)$ with variance V and T_1 and T_2 are two unbiased estimators of $T(\theta)$. With finite variance. If either T_1 or T_2 is MV estimator then show that correlation coefficient between T_1 or T_2 , ρ is $E^{\frac{1}{2}}$, where E is reciprocal of relative variance of estimator.

Or

If E_1 and E_2 are the efficiency of two unbiased estimators and ρ is the correlation coefficient between them then show that

$$\rho = \sqrt{E_1 E_2} \pm \sqrt{(1 - E_1)(1 - E_2)}$$

Hence, if E_1 or $E_2 = 1$ i.e. T_1 or T_2 is MVUE of $T(\theta)$ then $\rho = E^{\frac{1}{2}}$

Solution:-

Let for unbiased estimators T_1 and T_2 , T is most efficient estimator.

$$Var(T) = V$$

Efficiency of T_1

$$E_1 = \frac{Var(T)}{Var(T_1)} = \frac{V}{Var(T_1)} \Rightarrow Var(T_1) = \frac{V}{E_1} \dots \dots (A)$$

Efficiency of T_2

$$E_2 = \frac{Var(T)}{Var(T_2)} = \frac{V}{Var(T_2)} \Rightarrow Var(T_2) = \frac{V}{E_2} \dots \dots (B)$$

Let us define another estimator

$$T' = \lambda T_1 + \mu T_2, \text{ where } \lambda + \mu = 1$$

Variance of T'

$$Var(T') = Var(\lambda T_1 + \mu T_2) = \lambda^2 Var(T_1) + \mu^2 Var(T_2) + 2\lambda\mu Cov(T_1, T_2) \dots \dots (1)$$

We know that

$$\rho = \frac{Cov(T_1, T_2)}{\sqrt{Var(T_1)Var(T_2)}} \Rightarrow Cov(T_1, T_2) = \rho \sqrt{Var(T_1)Var(T_2)}$$

From (A) and (B)

$$Cov(T_1, T_2) = \rho \sqrt{\frac{V}{E_1} \cdot \frac{V}{E_2}} = \frac{\rho V}{\sqrt{E_1 E_2}} \dots \dots (C)$$

Put (A) and (B) and (C) in (1)

$$Var(T') = \frac{\lambda^2 V}{E_1} + \frac{\mu^2 V}{E_2} + 2\lambda\mu \frac{\rho V}{\sqrt{E_1 E_2}}$$

By Definition of MVB

$$\begin{aligned} Var(T') &\geq Var(T) \\ \frac{\lambda^2 V}{E_1} + \frac{\mu^2 V}{E_2} + 2\lambda\mu \frac{\rho V}{\sqrt{E_1 E_2}} &\geq V \\ \frac{\lambda^2}{E_1} + \frac{\mu^2}{E_2} + \frac{2\lambda\mu\rho}{\sqrt{E_1 E_2}} &\geq 1 \end{aligned}$$

Using $\lambda + \mu = 1$

$$\begin{aligned} \frac{\lambda^2}{E_1} + \frac{\mu^2}{E_2} + \frac{2\lambda\mu\rho}{\sqrt{E_1 E_2}} &\geq (\lambda + \mu)^2 \\ \frac{\lambda^2}{E_1} + \frac{\mu^2}{E_2} + \frac{2\lambda\mu\rho}{\sqrt{E_1 E_2}} - \lambda^2 - \mu^2 - 2\lambda\mu &\geq 0 \\ \frac{\lambda^2}{E_1} - \lambda^2 + \frac{\mu^2}{E_2} - \mu^2 + \frac{2\lambda\mu\rho}{\sqrt{E_1 E_2}} - 2\lambda\mu &\geq 0 \\ \left(\frac{1}{E_1} - 1\right)\lambda^2 + \left(\frac{1}{E_2} - 1\right)\mu^2 + 2\lambda\mu\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right) &\geq 0 \dots\dots\dots (2) \end{aligned}$$

Divide by μ^2

$$\left(\frac{1}{E_1} - 1\right)\frac{\lambda^2}{\mu^2} + \left(\frac{1}{E_2} - 1\right) + \frac{2\lambda}{\mu}\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right) \geq 0 \dots\dots\dots (3)$$

Consider (2)

$$\phi = \left(\frac{1}{E_1} - 1\right)\lambda^2 + \left(\frac{1}{E_2} - 1\right)\mu^2 + 2\lambda\mu\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)$$

Diff w.r.t λ for minimization

$$\frac{\partial \phi}{\partial \lambda} = \left(\frac{1}{E_1} - 1\right)(2\lambda) + 2\mu\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)$$

Put $\frac{\partial \phi}{\partial \lambda} = 0$

$$\left(\frac{1}{E_1} - 1\right)(2\lambda) + 2\mu\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right) = 0$$

Divide by 2μ

$$\left(\frac{1}{E_1} - 1\right) \frac{\lambda}{\mu} + \left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right) = 0$$

$$\frac{\lambda}{\mu} = \frac{-\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)}{\left(\frac{1}{E_1} - 1\right)}$$

Also

$$\frac{\partial^2 \phi}{\partial \lambda^2} = 2 \left(\frac{1}{E_1} - 1\right) > 0, \text{ since } E_1 < 1, \text{ and } \frac{1}{E_1} > 1$$

Putting $\frac{\lambda}{\mu}$ in (3)

$$\begin{aligned} \left(\frac{1}{E_1} - 1\right) \left[\frac{-\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)^2}{\left(\frac{1}{E_1} - 1\right)} + \left(\frac{1}{E_2} - 1\right) + 2 \frac{-\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)}{\left(\frac{1}{E_1} - 1\right)} \left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right) \right] &\geq 0 \\ \frac{\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)^2}{\left(\frac{1}{E_1} - 1\right)} + \left(\frac{1}{E_2} - 1\right) - 2 \frac{\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)^2}{\left(\frac{1}{E_1} - 1\right)} &\geq 0 \\ -\frac{\left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)^2}{\left(\frac{1}{E_1} - 1\right)} + \left(\frac{1}{E_2} - 1\right) &\geq 0 \end{aligned}$$

Multiply by $-\left(\frac{1}{E_1} - 1\right)$

$$\begin{aligned} \left(\frac{\rho}{\sqrt{E_1 E_2}} - 1\right)^2 - \left(\frac{1}{E_1} - 1\right) \left(\frac{1}{E_2} - 1\right) &\leq 0 \\ \frac{\rho^2}{E_1 E_2} + 1 - \frac{2\rho}{\sqrt{E_1 E_2}} - \frac{1}{E_1 E_2} + \frac{1}{E_1} + \frac{1}{E_2} - 1 &\leq 0 \\ \frac{\rho^2}{E_1 E_2} - \frac{2\rho}{\sqrt{E_1 E_2}} - \frac{1}{E_1 E_2} + \frac{1}{E_1} + \frac{1}{E_2} &\leq 0 \end{aligned}$$

Multiply by $E_1 E_2$

$$\rho^2 - 2\rho\sqrt{E_1E_2} - 1 + E_1 + E_2 \geq 0$$

$$\rho^2 - (2\sqrt{E_1E_2})\rho + (E_1 + E_2 - 1) \geq 0$$

Which is quadratic in ρ

$$\rho = \frac{-[-(2\sqrt{E_1E_2})] \pm \sqrt{[-(2\sqrt{E_1E_2})]^2 - 4(1)(E_1 + E_2 - 1)}}{2(1)}$$

$$\rho = \frac{2\sqrt{E_1E_2} \pm \sqrt{4E_1E_2 - 4(E_1 + E_2 - 1)}}{2} = \frac{2[\sqrt{E_1E_2} \pm \sqrt{E_1E_2 - E_1 - E_2 + 1}]}{2}$$

$$\rho = \sqrt{E_1E_2} \pm \sqrt{E_1E_2 - E_1 - E_2 + 1} = \sqrt{E_1E_2} \pm \sqrt{E_1(E_2 - 1) - 1(E_2 - 1)}$$

$$\rho = \sqrt{E_1E_2} \pm \sqrt{(E_1 - 1)(E_2 - 1)} \dots \dots \dots (4)$$

If either of T_1 or T_2 is MVB estimator then either $E_1 = 1$ or $E_2 = 1$

So, Equation (4)

$$\rho = \sqrt{E_1} \quad \text{or} \quad \rho = \sqrt{E_2}$$

So, we can state

$$\rho = E^{\frac{1}{2}}$$

5.5 Best Linear Combination of Unbiased Estimators for Minimum Variance

Question 53: (a) If T_1 and T_2 be two unbiased estimators of $T(\theta)$ with variances σ_1^2 and σ_2^2 and correlation co-efficient ρ . What is the best unbiased linear combination of T_1 and T_2 and what is the variance of the combination

(b) Suppose T_1 in above case is an unbiased minimum variance estimate (σ^2) and T_2 is any other unbiased estimate with variance $\frac{\sigma^2}{E}$, then prove that the correlation co-efficient between T_1 and T_2 is $\sqrt{E}$.

Solution:-

Given that

$$E(T_1) = E(T_2) = T(\theta)$$

And

$$Var(T_1) = \sigma_1^2, Var(T_2) = \sigma_2^2$$

Let T be another estimator which is linear combination of T_1 and T_2

$$T = L_1 T_1 + L_2 T_2$$

Where, L_1 and L_2 are any arbitrary constants

$$E(T) = L_1 E(T_1) + L_2 E(T_2) = (L_1 + L_2) T(\theta)$$

T will be unbiased estimator for $T(\theta)$ when

$$L_1 + L_2 = 1 \dots \dots \dots (A)$$

Now for Variance of T

$$Var(T) = Var(L_1 T_1 + L_2 T_2) = L_1^2 Var(T_1) + L_2^2 Var(T_2) + 2L_1 L_2 Cov(T_1, T_2)$$

$$Var(T) = L_1^2 \sigma_1^2 + L_2^2 \sigma_2^2 + 2L_1 L_2 Cov(T_1, T_2)$$

$$\therefore Cov(T_1, T_2) = \rho \sqrt{Var(T_1) Var(T_2)} = \rho \sqrt{\sigma_1^2 \sigma_1^2} = \rho \sigma_1 \sigma_2$$

$$Var(T) = L_1^2 \sigma_1^2 + L_2^2 \sigma_2^2 + 2L_1 L_2 \rho \sigma_1 \sigma_2 \dots \dots \dots (B)$$

Now we have to minimize $Var(T)$,

Diff (B) w.r.t L_1

$$\frac{\partial Var(T)}{\partial L_1} = 2\sigma_1^2 L_1 + 2\rho \sigma_1 \sigma_2 L_2$$

Put $\frac{\partial Var(T)}{\partial L_1} = 0$

$$\begin{aligned} 2\sigma_1^2 L_1 + 2\rho\sigma_1\sigma_2 L_2 &= 0 \\ \sigma_1^2 L_1 + \rho\sigma_1\sigma_2 L_2 &= 0 \dots \dots (1) \end{aligned}$$

Diff (B) w.r.t L_2

$$\frac{\partial Var(T)}{\partial L_2} = 2\sigma_2^2 L_2 + 2\rho\sigma_1\sigma_2 L_1$$

Put $\frac{\partial Var(T)}{\partial L_2} = 0$

$$\begin{aligned} 2\sigma_2^2 L_2 + 2\rho\sigma_1\sigma_2 L_1 &= 0 \\ \sigma_2^2 L_2 + \rho\sigma_1\sigma_2 L_1 &= 0 \dots \dots (2) \end{aligned}$$

Subtracting eq(1) and eq(2)

$$\begin{aligned} \sigma_1^2 L_1 + \rho\sigma_1\sigma_2 L_2 - (\sigma_2^2 L_2 + \rho\sigma_1\sigma_2 L_1) &= 0 \\ (\sigma_1^2 - \rho\sigma_1\sigma_2) L_1 - (\sigma_2^2 - \rho\sigma_1\sigma_2) L_2 &= 0 \\ (\sigma_1^2 - \rho\sigma_1\sigma_2) L_1 &= (\sigma_2^2 - \rho\sigma_1\sigma_2) L_2 \\ \frac{L_1}{L_2} &= \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 - \rho\sigma_1\sigma_2} \end{aligned}$$

Using Componendo property

$$\text{if } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a+b}{b} = \frac{c+d}{d}$$

$$\frac{L_1 + L_2}{L_2} = \frac{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}{\sigma_1^2 - \rho\sigma_1\sigma_2}$$

As $L_1 + L_2 = 1$

$$\begin{aligned} \frac{1}{L_2} &= \frac{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}{\sigma_1^2 - \rho\sigma_1\sigma_2} \\ L_2 &= \frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \end{aligned}$$

Using Componendo property

$$\text{if } \frac{a}{b} = \frac{c}{d}, \text{ then } \frac{a}{a+b} = \frac{c}{c+d}$$

$$\frac{L_1}{L_1 + L_2} = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

As $L_1 + L_2 = 1$

$$L_1 = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

Hence,

$$L_1 = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \text{ and } L_2 = \frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

These are the values of L_1 and L_2 for which the variance of T in (1) is minimum

$$\begin{aligned} \text{Var}(T) &= \left(\frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \right)^2 \sigma_1^2 + \left(\frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \right)^2 \sigma_2^2 \\ &\quad + 2 \left(\frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \right) \left(\frac{\sigma_1^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \right) \rho\sigma_1\sigma_2 \end{aligned}$$

(ii)

Given that

$$\text{Var}(T_1) = \sigma^2.$$

The efficiency of T_2

$$E = \frac{\text{Var}(T_1)}{\text{Var}(T_2)} = \frac{\sigma^2}{\sigma_2^2} \Rightarrow \sigma_2^2 = \frac{\sigma^2}{E}$$

Putting σ_2^2 in L_1 and L_2

$$L_1 = \frac{\frac{\sigma^2}{E} - \rho \frac{\sigma^2}{\sqrt{E}}}{\sigma^2 + \frac{\sigma^2}{E} - 2\rho \frac{\sigma^2}{\sqrt{E}}} = \frac{(1 - \rho\sqrt{E}) \frac{\sigma^2}{E}}{(E + 1 - 2\rho\sqrt{E}) \frac{\sigma^2}{E}} = \frac{1 - \rho\sqrt{E}}{1 + E - 2\rho\sqrt{E}} = \frac{1 - \rho\sqrt{E}}{D}$$

Where, $1 + E - 2\rho\sqrt{E} = D$

$$L_2 = \frac{\sigma^2 - \rho \frac{\sigma^2}{\sqrt{E}}}{\sigma^2 + \frac{\sigma^2}{E} - 2\rho \frac{\sigma^2}{\sqrt{E}}} = \frac{(E - \rho\sqrt{E}) \frac{\sigma^2}{E}}{(E + 1 - 2\rho\sqrt{E}) \frac{\sigma^2}{E}} = \frac{E - \rho\sqrt{E}}{1 + E - 2\rho\sqrt{E}} = \frac{E - \rho\sqrt{E}}{D}$$

As,

$$\text{Var}(T) = L_1^2 \sigma_1^2 + L_2^2 \sigma_2^2 + 2L_1 L_2 \rho\sigma_1\sigma_2$$

Putting values

$$Var(T) = \left(\frac{1 - \rho\sqrt{E}}{D} \right)^2 \sigma^2 + \left(\frac{E - \rho\sqrt{E}}{D} \right)^2 \frac{\sigma^2}{E} + 2 \left(\frac{1 - \rho\sqrt{E}}{D} \right) \left(\frac{E - \rho\sqrt{E}}{D} \right) \frac{\rho\sigma^2}{\sqrt{E}}$$

$$Var(T) = \frac{\sigma^2}{D^2} \left[1 + \rho^2 E - 2\rho\sqrt{E} + \frac{E^2}{E} + \frac{\rho^2 E}{E} - \frac{2\rho\sqrt{E}}{E} + \frac{2\rho E}{\sqrt{E}} - \frac{2\rho^2\sqrt{E}}{\sqrt{E}} - \frac{2\rho^2 E\sqrt{E}}{\sqrt{E}} + \frac{2\rho^3 E}{\sqrt{E}} \right]$$

$$Var(T) = \frac{\sigma^2}{D^2} [1 + \rho^2 E - 2\rho\sqrt{E} + E + \rho^2 - 2\rho\sqrt{E} + 2\rho\sqrt{E} - 2\rho^2 - 2\rho^2 E + 2\rho^3\sqrt{E}]$$

$$Var(T) = \frac{\sigma^2}{D^2} [1 + E - 2\rho\sqrt{E} - \rho^2 E - \rho^2 + 2\rho^3\sqrt{E}]$$

$$Var(T) = \frac{\sigma^2}{D^2} [1(1 + E - 2\rho\sqrt{E}) - \rho^2(1 + E - 2\rho\sqrt{E})]$$

$$Var(T) = \frac{\sigma^2}{D^2} [(1 - \rho^2)(1 + E - 2\rho\sqrt{E})] = \frac{\sigma^2}{(1 + E - 2\rho\sqrt{E})^2} [(1 - \rho^2)(1 + E - 2\rho\sqrt{E})]$$

$$Var(T) = \frac{(1 - \rho^2)\sigma^2}{1 + E - 2\rho\sqrt{E}} = \frac{(1 - \rho^2)\sigma^2}{(1 - \rho^2) + (\rho^2 + E - 2\rho\sqrt{E})} = \frac{(1 - \rho^2)\sigma^2}{(1 - \rho^2) + (\sqrt{E} - \rho)^2}$$

$$\frac{Var(T)}{\sigma^2} = \frac{1 - \rho^2}{(1 - \rho^2) + (\sqrt{E} - \rho)^2} \leq 1, \text{ As } 0 < \rho^2 < 1$$

$$\Rightarrow Var(T) \leq \sigma^2$$

Since, T_1 is most efficient with variance σ^2 .

$$\frac{Var(T)}{\sigma^2} = 1$$

$$\frac{1 - \rho^2}{(1 - \rho^2) + (\sqrt{E} - \rho)^2} = 1$$

$$1 - \rho^2 = (1 - \rho^2) + (\sqrt{E} - \rho)^2$$

$$(\sqrt{E} - \rho)^2 = 0$$

$$\sqrt{E} - \rho = 0$$

$$\rho = \sqrt{E}$$

5.6 Variance of Most Efficient and Less Efficient Estimators

Question 54: (a) Show that if a most efficient estimator A and less efficient estimator B with efficiency E tends to joint normally for sample, B-A tends to zero correlation with A.

(b) show that the Error in B may be regarded as composed of (for large sample) two parts which are independent of error in A and error in B-A.

(c) Show that $V(A - B) = \left(\frac{1}{E} - 1\right) V(A)$

Solution:-

(a) Let $V(A) = \sigma_A^2$, $V(B) = \sigma_B^2$ and $\rho = \text{corr}(A, B)$

We have to prove that

$$\text{corr}(A, B - A) = 0$$

$$\Rightarrow \text{Cov}(A, B - A) = 0$$

As

$$\text{Cov}(A, B - A) = \text{Cov}(A, B) - V(A) = \rho\sigma_A\sigma_B - \sigma_A^2 \dots \dots (1)$$

The efficiency of B

$$E = \frac{V(A)}{V(B)} = \frac{\sigma_A^2}{\sigma_B^2} \Rightarrow \sigma_B^2 = \frac{\sigma_A^2}{E} \Rightarrow \sigma_B = \frac{\sigma_A}{\sqrt{E}}$$

Also we know that

$$\rho = \sqrt{E} \dots \dots (2)$$

Putting values in (1)

$$\text{Cov}(A, B - A) = \sqrt{E}\sigma_A \frac{\sigma_A}{\sqrt{E}} - \sigma_A^2 = \sigma_A^2 - \sigma_A^2 = 0$$

(b)

$$B = A + (B - A)$$

$$V(B) = V[A + (B - A)] = V(A) + V(B - A) + 2\text{Cov}(A, B - A)$$

$$V(B) = V(A) + V(B - A) + 2(0)$$

$$V(B) = V(A) + V(B - A)$$

$$\text{Error in } B = \text{Error in } A + \text{Error in } (B - A)$$

(c)

$$V(A - B) = V(A) + V(B) - 2\text{Cov}(A, B) = \sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B$$

$$V(A - B) = \sigma_A^2 + \frac{\sigma_A^2}{E} - \frac{2\sqrt{E}\sigma_A\sigma_A}{\sqrt{E}} = \sigma_A^2 + \frac{\sigma_A^2}{E} - 2\sigma_A^2 = \frac{\sigma_A^2}{E} - \sigma_A^2$$

$$V(A - B) = \left(\frac{1}{E} - 1\right)\sigma_A^2 = \left(\frac{1}{E} - 1\right)V(A)$$

5.7 Correlation of Two Unbiased Estimators having Same Variance

Question 55: If T_1 and T_2 are two unbiased estimators of $T(\theta)$. Having same variance and ρ is correlation between them, then show that

$$\rho \geq 2E - 1$$

Where E is the efficiency of each estimator.

Solution:

Let T be the MVUE of $T(\theta)$.

Given that

$$E(T_1) = E(T_2) = T(\theta) \quad \text{and} \quad V(T_1) = V(T_2)$$

Efficiency of T_1

$$E = \frac{V(T)}{V(T_1)} \Rightarrow V(T_1) = \frac{V(T)}{E}$$

Efficiency of T_2

$$E = \frac{V(T)}{V(T_2)} \Rightarrow V(T_2) = \frac{V(T)}{E}$$

Consider another unbiased estimator of $T(\theta)$

$$T_3 = \frac{1}{2}(T_1 + T_2)$$

$$V(T_3) = \frac{1}{4}[V(T_1) + V(T_2) + 2Cov(T_1, T_2)] = \frac{1}{4}\left[\frac{V(T)}{E} + \frac{V(T)}{E} + 2\rho\sqrt{\frac{V(T)}{E}\frac{V(T)}{E}}\right]$$

$$V(T_3) = \frac{1}{4} \left[2 \frac{V(T)}{E} + 2\rho \frac{V(T)}{E} \right] = \frac{1}{2} \left[\frac{1+\rho}{E} \right] V(T)$$

As T is MVUE so

$$V(T_3) \geq V(T)$$

$$\frac{1}{2} \left[\frac{1+\rho}{E} \right] V(T) \geq V(T)$$

$$\frac{1}{2} \left[\frac{1+\rho}{E} \right] \geq 1$$

$$1 + \rho \geq 2E$$

$$\rho \geq 2E - 1$$

Hence proved.

5.8 Linear Combination of MVUE Estimator and Less Efficient Estimator

Question 56: If T_1 is MVUE of $T(\theta)$ and T_2 is any other unbiased estimator of $T(\theta)$ with efficiency $E < 1$, then the unbiased linear combination of T_1 and T_2 can be MVUE of $T(\theta)$ under what condition.

Solution;-

Given that

$$E(T_1) = E(T_2) = T(\theta)$$

Efficiency of T_1

$$E = \frac{V(T_1)}{V(T_2)} < 1 \Rightarrow V(T_2) = \frac{V(T_1)}{E}$$

Consider linear combination of T_1 and T_2

$$T_3 = L_1 T_1 + L_2 T_2$$

$$E(T_3) = L_1 E(T_1) + L_2 E(T_2) = (L_1 + L_2) T(\theta)$$

Now, T_3 will be unbiased estimator of $T(\theta)$, if

$$L_1 + L_2 = 1 \dots \dots (A)$$

Now, Variance of T_3

$$V(T_3) = L_1^2 V(T_1) + L_2^2 V(T_2) + 2L_1 L_2 \text{Cov}(T_1, T_2)$$

$$V(T_3) = L_1^2 V(T_1) + L_2^2 \frac{V(T_1)}{E} + 2L_1 L_2 \rho \sqrt{V(T_1) \cdot \frac{V(T_1)}{E}}$$

$$V(T_3) = V(T_1) \left[L_1^2 + \frac{L_2^2}{E} + \frac{2L_1 L_2 \rho}{\sqrt{E}} \right]$$

Putting $\rho = \sqrt{E}$

$$V(T_3) = V(T_1) \left[L_1^2 + L_2^2 \left(\frac{1}{E} \right) + 2L_1 L_2 \right]$$

Since $E < 1, \frac{1}{E} > 1$

$$V(T_3) \geq V(T_1) [L_1^2 + L_2^2 + 2L_1 L_2] \geq V(T_1) (L_1 + L_2)^2$$

$$V(T_3) = V(T) \quad \text{if } L_1 + L_2 = 1$$



6.0 Questions Related to Sufficiency

6.1 Sample Mean Sufficient Estimator in Poisson Distribution

Question 57: Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson Distribution with parameter λ . Show that sample mean $\bar{X}$ is sufficient estimator of λ .

Solution:-

The probability Mass Function of Poisson Distribution is

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$L(\lambda) = \frac{e^{-\lambda n} \lambda^{\sum x_i}}{x_1! \cdot x_2! \cdot \dots \cdot x_n!} = [e^{-\lambda n} \lambda^{n\bar{X}}] \left[\frac{1}{\prod x_i!} \right] = g(\bar{X}, \lambda) h(x_1, x_2, \dots, x_n)$$

By Neyman Factorization

$$g(\bar{X}, \lambda) = e^{-\lambda n} \lambda^{n\bar{X}}, \text{ Function of Parameter } \lambda \text{ and Statistic } \bar{X}$$

$$h(x_1, x_2, \dots, x_n) \text{ Function independent of parameter}$$

Hence, Sample Mean $\bar{X}$ is sufficient Estimator of λ .

6.2 Sufficient Estimator in Bernoulli Distribution

Question 58: Let $X_1, X_2, \dots, X_n$ be a random sample from the Bernoulli Distribution with parameter θ . Show that sample mean $t = \frac{\sum X_i}{n}$ is sufficient estimator of θ .

Solution:-

$$L(\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i}$$

$$L(\theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$$

$$L(\theta) = \theta^{n \frac{\sum x_i}{n}} (1 - \theta)^{n \left(1 - \frac{\sum x_i}{n}\right)}$$

$$L(\theta) = \theta^{nt}(1 - \theta)^{n(1-t)}$$

$$L(\theta) = [\theta^{nt}(1 - \theta)^{n(1-t)}][1] = g(t, \theta)h(x)$$

By Neyman Factorization

$$g(t, \theta) = \theta^{nt}(1 - \theta)^{n(1-t)}, \text{Function of Parameter } \theta \text{ and Statistic } t$$

$$h(x) = 1 \text{ Function independent of parameter}$$

Hence, t is sufficient Estimator of θ .

6.3 Sufficient Statistic in Exponential Distribution

Question 59: Show that a random sample of size n from the density function

$$f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} ; x > 0$$

$\sum X_i$ is sufficient statistic.

Solution:-

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} e^{-\frac{x}{\theta}}$$

$$L(\theta) = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum x_i}$$

$$L(\theta) = \left[\left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum x_i}\right] (1) = g(\theta, \sum x_i) h(x_1, x_2, \dots, x_n)$$

By Neyman Factorization

$$g(\theta, \sum x_i) = \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum x_i} \text{ Function of parameter } \theta \text{ and Statistic } \sum x_i$$

$$h(x_1, x_2, \dots, x_n) = 1 \text{ Function independent of parameter}$$

so, $\sum x_i$, is sufficient estimator of θ .

6.4 Product of Observation as Sufficient Estimator

Question 60: Let $X_1, X_2, \dots, X_n$ be a random sample from the density function

$$f(x, p) = px^{p-1}, \quad 0 < x < 1, \quad p > 0$$

Show that the product $X_1 \cdot X_2 \dots \dots X_n$ is sufficient statistic of p

Solution:-

The joint p.d.f

$$L(p) = \prod_{i=1}^n px_i^{p-1} = p^n (x_1 \cdot x_2 \dots \dots x_n)^{p-1}$$

$$L(p) = p^n (x_1 \cdot x_2 \dots \dots x_n)^p (x_1 \cdot x_2 \dots \dots x_n)^{-1}$$

$$L(p) = g(p, \prod x_i) h(x_1, x_2 \dots \dots x_n)$$

By Neyman Factorization

$g(p, \prod x_i) = p^n (x_1 \cdot x_2 \dots \dots x_n)^p$ Function of parameter p and Statistic $\prod x_i$

$h(x_1, x_2 \dots \dots x_n) = (x_1 \cdot x_2 \dots \dots x_n)^{-1}$ Function independent of parameter

so, $\prod x_i$, is sufficient estimator of p .

6.5 Product of Observation as Sufficient Estimator

Question 61: Let $X_1, X_2, \dots, X_n$ be a random sample from the density function

$$f(x, \alpha) = \alpha x^{\alpha-1} (1-x), \quad 0 < x < 1,$$

Show that the product $X_1 \cdot X_2 \dots \dots X_n$ is sufficient statistic of α

Solution:-

The joint p.d.f

$$L(\alpha) = \prod_{i=1}^n \alpha x_i^{\alpha-1} (1-x_i) = \alpha^n (x_1 \cdot x_2 \dots \dots x_n)^{\alpha-1} (1-x_1)(1-x_2) \dots \dots (1-x_n)$$

$$L(\alpha) = [\alpha^n (x_1 \cdot x_2 \dots x_n)^\alpha] \left[\frac{(1-x_1)(1-x_2) \dots (1-x_n)}{(x_1 \cdot x_2 \dots x_n)} \right]$$

$$L(\alpha) = g(\alpha, \prod x_i) h(x_1, x_2 \dots x_n)$$

By Neyman Factorization

$g(\alpha, \prod x_i) = \alpha^n (x_1 \cdot x_2 \dots x_n)^\alpha$ Function of parameter α and Statistic $\prod x_i$

$h(x_1, x_2 \dots x_n) = \frac{(1-x_1)(1-x_2) \dots (1-x_n)}{(x_1 \cdot x_2 \dots x_n)}$ Function independent of parameter

so, $\prod x_i$, is sufficient estimator of α .

6.6 Sample Mean Sufficient Estimator in Normal Distribution

Question 62 : Let $X_1, X_2 \dots X_n$ be a random sample from Normal distribution with mean μ and variance σ^2 . Show that sample mean $\bar{X} = \frac{\sum X}{n}$ is sufficient statistic for parameter μ .

Solution:-

The pdf of normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

The likelihood function

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$$

$$L(\mu) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} \sum (x_i-\mu)^2} \dots \dots (1)$$

Consider

$$-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 = -\frac{1}{2\sigma^2} \sum (x_i + \bar{x} - \bar{x} - \mu)^2 = -\frac{1}{2\sigma^2} \sum [(x_i - \bar{x}) + (\bar{x} - \mu)]^2$$

$$-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 = -\frac{1}{2\sigma^2} \sum [(x_i - \bar{x})^2 + (\bar{x} - \mu)^2 + 2(x_i - \bar{x})(\bar{x} - \mu)]$$

$$-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 = -\frac{1}{2\sigma^2} \left[\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 + 2(\bar{x} - \mu) \sum (x_i - \bar{x}) \right]$$

$$-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 = -\frac{1}{2\sigma^2} \left[\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 \right]$$

put in (1)

$$L(\mu) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} [\sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2]}$$

$$L(\mu) = \frac{e^{-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2}}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2\sigma^2} n(\bar{x} - \mu)^2}$$

$$L(\mu) = \left[e^{-\frac{1}{2\sigma^2} n(\bar{x} - \mu)^2} \right] \left[\frac{e^{-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2}}{\sigma^n (2\pi)^{\frac{n}{2}}} \right] = g(\bar{x}, \mu) h(x_1, \dots, x_n)$$

By Neyman Factorization

$g(\bar{x}, \mu) = e^{-\frac{1}{2\sigma^2} n(\bar{x} - \mu)^2}$ Function of parameter μ and Statistic $\bar{x}$

$h(x_1, x_2, \dots, x_n) = \frac{e^{-\frac{1}{2\sigma^2} \sum (x_i - \bar{x})^2}}{\sigma^n (2\pi)^{\frac{n}{2}}}$ Function independent of parameter μ

so, $\bar{x} = \frac{\sum x}{n}$ is sufficient estimator of μ .

Note: For normal distribution with mean μ and variance 1

$$L(\mu) = \left[e^{-\frac{1}{2} n(\bar{x} - \mu)^2} \right] \left[\frac{e^{-\frac{1}{2} \sum (x_i - \bar{x})^2}}{(2\pi)^{\frac{n}{2}}} \right] = g(\bar{x}, \mu) h(x_1, \dots, x_n)$$

6.7 Sufficient Estimator of variance in Normal Distribution

Question 63 : Let $X_1, X_2, \dots, X_n$ be a random sample from Normal distribution with mean $\mu = 0$ and variance σ^2 . Show that $s^2 = \frac{1}{n} \sum x_i^2$ is sufficient statistic for parameter σ^2 .

Solution:-

The pdf of normal distribution

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$

The likelihood function

$$L(\sigma^2) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$

$$L(\sigma^2) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{\sum x^2}{2\sigma^2}}$$

$$L(\sigma^2) = \left[\frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{\sigma^{*2}}{2\sigma^2}} \right] (1) = g(\sigma^2, \sigma^{*2}) h(x_1, x_2 \dots x_n)$$

By Neyman Factorization

$$g(\sigma^2, \sigma^{*2}) = \frac{1}{\sigma^n (2\pi)^{\frac{n}{2}}} e^{-\frac{\sigma^{*2}}{2\sigma^2}} \text{ Function of parameter } \sigma^2 \text{ and Statistic } \sigma^{*2}$$

$$h(x_1, x_2 \dots x_n) = 1 \text{ Function independent of parameter}$$

so, $\sigma^{*2} = \sum x^2$ is sufficient estimator of σ^2 .

6.8 Jointly Sufficient Statistic in Gamma Distribution

Question 63 : Let $X_1, X_2 \dots \dots X_n$ be a random sample from Gamma distribution with p.d.f

$$f(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha + 1) \beta^{\alpha+1}} e^{-\frac{x}{\beta}} x^\alpha, x > 0$$

Show that

$$\prod_{i=1}^n x_i \sum_{i=1}^n x_i$$

Is a jointly sufficient statistic for the vector parameter (α, β) .

Solution:

The likelihood function

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{1}{\Gamma(\alpha + 1) \beta^{\alpha+1}} e^{-\frac{x_i}{\beta}} x_i^\alpha$$

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{1}{(\alpha)! \beta^\alpha} e^{-\frac{x_i}{\beta}} x_i^\alpha$$

$$L(\alpha, \beta) = \left[\prod_{i=1}^n x_i^\alpha \frac{1}{(\alpha! \beta^{\alpha+1})^n} e^{-\frac{\sum x_i}{\beta}} \right] (1)$$

$$L(\alpha, \beta) = g\left(\prod_{i=1}^n x_i \sum_{i=1}^n x_i, \alpha, \beta\right) h(x_1, x_2 \dots x_n)$$

By Neyman Factorization

$$g\left(\prod_{i=1}^n x_i \sum_{i=1}^n x_i, \alpha, \beta\right) = \prod_{i=1}^n x_i^\alpha \frac{1}{(\alpha! \beta^{\alpha+1})^n} e^{-\frac{\sum x_i}{\beta}} \quad \text{Function of parameter vector } (\alpha, \beta) \text{ and}$$

joint Statistic $(\prod_{i=1}^n x_i \sum_{i=1}^n x_i)$

$$h(x_1, x_2 \dots x_n) = 1 \quad \text{Function independent of parameter}$$

so, $(\prod_{i=1}^n x_i \sum_{i=1}^n x_i)$ is jointly sufficient estimator of parameter vector (α, β) .

6.9 Sufficient Statistic in Two Independently and Identically Distributed Poisson

R.Vs

Question 64: Let X_1, X_2 be identically and independently distributed with Poisson distribution having parameter λ . Show that for $T_1 = X_1 + X_2$ and $T_2 = X_1 + 2X_2$, T_1 is sufficient statistic, but T_2 is not sufficient statistic for λ .

Solution:-

The probability Mass function of Poisson distribution is

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, x > 0$$

The joint probability function for X_1, X_2

$$f(x_1, x_2; \lambda) = f(x_1; \lambda) f(x_2; \lambda) = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \frac{e^{-\lambda} \lambda^{x_2}}{x_2!} = \frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} \dots \dots (A)$$

Consider

$$T_1 = X_1 + X_2$$

$$E(T_1) = E(X_1) + E(X_2) = \lambda + \lambda = 2\lambda$$

The sum of Poisson random variates is also Poisson random variable so.

$$f(T_1; 2\lambda) = \frac{e^{-2\lambda} (2\lambda)^{T_1}}{T_1!}$$

Now, we find the conditional Mass function of X_1 and X_2 given T_1

$$f(X_1, X_2 | T_1 = X_1 + X_2) = \frac{f(x_1, x_2; \lambda)}{f(T_1)}$$

$$f(X_1, X_2 | T_1 = X_1 + X_2) = \left(\frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{T_1!}{e^{-2\lambda} (2\lambda)^{T_1}} \right)$$

$$f(X_1, X_2 | T_1 = X_1 + X_2) = \left(\frac{\lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{T_1!}{2^{T_1} \lambda^{x_1+x_2}} \right) = \left(\frac{T_1!}{2^{T_1} x_1! x_2!} \right)$$

The conditional distribution is independent of parameter, hence T_1 is sufficient Statistic

Now take

$$T_2 = X_1 + 2X_2$$

$$E(T_2) = E(X_1) + 2E(X_2) = \lambda + 2\lambda = 3\lambda$$

The probability Mass function of T_2

$$f(T_2; 3\lambda) = \frac{e^{-3\lambda} (3\lambda)^{T_2}}{T_2!}$$

Now, we find the conditional Mass function of X_1 and X_2 given T_2

$$f(X_1, X_2 | T_2 = X_1 + 2X_2) = \frac{f(x_1, x_2; \lambda)}{f(T_2)}$$

$$f(X_1, X_2 | T_2 = X_1 + 2X_2) = \left(\frac{e^{-2\lambda} \lambda^{x_1+x_2}}{x_1! x_2!} \right) \left(\frac{T_2!}{e^{-3\lambda} (3\lambda)^{T_2}} \right)$$

$$f(X_1, X_2 | T_2 = X_1 + 2X_2) = \left(\frac{e^{-2\lambda+3\lambda} \lambda^{x_1+x_2-x_1-2x_2}}{x_1! x_2!} \right) \left(\frac{T_2!}{(3)^{T_2}} \right)$$

$$f(X_1, X_2 | T_2 = X_1 + 2X_2) = \left(\frac{e^{\lambda} \lambda^{-x_2}}{x_1! x_2!} \right) \left(\frac{T_2!}{(3)^{T_2}} \right)$$

The conditional distribution is not independent of the parameter λ . So T_2 is not sufficient statistic.

6.10 Sufficient Statistic in Order Statistics

Question 65: Let $Y_1 < Y_2 \dots \dots Y_n$ is an order statistic. Let the distribution be

$$f(x; \theta) = e^{-(x-\theta)}, \theta < x < \infty, -\infty < \theta < \infty$$

Show that Y_1 is a sufficient statistic and Y_n is not sufficient statistic of θ .

Solution:-

The distribution function

$$F(y) = \int_{\theta}^y e^{-(y-\theta)} dy = -e^{-(y-\theta)} \Big|_{\theta}^y = -[e^{-(y-\theta)} - e^{-(\theta-\theta)}] = 1 - e^{-(y-\theta)}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_1 of random sample of n $n = n, r = 1$

$$g(y_1) = \frac{n!}{(1-1)!(n-1)!} [F(y_1^n)]^{1-1} [1 - F(y_1^n)]^{n-1} f(y_1^n)$$

$$g(y_1) = n[1 - 1 + e^{-(y_1-\theta)}]^{n-1} e^{-(y_1-\theta)}$$

$$g(y_1) = n[e^{-(y_1-\theta)}]^{n-1} e^{-(y_1-\theta)} = n[e^{-(y_1-\theta)}]^n = ne^{-n(y_1-\theta)} = ne^{-ny_1} e^{n\theta}$$

The joint p.d.f is

$$f(x_1, x_2 \dots x_n, \theta) = L(\theta) = \prod_{i=1}^n e^{-(x_i-\theta)} = e^{-\sum x_i + n\theta} = e^{-\sum x_i} e^{n\theta}$$

now, the conditional distribution is

$$\frac{f(x_1, x_2 \dots x_n, \theta)}{g(y_1)} = \frac{e^{-\sum x_i} e^{n\theta}}{ne^{-ny_1} e^{n\theta}} = \frac{e^{-\sum x_i}}{ne^{-ny_1}}$$

Which is independent of θ . Hence y_1 is sufficient statistic of θ .

Now Distribution of y_n of random sample of n $n = n, r = n$

$$g(y_n) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1 - F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n[1 - e^{-(y_n-\theta)}]^{n-1} e^{-(y_n-\theta)} = n[1 - e^{-(y_n-\theta)}]^{n-1} e^{-y_n} e^{\theta}$$

The conditional distribution is

$$\frac{f(x_1, x_2 \dots x_n, \theta)}{g(y_n)} = \frac{e^{-\sum x_i} e^{n\theta}}{n[1 - e^{-(y_n - \theta)}]^{n-1} e^{-y_n} e^\theta} = \frac{e^{-\sum x_i + n\theta - \theta}}{n[1 - e^{-(y_n - \theta)}]^{n-1} e^{-y_n}}$$

Which is not independent of θ . Hence y_n is not sufficient statistic of θ

6.11 Sufficient Statistic in Order Statistics

Question 66: Let $Y_1 < Y_2 \dots \dots Y_n$ is an order statistic. Let the distribution be

$$f(x; \theta) = e^{-(x-\theta)}, \theta < x < \infty, -\infty < \theta < \infty$$

Assume $n=3$ Show that $\text{Max}(x_i)$ is not sufficient statistic

Solution:-

The distribution function

$$F(y) = \int_{\theta}^y e^{-(y-\theta)} dy = -e^{-(y-\theta)} \Big|_{\theta}^y = -[e^{-(y-\theta)} - e^{-(\theta-\theta)}] = 1 - e^{-(y-\theta)}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_1 of random sample of 3, $n = 3, r = 3$

$$\max(x_i) = g(y_3) = \frac{3!}{(3-1)!(3-3)!} [F(y_3^3)]^{3-1} [1 - F(y_3^3)]^{3-3} f(y_3^3)$$

$$g(y_3) = 3[1 - e^{-(y_n - \theta)}]^2 e^{-(y_3 - \theta)} = 3[1 - e^{-(y_3 - \theta)}]^2 e^{-y_3} e^\theta$$

The joint p.d.f is

$$f(x_1, x_2, x_3, \theta) = L(\theta) = e^{-(x_1 - \theta)} e^{-(x_2 - \theta)} e^{-(x_3 - \theta)} = e^{-(x_1 + x_2 + x_3) + 3\theta} = e^{-(x_1 + x_2 + x_3)} e^{3\theta}$$

now, the conditional distribution is

$$\frac{f(x_1, x_2, x_3, \theta)}{g(y_3)} = \frac{e^{-(x_1 + x_2 + x_3)} e^{3\theta}}{3[1 - e^{-(y_3 - \theta)}]^2 e^{-y_3} e^\theta} = \frac{e^{-(x_1 + x_2 + x_3)} e^{2\theta}}{3[1 - e^{-(y_3 - \theta)}]^2 e^{-y_3}}$$

Which is not independent of θ . Hence $y_3 = \max(x_i)$ is not sufficient y of θ .

6.12 Sufficient Statistic of Order Statistic from Uniform Distribution

Question 67: Let $X_1, X_2, \dots, X_n$ be independent random sample from

$$f(x; \theta) = \frac{1}{\theta}, \quad 0 < x < \theta$$

Show $Y_1 = \min(x_i)$ is not sufficient statistic for θ , but $Y_n = \max(x_i)$ is sufficient statistic for θ

Solution:-

The joint p.d.f

$$f(x_1, x_2, \dots, x_n; \theta) = f(x_1, \theta) f(x_2, \theta) \dots f(x_n, \theta) = \left(\frac{1}{\theta}\right) \left(\frac{1}{\theta}\right) \dots \left(\frac{1}{\theta}\right) = \frac{1}{\theta^n}$$

The distribution function

$$F(y) = \int_0^y \frac{1}{\theta} dy = \frac{y}{\theta} \Big|_0^y = \frac{y - 0}{\theta} = \frac{y}{\theta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of $y_1 = \min(x_i)$ of random sample of n $n = n, r = 1$

$$g(y_1) = \frac{n!}{(1-1)!(n-1)!} [F(y_1^n)]^{1-1} [1 - F(y_1^n)]^{n-1} f(y_1^n)$$

$$g(y_1) = n \left[1 - \frac{y_1}{\theta}\right]^{n-1} \frac{1}{\theta} = \frac{n}{\theta^{n-1}} \frac{1}{\theta} (\theta - y_1)^{n-1} = \frac{n}{\theta^n} (\theta - y_1)^{n-1}$$

Now, the conditional distribution given y_1

$$\frac{f(x_1, x_2, \dots, x_n, \theta)}{g(y_1)} = \frac{\frac{1}{\theta^n}}{\frac{n}{\theta^n} (\theta - y_1)^{n-1}} = \frac{1}{n(\theta - y_1)^{n-1}}$$

which is not independent of θ . So y_1 is not sufficient statistic for θ .

Now Distribution of $y_n = \max(x_i)$ of random sample of n $n = n, r = n$

$$g(y_n) = \frac{n!}{(n-1)!(n-n)!} [F(y_n^n)]^{n-1} [1 - F(y_n^n)]^{n-n} f(y_n^n)$$

$$g(y_n) = n \left[\frac{y_n}{\theta} \right]^{n-1} \frac{1}{\theta} = \frac{n}{\theta^{n-1}} \frac{1}{\theta} (y_n)^{n-1} = \frac{n}{\theta^n} (y_n)^{n-1}$$

Now, the conditional distribution given y_n

$$\frac{f(x_1, x_2, \dots, x_n, \theta)}{g(y_n)} = \frac{\frac{1}{\theta^n}}{\frac{n}{\theta^n} (y_n)^{n-1}} = \frac{1}{n(y_n)^{n-1}}$$

which is independent of θ . So y_n is sufficient statistic for θ .

6.13 Joint Pdf of Sufficient Statistic of order Statistic From Uniform n=5

Question 68: Let $Y_1 < Y_2 < Y_3 < Y_4 < Y_5$ be the order statistic of a random sample of size 5 from uniform distribution with p.d.f

$$f(x; \theta) = \frac{1}{\theta}, \quad 0 < x < \theta \text{ and } \theta > 0$$

- (a) Show that $2Y_3$ is an unbiased estimator of θ .
- (b) Determine the joint p.d.f of Y_3 and sufficient statistic Y_5 of θ .
- (c) Find $E(2Y_3 | Y_5) = \phi(Y_5)$.
- (d) Compare the variances of $2Y_3$ and $\phi(Y_5)$.

Solution.

- (a) The distribution function

$$F(y) = \int_0^y \frac{1}{\theta} dy = \frac{y}{\theta} \Big|_0^y = \frac{y-0}{\theta} = \frac{y}{\theta}$$

As Distribution of order statistic y_r^n is

$$g[y_r^n] = \frac{n!}{(r-1)!(n-r)!} [F(y_r^n)]^{r-1} [1 - F(y_r^n)]^{n-r} dF(F(y_r^n))$$

Now Distribution of y_3 of random sample of $n = 5, r = 3$

$$g(y_3) = \frac{5!}{(3-1)!(5-3)!} [F(y_3^5)]^{3-1} [1 - F(y_3^5)]^{5-3} f(y_3^5)$$

$$g(y_3) = 30 \left(\frac{y_3}{\theta}\right)^2 \left[1 - \frac{y_3}{\theta}\right]^2 \frac{1}{\theta}$$

$$E(2Y_3) = 2E(Y_3) = 2 \int_0^{\theta} y_3 30 \left(\frac{y_3}{\theta}\right)^2 \left[1 - \frac{y_3}{\theta}\right]^2 \frac{1}{\theta} dy_3$$

Let

$$t = \frac{y_3}{\theta} \Rightarrow t\theta = y_3 \Rightarrow dt\theta = dy_3$$

When $y_3 = 0$, then $t = 0$, When $y_3 = \theta$, then $t = 1$

$$E(2Y_3) = 60 \int_0^1 (t\theta) t^2 [1-t]^2 \frac{1}{\theta} (\theta dt) = 60\theta \int_0^1 t^3 (1-t)^2 dt$$

$$E(2Y_3) = 60\theta \int_0^1 t^{4-1} (1-t)^{3-1} dt$$

Using beta function

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$E(2Y_3) = 60\theta \frac{\Gamma 4 \Gamma 3}{\Gamma(4+3)} = 60\theta \frac{3!.2!}{6!} = 60\theta \frac{3!.2!}{6.5.4.3!} = \frac{120\theta}{120} = \theta$$

Hence $2Y_3$ is unbiased estimator of θ .

(b) The joint p.d.f of r^{th} and s^{th} order statistic is

$$g[y_r, y_s] = \frac{n!}{(r-1)!(s-r-1)!(n-s)!} [F(y_r^n)]^{r-1} [1 - F(y_s^n)]^{n-s} [F(y_s^n) - F(y_r^n)]^{s-r-1} f(y_r) f(y_s)$$

Put $n=5, r=3, s=5$

$$g[y_3, y_5] = \frac{5!}{(3-1)!(5-3-1)!(5-5)!} [F(y_3^5)]^{3-1} [1 - F(y_5^5)]^{5-5} [F(y_5^5) - F(y_3^5)]^{5-3-1} f(y_3) f(y_5)$$

$$g(y_3, y_5) = \frac{5!}{2! \cdot 1! \cdot 0!} \left[\frac{y_3}{\theta} \right]^2 \left[\frac{y_5}{\theta} - \frac{y_3}{\theta} \right] \frac{1}{\theta} \cdot \frac{1}{\theta} = 60 \left[\frac{y_3}{\theta} \right]^2 \left[\frac{y_5}{\theta} - \frac{y_3}{\theta} \right] \frac{1}{\theta^2}$$

(c)

$$E(2Y_3|Y_5) = 2E(Y_3|Y_5) = 2 \int_0^{y_5} y_3 \frac{g(y_3, y_5)}{h(y_5)} dy_3$$

Now Distribution of y_5 of random sample of $n = 5, r = 5$

$$h(y_5) = \frac{5!}{(5-1)!(5-5)!} [F(y_5^5)]^{5-1} [1 - F(y_5^5)]^{5-5} f(y_5^5)$$

$$h(y_5) = 5 \left(\frac{y_5}{\theta} \right)^4 \left(\frac{1}{\theta} \right)$$

$$E(2Y_3|Y_5) = 2 \int_0^{y_5} y_3 \frac{60 \left[\frac{y_3}{\theta} \right]^2 \left[\frac{y_5}{\theta} - \frac{y_3}{\theta} \right] \frac{1}{\theta^2}}{5 \left(\frac{y_5}{\theta} \right)^4 \left(\frac{1}{\theta} \right)} dy_3 = \frac{24}{(y_5)^4} \theta^5 \cdot \frac{1}{\theta^5} \int_0^{y_5} (y_3)^3 (y_5 - y_3) dy_3$$

$$E(2Y_3|Y_5) = \frac{24}{(y_5)^4} \left[y_5 \int_0^{y_5} (y_3)^3 dy_3 - \int_0^{y_5} (y_3)^4 dy_3 \right] = \frac{24}{(y_5)^4} \left[y_5 \frac{(y_3)^4}{4} \Big|_0^{y_5} - \frac{(y_3)^5}{5} \Big|_0^{y_5} \right]$$

$$E(2Y_3|Y_5) = \frac{24}{(y_5)^4} \left[y_5 \frac{(y_5)^4}{4} - \frac{(y_5)^5}{5} \right] = 24 \left[\frac{1}{4} y_5 - \frac{1}{5} y_5 \right] = \frac{24}{20} y_5 = \frac{6}{5} y_5$$

So,

$$E(2Y_3|Y_5) = \frac{6}{5} y_5 = \phi(y_5)$$

(d) Variance of $2Y_3$

$$E(2Y_3^2)^2 = \int_0^{\theta} (2y_3)^2 30 \left(\frac{y_3}{\theta}\right)^2 \left[1 - \frac{y_3}{\theta}\right]^2 \frac{1}{\theta} dy_3$$

Let

$$t = \frac{y_3}{\theta} \Rightarrow t\theta = y_3 \Rightarrow dt\theta = dy_3$$

When $y_3 = 0$, then $t = 0$, When $y_3 = \theta$, then $t = 1$

$$E(2Y_3^2)^2 = 120 \int_0^1 (t\theta)^2 t^2 [1-t]^2 \frac{1}{\theta} (\theta dt) = 120\theta^2 \int_0^1 t^4 (1-t)^2 dt$$

$$E(2Y_3^2)^2 = 120\theta^2 \int_0^1 t^{5-1} (1-t)^{3-1} dt$$

Using beta function

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$$

$$E(2Y_3^2)^2 = 120\theta^2 \frac{\Gamma 5 \Gamma 3}{\Gamma(5+3)} = 120\theta^2 \frac{4! \cdot 2!}{7!} = 120\theta^2 \frac{4! \cdot 2!}{7 \cdot 6 \cdot 5 \cdot 4!} = \frac{240\theta^2}{210} = \frac{8\theta^2}{7}$$

So,

$$Var(2Y_3) = E(2Y_3)^2 - [E(2Y_3)]^2 = \frac{8\theta^2}{7} - \theta^2 = \frac{1}{7}\theta^2$$

and

$$Var[\phi(y_5)] = Var\left(\frac{6}{5}y_5\right) = \frac{36}{25}Var(y_5) \dots \dots (A)$$

$$E(Y_5) = \int_0^{\theta} y_5 h(y_5) dy_5 = \int_0^{\theta} y_5 5 \left(\frac{y_5}{\theta}\right)^4 \left(\frac{1}{\theta}\right) dy_5 = \frac{5}{\theta^5} \int_0^{\theta} (y_5)^5 dy_5 = \frac{5}{\theta^5} \frac{(y_5)^6}{6} \Big|_0^{\theta}$$

$$E(Y_5) = \frac{5}{\theta^5} \cdot \frac{\theta^6}{6} = \frac{5}{6}\theta$$

$$E(Y_5)^2 = \int_0^{\theta} y_5^2 h(y_5) dy_5 = \int_0^{\theta} y_5^2 5 \left(\frac{y_5}{\theta}\right)^4 \left(\frac{1}{\theta}\right) dy_5 = \frac{5}{\theta^5} \int_0^{\theta} (y_5)^6 dy_5 = \frac{5}{\theta^5} \frac{(y_5)^7}{7} \Big|_0^{\theta}$$

$$E(Y_5)^2 = \frac{5}{\theta^5} \cdot \frac{\theta^7}{7} = \frac{5}{7} \theta^2$$

$$Var(Y_5) = E(Y_5)^2 - [E(Y_5)]^2 = \frac{5}{7} \theta^2 - \left(\frac{5}{6} \theta\right)^2 = \left[\frac{5}{7} - \frac{25}{36}\right] \theta^2 = \left[\frac{(5)(36) - (7)(35)}{(7)(36)}\right] \theta^2$$

Put in A

$$Var[\phi(y_5)] = \frac{36}{25} \left[\frac{(5)(36) - (7)(35)}{(7)(36)}\right] \theta^2 = \frac{1}{35} \theta^2$$

Now comparing $Var[\phi(y_5)]$ and $Var(2Y_3)$

$$\frac{1}{35} \theta^2 < \frac{1}{7} \theta^2 \Rightarrow \frac{1}{5} < 1$$

Hence

$$Var[\phi(y_5)] < Var(2Y_3)$$

Hence this shows that the sufficient statistic has less variance.

In fact from the current evidence the variance of sufficient statistic is 5 times less than the variance of $2Y_3$.

6.14 Sufficient Estimator of Parameter of Bernoulli Distribution n=3

Question 69: Let X_1, X_2, X_3 be a random sample of size 3 from Bernoulli distribution consider two estimators

$$T_1 = t_1(X_1, X_2, X_3) = X_1 + X_2 + X_3$$

$$T_2 = t_2(X_1, X_2, X_3) = X_1 X_2 + X_3$$

Check T_1 and T_2 for sufficiency.

Solution:-

Since X_1, X_2, X_3 are Bernoulli random variables with possible values 0 and 1, to compute the statistics T_1 and T_2 based upon values of random variates there will be eight possible combinations.

The probability Mass function of Bernoulli Distribution is

$$p(x) = p^x(1-p)^{1-x}, x = 0, 1$$

Now we compute conditional Probabilities for each T_1 and T_2

For $T_1 = 0$

$$P(X_1 = 0, X_2 = 0, X_3 = 0 | T_1 = 0) = \frac{P(X_1 = 0, X_2 = 0, X_3 = 0)}{P(T_1 = 0) = P(X_1 = 0, X_2 = 0, X_3 = 0)} = 1$$

For $T_1 = 1$, possible combinations (0,0,1), (0,1,0), (1,0,0),

$$P(X_1 = 0, X_2 = 0, X_3 = 1) = P(X_1 = 0, X_2 = 1, X_3 = 0) = P(X_1 = 1, X_2 = 0, X_3 = 0) = p(1-p)^2$$

And

$$P(T_1 = 1) = p(1-p)^2 + p(1-p)^2 + p(1-p)^2 = 3p(1-p)^2$$

$$P(1,0,0 | T_1 = 1) = P(0,1,0 | T_1 = 1) = P(0,0,1 | T_1 = 1) = \frac{p(1-p)^2}{3p(1-p)^2} = \frac{1}{3}$$

For $T_1 = 2$, possible combinations (0,1,1), (1,0,1), (1,1,0),

$$P(X_1 = 0, X_2 = 1, X_3 = 1) = P(X_1 = 1, X_2 = 0, X_3 = 1) = P(X_1 = 1, X_2 = 1, X_3 = 0) = p^2(1-p)$$

And

$$P(T_1 = 2) = p^2(1-p) + p^2(1-p) + p^2(1-p) = 3p^2(1-p)$$

$$P(0,1,1 | T_1 = 2) = P(1,0,1 | T_1 = 2) = P(1,1,0 | T_1 = 2) = \frac{p^2(1-p)}{3p^2(1-p)} = \frac{1}{3}$$

For $T_1 = 3$

$$P(X_1 = 1, X_2 = 1, X_3 = 1 | T_1 = 3) = \frac{P(X_1 = 1, X_2 = 1, X_3 = 1)}{P(T_1 = 3) = P(X_1 = 1, X_2 = 1, X_3 = 1)} = 1$$

For $T_2 = 0$, possible combinations (0,0,0), (0,1,0), (1,0,0),

$$P(X_1 = 0, X_2 = 0, X_3 = 0) = (1 - p)^3$$

$$P(X_1 = 0, X_2 = 1, X_3 = 0) = P(X_1 = 1, X_2 = 0, X_3 = 0) = p(1 - p)^2$$

And

$$P(T_2 = 0) = (1 - p)^3 + p(1 - p)^2 + p(1 - p)^2 = (1 - p)^2[1 - p + p + p] = (1 - p)^2(1 + p)$$

$$P(0,0,0|T_2 = 0) = \frac{(1 - p)^3}{(1 - p)^2(1 + p)} = \frac{1 - p}{1 + p}$$

$$P(0,1,0|T_2 = 0) = P(1,0,0|T_2 = 0) = \frac{p(1 - p)^2}{(1 - p)^2(1 + p)} = \frac{p}{1 + p}$$

For $T_2 = 1$, possible combinations (0,0,1), (0,1,1), (1,0,1), (1,1,0)

$$P(X_1 = 0, X_2 = 0, X_3 = 1) = p(1 - p)^2$$

$$P(X_1 = 0, X_2 = 1, X_3 = 1) = P(X_1 = 1, X_2 = 0, X_3 = 1) = P(X_1 = 1, X_2 = 1, X_3 = 0) = p^2(1 - p)$$

$$P(T_2 = 1) = p(1 - p)^2 + 3p^2(1 - p) = p(1 - p)[1 - p + 3p] = p(1 - p)(1 + 2p)$$

$$P(0,0,0|T_2 = 1) = \frac{p(1 - p)^2}{p(1 - p)(1 + 2p)} = \frac{1 - p}{1 + 2p}$$

$$P(0,1,1|T_2 = 1) = P(1,0,1|T_2 = 1) = P(1,1,0|T_2 = 1) = \frac{p^2(1 - p)}{p(1 - p)(1 + 2p)} = \frac{p}{1 + 2p}$$

For $T_2 = 2$, possible combinations (1,1,1)

$$P(X_1 = 1, X_2 = 1, X_3 = 1) = p^3$$

$$P(T_2 = 2) = p^3$$

$$P(1,1,1|T_2 = 2) = \frac{p^3}{p^3} = 1$$

The all possible combinations of variates along with the conditional probabilities is presented in the following table

Outcomes	T_1	T_2	$f(x_1, x_2, x_3 T_1)$	$f(x_1, x_2, x_3 T_2)$
(x_1, x_2, x_3)				
(0,0,0)	0	0	1	$\frac{1-p}{1+p}$
(0,0,1)	1	1	$\frac{1}{3}$	$\frac{1-p}{1+2p}$
(0,1,0)	1	0	$\frac{1}{3}$	$\frac{p}{1+p}$
(0,1,1)	2	1	$\frac{1}{3}$	$\frac{p}{1+2p}$
(1,0,0)	1	0	$\frac{1}{3}$	$\frac{p}{1+p}$
(1,0,1)	2	1	$\frac{1}{3}$	$\frac{p}{1+2p}$
(1,1,0)	2	1	$\frac{1}{3}$	$\frac{p}{1+2p}$
(1,1,1)	3	2	1	1

From the above table it is clear that all the conditional probabilities for given values of T_1 are independent of the parameter p , so T_1 is sufficient statistic.

For statistic T_2 only one conditional probability for combination (1,1,1) is independent of the parameter p . hence T_2 is not sufficient statistic.

6.17 Joint Sufficient Statistic (Maximum and Minimum order Statistics)

Question 70: Let $X_1, X_2, \dots, X_n$ be a random sample from a Uniform distribution with p.d.f

$$f(x; \theta_1, \theta_2) = \frac{1}{\theta_2 - \theta_1} I_{(\theta_1, \theta_2)}(X)$$

Show that the statistic $Y_1 = \min(X_1, X_2, \dots, X_n)$ and $Y_n = \max(X_1, X_2, \dots, X_n)$ are jointly sufficient for (θ_1, θ_2) .

Solution:-

The joint p.d.f is given as

$$L = \prod_{i=1}^n \frac{1}{\theta_2 - \theta_1} I_{(\theta_1, \theta_2)}(X_i) = \frac{1}{(\theta_2 - \theta_1)^n} \prod_{i=1}^n I_{(\theta_1, \theta_2)}(X_i)$$

As

$$\begin{aligned} \theta_1 &< X_i < \theta_2 \\ \theta_1 &< X_1, X_2, \dots, X_n < \theta_2 \\ \theta_1 &< Y_1, Y_2, \dots, Y_n < \theta_2 \\ \theta_1 &< Y_1 < Y_n < \theta_2 \end{aligned}$$

$$L = \frac{1}{(\theta_2 - \theta_1)^n} I_{(\theta_1, Y_n)}(Y_1) I_{(Y_1, \theta_2)}(Y_n)$$

It can be factorize as

$$\begin{aligned} L &= \left[\frac{1}{(\theta_2 - \theta_1)^n} I_{(\theta_1, Y_n)}(Y_1) I_{(Y_1, \theta_2)}(Y_n) \right] (1) \\ L &= h(u_1(X_1, \dots, X_n), u_2(X_1, \dots, X_n); \theta) k(X_1, \dots, X_n) \end{aligned}$$

Hence Y_1 and Y_n are jointly sufficient for (θ_1, θ_2)

If $\theta_1 = \theta$, $\theta_2 = \theta + 1$

$$f(x; \theta, \theta + 1) = \frac{1}{\theta + 1 - \theta} I_{(\theta, \theta+1)}(X) = I_{(\theta, \theta+1)}(X)$$

The joint p.d.f

$$L = \prod_{i=1}^n I_{(\theta, \theta+1)}(X_i)$$

As

$$\theta < X_i < \theta + 1$$

$$\theta < X_1, X_2, \dots, X_n < \theta + 1$$

$$\theta < Y_1, Y_2, \dots, Y_n < \theta + 1$$

$$\theta < Y_1 < Y_n < \theta + 1$$

$$L = I_{(\theta, Y_n)}(Y_1) I_{(Y_1, \theta+1)}(Y_n)$$

It can be factorize as

$$L = [I_{(\theta, Y_n)}(Y_1) I_{(Y_1, \theta+1)}(Y_n)](1)$$

Hence Y_1 and Y_n are jointly sufficient for $(\theta, \theta + 1)$

If $\theta_1 = 0$, $\theta_2 = \theta$

$$f(x; 0, \theta) = \frac{1}{\theta - 0} I_{(0, \theta)}(X) = \frac{1}{\theta} I_{(0, \theta)}(X)$$

The joint p.d.f

$$L = \prod_{i=1}^n \frac{1}{\theta} I_{(0, \theta)}(X) = \frac{1}{\theta^n} \prod_{i=1}^n I_{(0, \theta)}(X_i)$$

As

$$0 < X_i < \theta$$

$$0 < X_1, X_2, \dots, X_n < \theta$$

$$0 < Y_1, Y_2, \dots, Y_n < \theta$$

$$0 < Y_1 < Y_n < \theta$$

$$L = \frac{1}{\theta^n} I_{(0, Y_n)}(Y_1) I_{(Y_1, \theta)}(Y_n)$$

It can be factorize as

$$L = \left[\frac{1}{\theta^n} I_{(0, Y_n)}(Y_1) \right] [I_{(Y_1, \theta)}(Y_n)]$$

Hence Y_n alone is sufficient statistic.

Similarly if $\theta_1 = -\theta$, $\theta_2 = 0$

$$L = \left[\frac{1}{(2\theta)^n} I_{(-\theta, Y_n)}(Y_1) \right] [I_{(Y_1, \theta)}(Y_n)]$$

Hence, Y_1 and Y_n are jointly sufficient

6.18 Sufficient Statistic is Unique

Question 71: Prove that apart from functional relationship, the sufficient statistic is unique.

Solution:

if there are two distinct sufficient statistic t_1 and t_2 then we have

$$f(t_1, t_2; \theta) = K_1(t_1, \theta) K_2(t_2 | t_1) = K_1'(t_2, \theta) K_2'(t_1 | t_2)$$

$$K_1(t_1, \theta) = \frac{f(t_1, t_2; \theta)}{K_2(t_2 | t_1)} = \frac{K_1'(t_2, \theta) K_2'(t_1 | t_2)}{K_2(t_2 | t_1)}$$

Writing

$$K_3(t_1, t_2) = \frac{K_2'(t_1 | t_2)}{K_2(t_2 | t_1)}$$

We obtain

$$K_1(t_1, \theta) = K_1'(t_2, \theta) K_3(t_1, t_2)$$

Identically in θ , this cannot hold unless t_1 and t_2 are functionally related.

6.19 Rao Blackwell Theorem

Question 72: State and prove Rao Blackwell Theorem.

Statement: Let X and Y be two random variables such that

$$E(Y) = \mu, \quad V(Y) = \sigma_Y^2 > 0$$

Let $E(Y|X = x) = \phi(x)$ then (i) $E[\phi(x)] = \mu$ (ii) $V[\phi(x)] < V(Y)$

Proof:

Let $f(x, y)$ be the joint p.d.f of two random variates X and Y . Let $f_1(\cdot)$ and $f_2(\cdot)$ are the marginal distributions of X and Y respectively and let $h(y|x)$ be the conditional distribution of Y given $X = x$ such that

$$h(y|x) = \frac{f(x, y)}{f_1(x)}$$

take $E(Y|X = x) = \phi(x)$

$$E(Y|X = x) = \int_{-\infty}^{\infty} y h(y|x) dy = \int_{-\infty}^{\infty} y \frac{f(x, y)}{f_1(x)} dy = \phi(x)$$

$$E(Y|X = x) = \frac{1}{f_1(x)} \int_{-\infty}^{\infty} y f(x, y) dy = \phi(x) \dots \dots \dots (1)$$

$$\int_{-\infty}^{\infty} y f(x, y) dy = \phi(x) f_1(x) \dots \dots \dots (2)$$

From equation (1) we observe that the conditional distribution of Y given $X = x$ is independent of the parameter μ so X is a sufficient statistic.

Now,

$$E[\phi(x)] = E(Y|X = x) = E(Y) = \mu$$

Also we have

$$V(Y) = E[Y - E(Y)]^2 = E[Y - \mu]^2 = E[Y - \phi(x) + \phi(x) - \mu]^2$$

$$V(Y) = E[(Y - \phi(x)) + (\phi(x) - \mu)]^2$$

$$V(Y) = E(Y - \phi(x))^2 + E(\phi(x) - \mu)^2 + 2E[(Y - \phi(x))(\phi(x) - \mu)] \dots \dots (3)$$

$$E[(Y - \phi(x))(\phi(x) - \mu)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [(y - \phi(x))(\phi(x) - \mu)] f(x, y) dx dy$$

$$E[(Y - \phi(x))(\phi(x) - \mu)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} [(y - \phi(x))(\phi(x) - \mu)] f_1(x) h(y|x) dx dy$$

$$E[(Y - \phi(x))(\phi(x) - \mu)] = \int_{-\infty}^{\infty} (\phi(x) - \mu) \left[\int_{-\infty}^{\infty} (y - \phi(x)) f_1(x) h(y|x) dy \right] dx$$

But

$$\int_{-\infty}^{\infty} (y - \phi(x)) f_1(x) h(y|x) dy = 0, \quad \text{As } E(Y|X = x) = \phi(x)$$

So,

$$E[(Y - \phi(x))(\phi(x) - \mu)] = 0$$

Put in (3)

$$V(Y) = E(Y - \phi(x))^2 + E(\phi(x) - \mu)^2$$

$$V(Y) = E(Y - \phi(x))^2 + V(\phi(x))$$

$$\therefore E(Y - \phi(x))^2 \geq 0$$

$$V(Y) \geq V(\phi(x))$$

$$\Rightarrow V(\phi(x)) \leq V(Y)$$

Which completes the proof.

6.20 Neyman Factorization Theorem

Question 73: State and prove Neyman Factorization Theorem.

Statement: Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution that has a p.d.f $f(x; \theta)$ $\theta \in \Omega$. The statistic $T_1 = \phi_1(X_1, X_2, \dots, X_n)$ be a sufficient statistic iff we can find two non-negative functions “h” and “k” such that

$$f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) = h[\phi_1(X_1, X_2, \dots, X_n); \theta] k(X_1, X_2, \dots, X_n)$$

Where, $k(X_1, X_2, \dots, X_n)$ does not depend upon parameter θ .

Proof:

we will prove the theorem when the variables are of continuous type. The factorization as stated

$$f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta) = h[\phi_1(X_1, X_2, \dots, X_n); \theta] k(X_1, X_2, \dots, X_n) \dots (A)$$

Where, $k(X_1, X_2, \dots, X_n)$ does not depend upon parameter θ .

We shall make a one-to-one transformation

$$t_1 = \phi_1(X_1, X_2, \dots, X_n)$$

$$t_2 = \phi_2(X_1, X_2, \dots, X_n)$$

$$\vdots$$

$$t_n = \phi_n(X_1, X_2, \dots, X_n)$$

Having inverse function

$$X_1 = u_1(t_1, t_2, \dots, t_n)$$

$$X_2 = u_2(t_1, t_2, \dots, t_n)$$

$$\vdots$$

$$X_n = u_n(t_1, t_2, \dots, t_n)$$

And the Jacobian of the transformation

$$|J| = X_1 = \frac{\partial(X_1, X_2, \dots, X_n)}{\partial(t_1, t_2, \dots, t_n)}$$

The joint p.d.f of the statistics $T_1, T_2, \dots, T_n$ is given by

$$g(t_1, t_2, \dots, t_n; \theta) = h[\phi_1(X_1, X_2, \dots, X_n; \theta)]k[u_1(t_1, t_2, \dots, t_n), u_2(t_1, t_2, \dots, t_n) \dots u_1(t_1, t_2, \dots, t_n)]|J|$$

$$g(t_1, t_2, \dots, t_n; \theta) = h(t_1; \theta)k(u_1, u_2, \dots, u_n)|J|$$

Where t_1 and $|J|$ do not depend upon θ

The marginal p.d.f of T_1 is given by (say $g_1(t_1, \theta)$)

$$\begin{aligned} g_1(t_1, \theta) &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} g(t_1, t_2, \dots, t_n) dt_2 dt_3 \dots dt_n \\ g_1(t_1, \theta) &= \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} h(t_1; \theta) k(u_1, u_2, \dots, u_n) |J| dt_2 dt_3 \dots dt_n \\ g_1(t_1, \theta) &= h(t_1; \theta) \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} k(u_1, u_2, \dots, u_n) |J| dt_2 dt_3 \dots dt_n \dots \dots (1) \end{aligned}$$

Now, the function k for every fixed value $t_1 = \phi_1(X_1, X_2, \dots, X_n; \theta)$ does not depend on θ nor θ is involved in either the Jacobian or limit of integration.

Hence, the $(n-1)$ fold integrals in the R.H.S of eq(1) is function of t_1 alone say $m(t_1)$.

$$g_1(t_1; \theta) = h(t_1; \theta)m(t_1) \dots \dots \dots (2)$$

$$\text{If } m(t_1) = 0 \Rightarrow g_1(t_1; \theta) = 0$$

If $m(t_1) > 0$, then eq (2) can be written as;

$$G \quad \frac{g_1(t_1; \theta)}{m(t_1)} = h(t_1; \theta)$$

$$\because t_1 = \phi_1(X_1, X_2, \dots, X_n)$$

$$h[\phi_1(X_1, X_2, \dots, X_n; \theta)] = \frac{g_1[\phi_1(X_1, X_2, \dots, X_n; \theta)]}{m[\phi_1(X_1, X_2, \dots, X_n)]}$$

putting in eq(A)

$$f(x_1; \theta)f(x_2; \theta) \dots f(x_n; \theta) = \frac{g_1[\phi_1(X_1, X_2 \dots X_n; \theta)]}{m[\phi_1(X_1, X_2 \dots X_n)]} k(X_1, X_2 \dots X_n)$$

$$\frac{f(x_1; \theta)f(x_2; \theta) \dots f(x_n; \theta)}{g_1[\phi_1(X_1, X_2 \dots X_n; \theta)]} = \frac{k(X_1, X_2 \dots X_n)}{m[\phi_1(X_1, X_2 \dots X_n)]}$$

Since neither the function k nor the function m depends on θ , then in accordance with the definition T_1 is a sufficient statistic.

Conversely, if T_1 is a sufficient statistic for θ . Then the conditional distribution of $X_1, X_2 \dots X_n$ given $T_1 = t_1$ is independent of θ .

$$\frac{f(X_1; \theta) \dots f(X_n; \theta)}{g_1(t_1; \theta)} = k(X_1, X_2 \dots X_n)$$

$$f(X_1; \theta) \dots f(X_n; \theta) = g_1(t_1; \theta) k(X_1, X_2 \dots X_n)$$

$$f(X_1; \theta) \dots f(X_n; \theta) = g_1[\phi_1(X_1, X_2 \dots X_n; \theta)] k(X_1, X_2 \dots X_n)$$

The factorization can be realized by taking the function h to be the p.d.f of t_1 that is

$$g_1[\phi_1(X_1, X_2 \dots X_n; \theta)] = h[\phi_1(X_1, X_2 \dots X_n; \theta)]$$

Hence,

$$f(X_1; \theta) \dots f(X_n; \theta) = h[\phi_1(X_1, X_2 \dots X_n; \theta)] k(X_1, X_2 \dots X_n)$$

As required.

6.21 Theorem: Every One-One function of Sufficient Statistic is Also Sufficient

Question 74: If T is a sufficient statistic for θ every one-one function $Z = u(T)$ or $Z = u[t(X_1, X_2 \dots X_n)] = v(X_1, X_2 \dots X_n)$ is also sufficient.

Proof:

we write factorization theorem

$$f(X_1; \theta) \dots f(X_n; \theta) = k_1[t(X_1, X_2 \dots X_n); \theta] k_2(X_1, X_2 \dots X_n)$$

Let the inverse function be

$$t = WZ$$

$$t(X_1, X_2 \dots X_n) = W[v(X_1, X_2 \dots X_n)]$$

Which is not a function of θ

$$f(X_1; \theta) \dots \dots f(X_n; \theta) = k_1[W\{v(X_1, X_2 \dots X_n)\}; \theta]k_2(X_1, X_2 \dots X_n)$$

Since the first function of the R.H.S of above equation is a function $Z = v(X_1, X_2 \dots X_n)$ and θ while the second function does not depend upon θ .

The factorization theorem implies that $Z = u(T)$ is also a sufficient statistic for θ .

Note: In normal $N(\mu, \sigma^2)$, $\sum x_i$ is sufficient for μ . and also $\bar{X} = \frac{1}{n} \sum x_i$ is also sufficient for μ .

